Heterogeneity in Overnight-versus-Intraday Stock-Return Behavior with Asymmetric Information and 52-week Price Anchoring*

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Abstract

We document strong and robust cross-sectional heterogeneity in the overnight–intraday return anomaly along two distinct dimensions. First, a stock's asymmetric-information environment, summarized by our composite information-asymmetry metric (INFO-Z), predicts how last month's overnight return is propagated into this month's overnight and intraday returns. Higher INFO-Z stocks exhibit stronger overnight momentum and a stronger intraday cross-reversal. Second, a stock's price relative to its 52-week history, summarized by a composite price-to-52-week-anchor metric (PTA), predicts how last month's intraday return is propagated forward. High PTA stocks exhibit weaker intraday momentum and a weaker overnight cross-reversal. Our INFO-Z results complement the overnight-versus-intraday literature that emphasizes the role of differentially informed agents. Our PTA results introduce the empirical importance of price anchoring in this setting and are consistent with an extrapolative-belief role. Finally, higher INFO-Z stocks earn higher average overnight returns and lower average intraday returns, highlighting the time-of-day dimension of information asymmetry.

JEL Classification: G11, G12, G14

Keywords: overnight-intraday return anomaly, information asymmetry, anchoring bias, 52-week price anchor

1. Introduction

Lou, Polk, and Skouras (2019) document a striking pattern in stocks' daily returns: both the overnight and intraday return components display strong continuation, yet the two components also exhibit an offsetting cross-period reversal.[1] Hendershott, Livdan, and Rösch (2020) similarly show that stock returns are positively related to market beta overnight but negatively related during the trading day. Together, these findings suggest that the market open and the subsequent trading day embed different pricing forces, but the economic sources of those forces remain unsettled.[2]

Existing explanations emphasize investor heterogeneity, institutional frictions, information asymmetry, or belief distortions. Lou, Polk, and Skouras (2019) emphasize persistent demand from different investor clienteles. Bogousslavsky (2021) stresses higher funding and short-selling frictions overnight. Lu, Malliaris, and Qin (2023) develop a model in which information asymmetry increases the price of liquidity at the open. Jones, Pyun, and Wang (2024) highlight extrapolative beliefs and morning price-chasing. Akbas et al. (2022) show that the intensity of the daily tug of war predicts future stock returns. These studies provide important explanations for the anomaly, but the literature has not directly examined whether economically motivated firm-level characteristics systematically condition how prior overnight and intraday returns propagate across months.

Our primary contribution is to document two distinct empirical propagation patterns. First, a stock's asymmetric-information environment, summarized by our composite information-asymmetry measure (INFO-Z), organizes how last month's *overnight* return is related to next month's overnight return (momentum) and intraday return (cross-reversal). Second, a stock's 52-week price-anchoring state, summarized by a composite price-to-52-week-anchor (PTA) metric, organizes how last month's *intraday* return is related to next month's intraday return (momentum) and overnight return (cross-reversal). Thus, the familiar overnight–intraday anomaly contains substantial and systematic cross-sectional heterogeneity that is not apparent from the unconditional pattern alone.

The two findings are useful for different reasons. The INFO-Z evidence is empirically stronger and provides a direct quantitative characterization of a dimension that is fundamental to several existing explanations. Although its broad direction is economically intuitive, the magnitude, pervasiveness, and concentration on prior overnight-return propagation are new empirical facts. The PTA

[1] Continuation means that last month's return-component winners (losers) are more likely to exhibit relatively higher (lower) returns this month too. In this setting, we consider momentum and persistence as synonymous terms, and we generally use the term momentum in our study.

[2] Related aggregate-market evidence also highlights the importance of time-of-day segmentation. Using S&P 500 futures traded around the clock, Bondarenko and Muravyev (2023) show that average market returns are concentrated in a narrow window around the European open. Their setting differs from ours because they study aggregate index-futures returns rather than the cross section of individual stocks.

dynamic patterns are somewhat weaker statistically but more surprising from the perspective of the overnight–intraday literature. To our knowledge, 52-week price anchoring has not been emphasized as a conditioning variable in this literature. Its association with prior intraday-return propagation therefore introduces a new price-anchor dimension and provides a natural connection to explanations based on extrapolative beliefs.

The empirical motivation follows directly from the related literature. The overnight period is much less liquid, while after-hours trades can be especially informationally impactful (Barclay and Hendershott (2003)), suggesting that asymmetric information may be particularly relevant for the propagation of overnight returns. Separately, if salient 52-week price anchors affect how investors digest information and form extrapolative beliefs, they may help explain the propagation of intraday returns. Jones et al. (2024), for example, document extrapolative trading near the market open, particularly among retail investors. Sections 2.2 and 7 elaborate on these potential linkages.

We construct INFO-Z from a stock's probability of informed trading, idiosyncratic volatility, and bid–ask spread, while purging its common component with well-known illiquidity proxies. We measure 52-week price anchoring using the PTA metric of Chen et al. (2026), which equally weights information in a stock's price-to-52-week-high and price-to-52-week-low ratios. Section 3 develops these measures in detail.

Our main results use individual stock returns from 1997 to 2021.[3] Higher-INFO-Z stocks exhibit stronger overnight momentum and a stronger intraday cross-reversal following prior overnight returns. In double-quintile sorts, the overnight momentum spread rises from 1.08% per month for low-INFO-Z stocks to 3.09% for high-INFO-Z stocks (Table 1), while the intraday cross-reversal increases in magnitude from -1.06% to -2.79% (Table 4).[4]

The PTA results reveal a different pattern. Intraday momentum declines from 2.42% per month for low-PTA stocks to 1.04% for high-PTA stocks (Table 6), while the overnight cross-reversal following prior intraday returns weakens from -2.70% to -1.37% (Table 9). Thus, INFO-Z is most closely linked to prior overnight-return propagation, whereas PTA is most closely linked to prior intraday-return propagation.

The featured propagation patterns remain evident across alternative portfolio sorts and Fama–MacBeth regressions, survive controls motivated by Lu et al. (2023), Jones et al. (2024), and Akbas et al. (2022), and are stronger in higher-sentiment states. We do not claim that INFO-Z and PTA

[3]The sample period is dictated by the authors' availability of high-frequency intraday returns.

[4]By monthly overnight or intraday return, we refer to the cumulation of the corresponding daily return component over a calendar month. Throughout our portfolio analysis, we use value-weighted returns to mitigate concerns that small-cap stocks drive the results.

are the only firm characteristics that can condition day/night return dynamics. Rather, the evidence identifies two economically motivated, empirically robust, and differently oriented dimensions of that heterogeneity.

Interestingly, for these dynamic INFO-Z and PTA findings, stocks that were past-winners with high INFO-Z, or high PTA, are more influential in establishing the conditional patterns. This evidence assists in interpretation, while also suggesting that highly illiquid stocks are not overly influential in our results.[5] This liquidity view is also supported by our size-segmented analysis, where the conditional patterns remain strongly evident for stocks in the top-third of market capitalization, and by our focus on value-weighted portfolio returns in our main results.

Finally, we find that higher INFO-Z stocks have appreciably higher average overnight returns and appreciably lower average intraday returns. These patterns suggest that the relation between asymmetric information and average excess stock returns has an important time-of-day dependence.

Our primary contribution is therefore to the overnight–intraday return literature, with additional implications for two related literatures. The INFO-Z average-return results add a time-of-day perspective to the debate over whether asymmetric-information risk is priced in the cross section, while the PTA evidence extends the 52-week price-anchor literature into the day/night return setting. Section 7 discusses how our findings relate to the leading explanations in the overnight–intraday literature and highlights the empirical implications of the distinct INFO-Z and PTA patterns.

The remainder of the paper is organized as follows. Section 2 reviews leading explanations for overnight–intraday return patterns and motivates the roles of asymmetric information and 52-week price anchoring. Section 3 describes the data and variable construction. Sections 4 and 5 present the main INFO-Z and PTA results, respectively. Section 6 presents additional analyses and controls, Section 7 discusses implications for the related literature, and Section 8 concludes.

2. Explanations for the Overnight-Intraday Return Anomaly

Overnight-versus-intraday return behavior has attracted substantial attention in the recent finance literature because it indicates distinct pricing forces during trading and non-trading hours, with striking patterns of: (1) momentum in the successive returns of both monthly overnight returns and monthly intraday returns; and (2) cross tug-of-war reversals between the overnight and intraday returns. In this section, we review explanations and perspectives from five key studies in this

[5]This premise follows from the view that stocks that were past relative losers with a low-PTA are likely to have a smaller market-cap and/or reflect greater financial distress, and hence lower liquidity (on average); relative to past-winner stocks with a high PTA. See discussion and evidence in Section 6.5.

literature. From the perspective of these existing explanations, we then briefly discuss why the cross-sectional variation in asymmetric information and 52-week price anchoring influences might bear on this return behavior. A more detailed discussion of our findings' implications for the literature is deferred to Section 7, after presenting our empirical results.

2.1. Existing Explanations of the Overnight-Intraday Return Anomaly

The recent literature has provided important insights that have helped characterize and explain the overnight–intraday return anomaly, while also adding to the body of evidence in this area. To frame our study, we focus on the following five studies.

First, Lou, Polk, and Skouras (2019) interpret the anomaly through investor heterogeneity. Their main idea is that different clienteles tend to dominate different parts of the trading day, creating persistent price pressure in both overnight and intraday returns together with an offsetting same-day reversal. In their framework, the key empirical objects are past overnight and past intraday returns of characteristic-sorted portfolios, which reveal a recurring "tug of war" between these two segments of the trading cycle. This clientele-based explanation is an important starting point for the literature because it establishes that the anomaly is systematic and closely tied to time-of-day segmentation.

Second, Bogousslavsky (2021) studies the overnight-intraday anomaly with a focus on institutional constraints and overnight illiquidity risk. He argues that arbitrageurs may be reluctant to carry positions overnight because of higher margin requirements, stock-lending fees, and overnight risk, which can induce them to unwind some positions prior to the market close. Consistent with this intuition, he finds that a mispricing factor earns positive average returns over every half-hour trading period and the overnight period, except for the closing half-hour when it earns a sizably negative average return. This mechanism helps explain why some anomaly returns are realized during the day but reverse across the overnight period.

Third, Akbas et al. (2022) contribute with an explanation focusing on the intensity of the daily tug of war. Their key idea is that a stock is engaged in a more intense tug of war when it repeatedly experiences positive overnight returns followed by negative daytime reversals. They interpret this pattern as reflecting pressure from optimistic overnight noise traders that is subsequently offset by daytime arbitrageurs. Their main empirical variable is the *abnormal frequency* of such negative daytime reversals, denoted $ABNR$, which scales the current monthly frequency of positive-overnight/negative-day reversals by the stock's own recent history. Akbas et al. show that a higher value of this tug-of-war measure predicts higher future returns, mainly through the subsequent overnight component, and they interpret this as evidence that daytime arbitrageurs tend

to *overcorrect* persistent positive overnight price pressure.

Fourth, Lu et al. (2023) provide one of the leading rational theories. Their model emphasizes heterogeneous liquidity providers: 'fast arbitrageurs' (e.g., market makers or de facto market makers) possess informational advantages, while 'slow arbitrageurs' (e.g., large asset managers) have lower inventory costs. When information asymmetry is high around the market open, fast arbitrageurs can cream-skim uninformed flow and charge a high price of liquidity. Then, as information asymmetry subsides during the trading day, slow arbitrageurs increasingly undercut them, which compresses the initial price deviation and generates opposite-signed overnight and intraday returns. A key implication of their theory is that the anomaly should be linked to predictable open order imbalances and to variables that proxy for the price of liquidity supplied by market makers. Empirically, Lu et al. focus on expected order imbalances near the market open, especially retail order imbalance absorbed by market makers, and on measures of liquidity-provision costs and market-making conditions. Their explanation is fundamentally about *open-price formation* and the likelihood of trading associated with either asymmetric information or noise.

Finally, Jones et al. (2024) provide a behavioral alternative, where the central hypothesis is that differences between overnight and daytime returns arise from *return extrapolation*. They argue that investors extrapolate recent positive daytime returns, that such extrapolative trading is concentrated near the market open, and that the effect is asymmetric because short-selling constraints mute the response to negative past returns. In their empirical work, the main variables are lagged daytime returns, measures of morning order imbalance, return-dispersion proxies, and variables related to retail attention or retail trading intensity. They show that order imbalances in the first minutes of trading are strongly positively related to lagged daytime returns, and that this mechanism can account for a large part of the 'tug of war' between overnight and daytime returns.

Other recent evidence further highlights the importance of distinguishing overnight and intraday return signals. Barardehi, Bogousslavsky, and Muravyev (2026) document that medium-term momentum is concentrated in past intraday returns rather than overnight returns and interpret this pattern as consistent with underreaction to private information conveyed through trading. Our analysis addresses a complementary question: whether firm-level information asymmetry and price anchoring condition the persistence and cross-reversal of recent overnight and intraday returns. By examining these conditioning effects separately for subsequent overnight and intraday returns, we identify distinct sources of heterogeneity in short-term return dynamics.

2.2. Overnight-versus-Intraday Stock Return Behavior, with Cross-sectional Variation in Asymmetric Information and 52-week Price Anchoring

Our main findings point to a simple distinction. We document striking heterogeneity in how monthly *overnight* returns are propagated into next month's overnight returns (momentum) and intraday returns (cross-reversals), based on cross-sectional variation in a stock's asymmetric-information environment. Separately, we document substantial heterogeneity in how monthly *intraday* returns are propagated into next month's intraday returns (momentum) and overnight returns (cross-reversals), based on cross-sectional variation in a stock's 52-week price-anchoring state.

In our view, the existing overnight–intraday literature suggests why these two conditioning variables might organize different parts of the return dynamics. Asymmetric information is fundamental to explanations that rely on better-informed traders, such as the fast arbitrageurs in Lu et al. (2023), or on arbitrageurs trading on mispricing, as in Bogousslavsky (2021). The overnight period is characterized by appreciably lower liquidity, while Barclay and Hendershott (2003) find that after-hours trades, on average, contain more information than intraday trades when the market is open. Thus, cross-sectional variation in asymmetric information may be especially relevant for how overnight returns are propagated forward.

Price anchoring offers a different connection. Since Tversky and Kahneman (1974), a large literature has shown that agents often evaluate outcomes relative to salient reference values. In finance, the price-to-52-week-high ratio (PTH) of George and Hwang (2004) is one of the best-known anchor-based variables. Their evidence suggests that stocks near their 52-week highs tend to underreact to positive news, consistent with investors perceiving such stocks as already 'expensive' relative to a salient benchmark.[6]

In the overnight–intraday setting, this anchoring logic is particularly relevant to the return-extrapolation perspective of Jones et al. (2024). They find evidence of positive extrapolative trading around the open, especially among retail investors. Combined with the price-anchoring literature, this suggests that a high PTA could temper extrapolative demand following strong prior intraday performance, and thereby weaken subsequent intraday momentum. This possibility also provides a natural connection to the associated overnight cross-reversal. We return to these interpretations in Section 7, after establishing the empirical findings.

After our main findings are established, Section 7 discusses the implications of our collective evidence for the literatures on: (i) overnight-versus-intraday return behavior, (ii) asymmetric infor-

[6] See Li et al. (2023) and Chen et al. (2026), and the papers cited within, for more recent evident about an apparent bias in price formation linked to a stock's 52-week price anchor.

mation, and (iii) 52-week price anchoring.

3. Data and Variable Description

Our sample covers the period from 1997 to 2021 and includes all common stocks traded on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and NASDAQ. We obtain daily and monthly transaction data—such as stock returns, prices, and shares outstanding—from the Center for Research in Security Prices (CRSP). Intraday transaction data are sourced from the TAQ database, while financial statement data are retrieved from Compustat.

To measure intraday and overnight returns, we use the opening and closing prices from CRSP. Following Lou, Polk, and Skouras (2019), we define the intraday return as the price change between the market open and close on the same day. The overnight return is then calculated from the intraday return and the daily close-to-close return. Specifically, for stock i on day k, the intraday return is

$$r^i_{intraday,k} = \frac{P^i_{close,k}}{P^i_{open,k}} - 1,$$

and the overnight return is

$$r^i_{overnight,k} = \frac{1 + r^i_{close-to-close,k}}{1 + r^i_{intraday,k}} - 1.$$

Then, for each individual stock for each calendar month, we compute a monthly intraday return and a monthly overnight return, by cumulating the above daily intraday and overnight returns, respectively. See Lou, Polk, and Skouras (2019), page 196, for the equations.

We construct our primary measure of information asymmetry, the INFO-Z variable, as a composite of three widely used indicators of information asymmetry from prior studies (e.g., Venkatesh and Chiang, 1986; Durnev, Morck, Yeung, and Zarowin, 2003; Durnev, Morck, and Yeung, 2004; Chen, Goldstein, and Jiang, 2007): a stock's probability of informed trading (PIN), a stock's bid-ask spread, a stock's idiosyncratic return volatility. The estimation process for these measures and the calculation of our three-factor INFO-Z variable are outlined below.

Each month, we calculate the three asymmetric-information metrics based on the past 12 months of data. First, following Easley et al. (2002), we estimate PIN using transaction data from the TAQ database. We apply the data-cleaning procedure of Holden and Jacobsen (2014) and classify buy and sell orders using the method of Ellis et al. (2000). To mitigate numerical biases from underflow or overflow, we adopt the approaches proposed by Lin and Ke (2011) and Yan and Zhang (2014).

Second, we estimate idiosyncratic volatility using the Carhart (1997) four-factor model with daily returns. Third, the bid-ask spread is measured as in Corwin and Schultz (2012), using CRSP data. Next, each of the three metrics is winsorized at the 1% and 99% levels and then standardized into Z scores (mean = 0, standard deviation = 1). A stock's simple asymmetric-information score for the month is the average of the three Z scores from the stock's PIN, spread, and idiosyncratic volatility.

Our primary interest for this aspect of our empirical investigation is about cross-sectional variation in the likelihood of asymmetric-information trading across stocks. We acknowledge that a higher likelihood of asymmetric-information trading also suggests a less liquid stock, on average. Accordingly, to construct an asymmetric-information measure that is more distinct from illiquidity, we regress the simple average of the three Z scores against two measures that are more representative of a stock's illiquidity; a stock's Amihud's (2002) PIM illiquidity measure and a stock's turnover. PIM illiquidity is estimated using the past 12 months of CRSP data, while stock turnover is calculated as the average monthly turnover over the past 12 months. Then, the residuals from this regression are used as a stock's INFO-Z score for the month.[7]

The second major aspect of our empirical investigation is whether 52-week price anchoring has a role in understanding the momentum in monthly overnight and intraday stock returns. To measure whether a stock's current price seems expensive, relative to its 52-week price history, we use the 52-week price-to-anchor (PTA) metric from Chen et al. (2026). For a given month, a stock's PTA is constructed as the average of its standardized 52-week price-to-high ratio (PTH) and 52-week price-to-low ratio (PTL). The standardization is done every month by taking stocks' simple PTH (PTL) values for the month and converting them to have a mean of zero and a standard deviation of one in the usual way, based on the month's cross-sectional PTH (PTL) distribution.

When ranking stocks cross-sectionally, Chen et al. (2026) argue that this PTA should better indicate how expensive a stock is, relative to its 52-week price history; instead of using a stock's PTH by itself (or PTL by itself). In a cross-sectional ranking, a stock with a narrow 52-week trading band could have its current stock price indicate both a high PTH (indicating a relatively expensive stock) and a low PTL (indicating a relatively inexpensive stock). The PTA metric equally weights a stock's standardized PTH and PTL to mitigate this 'narrow-trading band' concern, where a stock's price could be relatively close to both its 52-week high and 52-week low. With the PTA, stocks with both a high PTH and high PTL would have a higher PTA and would rank as being relatively more expensive than stocks with only a high PTH (for example).[8]

[7] Our principal INFO-Z results are similar without controlling for the illiquidity variables.

[8] See Chen et al. (2026) for a more detailed discussion of the PTA and for a numerical example of the so-called 'narrow trading band' concern with PTH-only cross-sectional rankings.

In the Chen et al. study (2026), they find that the PTA metric outperforms a simpler PTH metric in their setting (in terms of the empirical importance of the 52-week price anchor in explaining the subsequent stock price behavior of interest). Following from their study, we use the lagged PTA as of the close of month $t-2$ to serve as a firm-specific state variable that indicates the expensiveness of a stock, relative to its 52-week price history. We then evaluate whether a stock's anchoring state has a role in understanding momentum variation in monthly overnight and intraday returns over the subsequent months $t-1$ and t. We also find that the PTA outperforms the PTH in our setting.

To examine night/day return patterns, we form 25 value-weighted portfolios (5×5) using sequential quintile sorts. At the end of month $t-1$, stocks are first sorted into quintiles based on either INFO-Z or PTA, measured as of the close of month $t-2$, using NYSE breakpoints. Within each INFO-Z or PTA quintile, stocks are then sorted into quintiles based on their past one-month overnight or intraday return in month $t-1$, again using NYSE breakpoints. We exclude microcap stocks, defined as stocks priced below \$5 or in the bottom NYSE size quintile.

For portfolio performance evaluation, we compute value-weighted one-month overnight/intraday returns at the end of month t. In our main tables, we report excess average returns for each of the 25 portfolios from the 5 x 5 double sorts, with excess returns formed by differencing from the 1-month T-bill rate. For the high-minus-low strategies reported in the tables formed from the 25 portfolios, we report excess average returns and alphas based on an extended Fama-French five-factor model (Fama–French, 2015) and an extended Q-factor model (Hou et al., 2015). The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). Following the Fama and French (2018) study that evaluates an extended Fama-French 5-factor model plus a momentum factor, we refer to this alpha as α_{FF6+2} in our tables. The Q-factor model is also extended by adding the same two behavioral factors, referred to as α_{Q+2} in our tables. Finally, t-statistics are computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags.

To examine the sensitivity of our results to investor sentiment, we split our full sample period into terciles based on the Baker and Wurgler (2006) investor sentiment index and replicate our portfolio analysis within each period.

Additionally, when estimating Fama–MacBeth (1973) cross-sectional regressions, we control for a standard set of firm characteristics. Specifically, following Lou et al. (2019), we include the following control variables measured in month $t-2$: the exponentially weighted moving average overnight/intraday returns with a half-life of 60 months, the lagged 12-month cumulative stock return, market capitalization, book-to-market ratio, 12-month market beta computed using daily

returns with three lags, 12-month stock turnover, return-on-equity, asset growth, equity issuance, and discretionary accruals. For our Fama-MacBeth regression analyses, we winsorize all the continuous variables at the 0.5% and 99.5% levels to mitigate the influence of extreme outliers or data errors.[9]

4. Information Asymmetry and Overnight & Intraday Return Behavior: Main Empirical Results

4.1. Main INFO-Z Momentum Results with Sequential Double-Sorted Portfolios

We begin by examining the relation between INFO-Z, our preferred proxy for information asymmetry, and the momentum in successive monthly overnight and intraday returns. Table 1 reports the results over our full sample from January 1997 through December 2021. Throughout, we focus on value-weighted portfolios to reduce the influence of illiquid stocks, and we exclude stocks priced below \$5 and stocks in the lowest NYSE size quintile to further limit the role of micro-cap firms.

Panel A of Table 1 studies whether the momentum in overnight returns varies systematically with information asymmetry. At the end of month $t-1$, we first sort stocks into INFO-Z quintiles using INFO-Z measured as of the close of month $t-2$, based on NYSE breakpoints. Within each INFO-Z quintile, we then sort stocks into quintiles according to their past one-month overnight return in month $t-1$, again using NYSE breakpoints. This timing convention helps separate the information-asymmetry signal from the return-sorting variable. We then examine the average month-t overnight returns of the resulting 25 portfolios, along with the high-minus-low spread portfolios, their excess returns, and alphas relative to the extended Fama–French six-factor plus two behavioral-factor model and the corresponding extended q-factor benchmark.

To begin, we note that our findings are consistent with the literature that documents strong momentum in both monthly overnight and intraday individual stock returns. For every INFO-Z quintile, we find that past-winner stocks reliably outperform past-loser stocks for both monthly overnight and intraday returns, indicating substantial momentum for both daily-return components.

Next, relating to cross-sectional variation in information asymmetry, the results in Panel A deliver two clear findings for the monthly overnight-return behavior. First, the overnight-return momentum increases substantially with asymmetric information. Specifically, the average excess overnight return of the high-minus-low past-overnight-return strategy increases from 1.08% per

[9]Unlike Lou et al. (2019), we did not include idiosyncratic volatility as a control variable because it is already included in our INFO-Z score. For additional robustness, we also include the industry fixed effects, using the Fama-French 48-industry definitions.

month in the lowest INFO-Z quintile (t=3.99) to 3.09% per month in the highest INFO-Z quintile (t=5.06). The corresponding alphas for the momentum returns also rise materially and reliably with INFO-Z. Thus, the difference between the overnight momentum spread in the highest versus lowest INFO-Z quintile is 2.01% per month in excess returns (t=5.05), with corresponding alpha differences of 1.64% (t=4.76) and 1.65% (t=4.87).

Second, we find that the average monthly overnight returns are increasing with information asymmetry for all five of the past-return groupings. For each past-overnight-return quintile, the average overnight returns rise with INFO-Z, and the INFO-Z spread is especially pronounced for the past-overnight-winner stocks. Specifically, within the highest past-overnight return quintile, the high-INFO-Z portfolio averages 3.47% per month, compared with 0.97% for the low-INFO-Z portfolio, yielding a difference of 2.50% (t=6.26). The comparable factor-adjusted alphas are 2.20% (t=6.24) and 2.21% (t=6.37). Overall, Panel A provides strong evidence that a higher information-asymmetry metric for a stock predicts both stronger momentum in monthly overnight returns and higher average monthly overnight returns in the cross section.

Next, Panel B of Table 1 turns to the intraday leg of the analysis. The construction is identical, except that the second-stage sort uses past one-month intraday returns rather than past one-month overnight returns, and the dependent variable is the month-t intraday return. We find very different results than those observed for the overnight returns.

To begin, unlike the overnight results, there is relatively little variation in intraday-return momentum with INFO-Z. The higher asymmetric-information stocks have marginally higher momentum, but the difference with lower asymmetric-information stocks lacks statistical significance.

Further, opposite to the pattern in monthly overnight returns, we find that the average excess intraday returns and alphas are decreasing in information asymmetry for each past-intraday-return quintile. Across the five past-return quintiles, the average difference between the high INFO-Z and low INFO-Z average intraday stock returns is -1.13% with t-stats of (-3.00) or more negative for every past-return quintile. The comparable alpha differences are also sizable and statistically significant.

Next to probe further the robustness and pervasiveness of these INFO-Z return patterns, we report comparable results for one-half subperiods in Table 2: 1997:01-2009:06 and 2009:07-2021:12. Panel A reports on monthly overnight returns and Panel B on monthly intraday returns.

First, for the monthly overnight returns in the subperiod analysis, we see the same qualitative patterns in both one-half subperiods: (1) the momentum in monthly overnight returns increases with INFO-Z; and (2) average overnight returns and alphas increase with INFO-Z. The patterns are generally stronger for the first-half subperiod, but they are sizable and statistically significant for

both subperiods.

Second, for the monthly intraday returns with our one-half subperiod analysis, we find both interesting consistencies and differences across the subperiods. Consistent with our full-sample results, we find that the average intraday returns and alphas are decreasing with INFO-Z for both one-half subperiods, with statistically reliable differences in most cases. However, the variation in monthly intraday-return momentum with INFO-Z is inconsistent between the two subperiods; sizably and reliably increasing with INFO-Z for the first-half subperiod versus a negative and statistically insignificant INFO-Z variation for the second-half subperiod. These subperiod differences align with the full-sample momentum results, which indicated no reliable difference for the monthly intraday return momentum as INFO-Z varied.

4.2. INFO-Z Momentum Results: Robustness with Methodology Variations

4.2.1. With Independently Double-Sorted Portfolios on INFO-Z and Past Returns

The evidence in Tables 1 and 2 is based on sequential double-quintile sorts, in which stocks are first sorted on INFO-Z and then, within each group, sorted on past one-month overnight or intraday returns. That dependent sorting procedure is well suited to our empirical question, because it directly asks whether the momentum behavior varies across the conditioning variable of interest. Nevertheless, it is useful to verify that the same qualitative conclusions also emerge under an alternative independent sorting design for the two conditioning variables.

We perform the independent-sort analysis and find qualitatively consistent results. First,the positive relation between INFO-Z and overnight-return momentum remains clearly evident. For the independent sorts, the difference in overnight-return momentum between the high INFO-Z stocks and low INFO-Z stocks is 1.54% (t=4.28). However, there is little variation in the intraday-return momentum with INFO-Z. Second, again we find that the average monthly overnight returns (intraday returns) are increasing (decreasing) with INFO-Z. We conclude that our main asymmetric-information results are robust with the alternative independent double-sorting procedure. Tabular results and additional details are provided in External Appendix A.1.

4.2.2. With an Alternative 3-by-10 Double-Sorting Granularity

Our main results are based on double-quintile sorting on a stock's lagged INFO-Z and prior month's overnight or intraday return. To evaluate robustness further, we next perform a similar empirical exercise, but with a tercile sort on the past INFO-Z and a decile sort on the past return (or 3-by-10).

The same qualitative INFO-Z patterns are also evident in this case, similar to results in Table 1. External Appendix A.2 provides details, along with tabular results.

4.2.3. With a Simpler INFO-Z that Does Not Control for Illiquidity Variables

As we explained in Section 3, recall that our featured INFO-Z variable controls for two common illiquidity variables; a stock's Amihud's (2002) PIM and turnover. Here, we evaluate a simpler INFO-Z that is defined simply as the average Z-score of the three asymmetric-information variables: PIN, spread, and idiosyncratic volatility. We repeat the empirical exercise of Table 1, but with this alternative simpler INFO-Z construct.

We find similar qualitative results with this simpler INFO-Z variable. The monthly overnight-return momentum is increasing with the simpler INFO-Z (difference in momentum of 1.93% (t=4.77) between the high and low INFO-Z quintiles). And, again, for each past-return quintile, the average excess overnight return (intraday return) is increasing (decreasing) with the simpler INFO-Z variable too. External Appendix A.3 provides details, along with tabular results.

4.2.4. With Each INFO-Z Input Separately - PIN, Spread, and Idio. Volatility

Recall that our featured INFO-Z is a composite 3-factor asymmetric-information variable that is constructed from a stock's PIN, spread, and idiosyncratic volatility. Here, rather than our composite INFO-Z variable, we similarly evaluate single-factor INFO-Z variables for each of these three inputs separately. This both probes robustness and helps evaluate which of the three inputs is empirically more important in our main results. We re-estimate different versions of Table 1 for each of the single-factor INFO-Z variables.

For each of the three single-factor INFO-Z variables, we find results quite similar to our composite 3-factor INFO-Z— just with modestly weaker INFO-Z contrasts. For each single-factor INFO-Z, we find that: (1) the monthly overnight-return momentum is increasing with INFO-Z, and (2) for each past-return quintile, the average excess monthly overnight (intraday) return is increasing (decreasing) with INFO-Z. These patterns are broadly similar for each single-factor INFO-Z variable, with the idiosyncratic-volatility metric having the largest INFO-Z difference in overnight momentum at 1.54% (t=4.71) versus 1.47% (t=4.62) for the PIN and 1.36% (t=5.54) for the spread. External Appendix A.4 provides details, along with tabular results.

4.3. INFO-Z Momentum Results for Large-cap, Mid-cap, and Small-cap Firms

Our main results in the prior two subsections evaluate value-weighted portfolio returns, ensuring that our results are not driven solely by small-cap firms. Next, to evaluate whether our main asymmetric-information results vary importantly with firm size, we repeat our main INFO-Z analysis (as in Table 1) for large-cap, mid-cap, and small-cap stocks separately.

Table 3 reports the results, classifying firms into large-cap, mid-cap, and small-cap terciles based on market capitalization measured at the end of month $t-2$. Here, we focus on the INFO-Z patterns in overnight returns, given the sizable INFO-Z variation in overnight-return momentum in our prior results. Panels A, B, and C report on large-cap, mid-cap, and small-cap stocks, respectively.

We find that the overnight-return momentum is increasing with INFO-Z for all three firm-size groups, with the variation being marginally stronger for the small-cap stocks. For large-cap stocks, the high-minus-low overnight-return spread is 1.10% for the low INFO-Z group and 2.57% for the high INFO-Z group, for a difference of 1.47% (t=4.26). The comparable INFO-Z-based differences in overnight-return momentum are 1.47% ($t = 3.55$) for mid-cap stocks and 1.69% ($t = 4.02$) for small-cap stocks. The risk-adjusted alphas for these momentum differences are also sizable and highly statistically significant. Thus, the role of information asymmetry on overnight-return momentum is pervasive across stocks of different market capitalization.

Next, Table 3 also shows the pattern between stocks' INFO-Z and their average excess overnight-returns. For all three size groupings and for both the past losers and past winners, the high INFO-Z stocks always have higher average monthly overnight-returns than the comparable low INFO-Z stocks. These differences are sizable with high statistical significance for all cases, except for the past-loser, small-cap grouping that lacks statistical significance.

4.4. Tug-of-War Cross-Reversal Results with INFO-Z

Table 4 reports on the tug-of-war cross-reversal patterns with INFO-Z. The tug-of-war terminology follows from Lou et al. (2019) and Akbas et al. (2022). Here, we perform the same INFO-Z first-stage sort, but the return evaluation is a cross effect, meaning that: (1) in Panel A, past overnight returns over month $t-1$ are used in the sort, but then we evaluate the subsequent intraday returns over month t; and (2) in Panel B, past intraday returns over month $t-1$ are used in the sort, but then we evaluate the subsequent overnight returns over month t.

Before reporting on the INFO-Z results, we note that cross-reversal behavior is evident in both cases above. When sorting stocks on their past month's overnight returns, then the subsequent

average intraday returns are decreasing with the past overnight return (opposite to the momentum in successive monthly overnight returns). And, when sorting stocks on their past month's intraday returns, then the subsequent average overnight returns are decreasing with the past intraday return (opposite to the momentum in successive monthly intraday returns).

For the monthly intraday returns in Table 4, Panel A, the cross-reversal behavior increases appreciably with INFO-Z. For the high INFO-Z stocks, the difference in average monthly intraday returns between past-overnight-return-winner stocks and past-overnight-return-loser stocks is -2.79% (t=-7.27). Conversely, for the low INFO-Z stocks, this same difference in average intraday returns is -1.06% (t=-4.93). Thus, the difference in cross-reversal behavior between the two INFO-Z groupings is -1.73% (t=-6.15), with this difference also having sizable and highly reliable alphas.

Next, Table 4, Panel B, reports on a comparable cross-reversal analysis between this month's overnight returns and last month's intraday returns. For this case, we find only a marginal negative relation between INFO-Z and the cross-reversal relation.

Shifting to how average returns vary with INFO-Z, we note the following. The average overnight returns are increasing with INFO-Z (Panel B), with statistically significant differences between the high and low INFO-Z groupings for all five past-intraday-return quintiles. And, the average intraday returns are decreasing with INFO-Z (Panel A), with statistically significant differences between the high and low INFO-Z groupings for four of the five past-overnight-return quintiles.

We perform additional analysis to probe robustness and pervasiveness. First, we find similar INFO-Z patterns in a tug-of-war cross-reversal analysis that uses independent sorts on the two conditioning variables, rather than sequential. External Appendix A.5 has details. Second, we similarly evaluate one-half subperiods for the cross-reversal INFO-Z patterns. For both one-half subperiods, we find that the cross-reversals between this month's intraday returns and last month's overnight returns become stronger (more negative) with INFO-Z. External Appendix A.6 provides detailed results.

4.5. INFO-Z Evidence from Fama-MacBeth Cross-sectional Regressions

To complement our prior portfolio-sorting evidence, we also estimate monthly Fama–MacBeth regressions in which the dependent variable is either the stock's overnight return or its intraday return in month t. The main explanatory terms are the stock's lagged overnight return and intraday returns over month $t-1$, as well as those lagged returns interacted with a stock's INFO-Z term from the end of month $t-2$. Here, our limited goals are to evaluate whether our main INFO-Z findings above are qualitatively consistently evident in this alternative empirical approach, both without and with

additional firm-specific control variables.[10]

Table 5 reports the results. In Panel A, the INFO-Z explanatory terms use continuous measures of INFO-Z. Panel B replaces these variables with high- and low-quintile INFO-Z indicators. In the empirical regression equations that follow, we denote monthly overnight and intraday returns by $r^n_{i,t}$ and $r^d_{i,t}$, respectively. These correspond to the data variables *ret_night* and *ret_day* reported in the regression tables.

In Panel A, we estimate the following specification:

$$\begin{aligned} R_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(INFO\text{-}Z_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\, r^d_{i,t-1} + \beta_4\big(INFO\text{-}Z_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_5\, INFO\text{-}Z_{i,t-2} + \gamma' CONTROLS_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{1}$$

where $R_{i,t}$ denotes either $r^n_{i,t}$ or $r^d_{i,t}$, and $CONTROLS$ contains the standard set of lagged firm characteristics used as explained in Section 3. All specifications also include Fama–French 48-industry fixed effects.

Panel B replaces the continuous INFO-Z variables with high- and low-quintile INFO-Z dummy variables. The corresponding specification is

$$\begin{aligned} R_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(Hi_INFO\text{-}Z_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\big(Lo_INFO\text{-}Z_{i,t-2} \times r^n_{i,t-1}\big) \\ + \beta_4\, r^d_{i,t-1} + \beta_5\big(Hi_INFO\text{-}Z_{i,t-2} \times r^d_{i,t-1}\big) + \beta_6\big(Lo_INFO\text{-}Z_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_7 Hi_INFO\text{-}Z_{i,t-2} + \beta_8 Lo_INFO\text{-}Z_{i,t-2} + \gamma' CONTROLS + FF48 + \varepsilon_{i,t}. \end{aligned} \tag{2}$$

To begin, we note that the benchmark specifications in Columns (1) and (4) of Table 5 reproduce the basic intrday–overnight return dynamics. When the dependent variable is the overnight return, the coefficient on the lagged overnight return $r^n_{i,t-1}$ is strongly positive at 8.03 (t=7.70), while the coefficient on the lagged intraday return $r^d_{i,t-1}$ is negative at -5.42 (t=-5.48). When the dependent variable is the intraday return, the signs reverse: the coefficient on lagged overnight return $r^n_{i,t-1}$ is -9.42 (t=-9.64), and the coefficient on lagged intraday return $r^d_{i,t-1}$ is 3.35 (t=3.87). Thus, the regression framework confirms both momentum, for each daily return component (both the overnight and intraday components) and tug-of-war cross-reversals for each daily return component, relative to the other (for the overnight returns, relative to the lagged intraday returns; and for the intraday returns, relative to the lagged overnight returns).

[10] In our previous double-sorting analysis, we evaluate momentum variation in terms of the dynamics of cross-sectional return behavior; are past relative winners (losers) likely to continue to be current winners (losers)? The Fama-MacBeth analysis considers momentum in the time-series; is last month's stock return positively related to the current month's stock return? These are related concepts, but not strictly the same.

For our study, we are primarily interested in the INFO-Z terms. First, for the monthly overnight stock-return momentum, we find that the relation between a stock's overnight return over month t and its lagged overnight return over month $t-1$ is increasing with INFO-Z (the β_2 term in Panel A, and the β_2 and β_3 terms in Panel B). We also find that the average overnight returns are increasing with INFO-Z by itself (the β_5 term in Panel A and the β_7 and β_8 terms in Panel B). These results are evident both without and with the additional firm-specific control variables; see columns 2 and 3 in both panels.

Second, for the monthly intraday stock-return momentum, we find that the relation between a stock's intraday return over month t and its lagged intraday return over month $t-1$ is also increasing with INFO-Z. While this pattern is more statistically reliable than the comparable increase in Table 1 (which is statistically insignificant), the pattern is smaller than the comparable case in overnight returns. See columns 5 and 6 in both panels. We also find that the average intraday returns are decreasing with INFO-Z in column-5 without the additional controls, but this result is not robust to the additional control variables in column 6.

For the tug-of-war cross-reversal behavior in Panel A, we also see that the relation between the overnight (intraday) returns and the lagged intraday (overnight) returns becomes more negative as INFO-Z increases, but this pattern is marginal for the intraday returns.

In sum, the Fama-MacBeth results are consistent with our earlier main INFO-Z results.

4.6. Summary of Main INFO-Z Results

We close this section by summarizing our main asymmetric-information findings. We focus on the following four main INFO-Z findings, which are reliably evident in both our double-portfolio-sort investigation and our Fama-MacBeth investigation.

We begin with the momentum and cross-reversal variation with INFO-Z. A stock's asymmetric-information environment predicts how last month's *overnight* returns are propagated into next month's overnight and intraday return. Higher asymmetric-information stocks exhibit stronger momentum in successive monthly overnight returns and a stronger tug-of-war, regarding how a stock's monthly intraday return is negatively related to last month's overnight return.

Next, we shift to variation in average overnight and intraday returns with INFO-Z. The average excess overnight returns, and their alphas, are increasing with INFO-Z. And, the average excess intraday returns, and their alphas, are generally decreasing with INFO-Z; although this pattern is weaker than the opposing INFO-Z pattern in average overnight returns and this pattern loses statistical significance with the full set of control variables in the Fama-MacBeth analysis.

In sum, our collective evidence indicates an important role for asymmetric information in understanding both the dynamics and the average values of the overnight and intraday stock returns.

5. 52-week Price Anchoring and Overnight & Intraday Return Behavior: Main Empirical Results

5.1. Main PTA Momentum Results with Sequential Double-Sorted Portfolios

We next examine whether 52-week price anchoring has a role in understanding the behavior of the monthly intraday and overnight stock returns. Our focus is whether price anchoring is related to the momentum in monthly overnight and intraday returns. Our empirical investigation is comparable to Section 4, but with our price-anchoring metric replacing INFO-Z.

We evaluate a stock's price-to-52-week-anchor (PTA) metric, which is our preferred proxy for how 52-week price anchoring might influence price incorporation and return behavior in this paper's setting. Recall that a stock's PTA is the average of a stock's standardized price-to-52-week-high (PTH) ratio and price-to-52-week-low (PTL) ratio.

Table 6 reports the results over our full sample from January 1997 through December 2021. Consistent with evidence in Table 1, we again see that the return behavior is consistent with recent literature that documents substantial momentum behavior in both monthly overnight and intraday individual stock returns. For every PTA quintile and for both monthly overnight returns and intraday returns, we find that the past-winners reliably outperform the past-losers, indicating substantial momentum in each return component.

Panel A of Table 6 studies the interaction between PTA and overnight-return momentum. The portfolio construction parallels that in Table 1. At the end of month $t-1$, stocks are first sorted into quintiles based on their PTA ranking measured at the close of month $t-2$ using NYSE breakpoints. Within each PTA quintile, stocks are then sorted into quintiles according to their past one-month overnight returns in month $t-1$, again using NYSE breakpoints. We report the average month-t overnight returns for the resulting 25 portfolios, together with the corresponding high-minus-low spread portfolios and their factor-model-adjusted alphas.

The evidence in Panel A indicates that the momentum in overnight returns is marginally increasing with a stock's PTA. However, relative to the strong INFO-Z comparable pattern, the magnitude is modest and the alphas for this difference are only statistically significant at the 10% p-value level. Next, for each past-overnight-return quintile in Panel A, we see that high-PTA stocks have

higher average overnight returns and alphas, relative to the comparable low-PTA stocks. While these PTA-based differences are sizable and statistically reliable, the magnitude is somewhat lower than the comparable result in our INFO-Z analysis. The average PTA-difference across the five past-overnight-return quintiles in Panel A is 0.89%, versus a comparable average INFO-Z difference of 1.29%. For both PTA and INFO-Z, this difference is the largest for the top quintile of past-overnight-return stocks.

We next turn to how the intraday-return momentum varies with a stock's PTA. We perform the same double-sorting exercise as we did for the overnight returns as described above, but for the monthly intraday returns instead. The results in Panel B show that the momentum in successive monthly intraday returns is decreasing with a stock's PTA. For the largest (smallest) PTA quintile, the past-winners outperform the past-losers by 1.04% (2.42%). This PTA-based difference of -1.38% has a p-value less than 0.1%, with similar and highly statistically-reliable alphas.

Finally, for the past intraday-return quintiles we note that the average returns and alphas tend to be declining with a stock's PTA, except for the lowest-past-return quintile. This result largely fits with a tug-of-war perspective, given that the average overnight returns are increasing appreciably with PTA (Panel A).

Next to probe the robustness and pervasiveness of these PTA return patterns, we report on a comparable analysis for one-half subperiods in Table 7: 1997:01-2009:06 and 2009:07-2021:12. First, for the monthly overnight returns in Panel A, we see that the momentum in monthly overnight returns reliably increases with PTA only in the second-half subperiod. Second, for both subperiods, we see that the average overnight returns are increasing with PTA for all five past-return quintiles, although these PTA differences lack statistical significance for two of the quintiles in the first-half subperiod. On average, across the five past-return quintiles, the average monthly overnight returns for the high PTA stocks are 0.86% (0.94%) greater than those for the low PTA stocks for the first-half (second-half) subperiod.

Second, Panel B reports on the monthly intraday returns. Consistent with our full-sample results, we find that the intraday-return momentum is decreasing with PTA for both one-half subperiods. While the momentum difference between the high-PTA stocks and low-PTA stocks is larger for the first-half at -2.08% (t=-3.20), it remains appreciable with statistical significance in the second-half too at -0.69% (t=-2.12). Finally, for both subperiods, we note that the average intraday returns generally decline with the PTA (for 9 of the 10 cases) with statistical significance for 4 of the 10 cases. This pattern fits with a tug-of-war intuition, given that the average overnight returns in Panel A were increasing appreciably with PTA.

In sum, we conclude that the key full-sample PTA findings are qualitatively similar in each one-half subperiod.

5.2. PTA Momentum Results: Robustness with Methodology Variations

5.2.1. With Independently Double-Sorted Portfolios on PTA and Past Returns

As we did for INFO-Z in Section 4.2.1, we next verify that the same qualitative PTA conclusions emerge under an alternative independent sorting design for the two conditioning variables. First, we find that the negative relation between PTA and the momentum in monthly intraday returns remains strongly evident with the independent sorting. With the independent sorting, the difference in intraday-return momentum between the high PTA stocks and low PTA stocks is -1.15% (t=-3.10). Second, across the past-return quintiles, we again find that the average overnight returns are increasing with PTA, while the average intraday returns tend to decline with PTA. Finally, the momentum in overnight returns is increasing with PTA, to a larger and more statistically-reliable degree than seen in Table 6 with the sequential sorts. In sum, we conclude that our main PTA results are robust with the alternative independent double-sorting procedure. Tabular results and additional details are provided in External Appendix A.7.

5.2.2. With an Alternative 3-by-10 Double-Sorting Granularity

Our main results are based on double-quintile sorting on a stock's lagged PTA and the prior month's overnight or intraday return. To evaluate robustness further, we next perform a similar empirical exercise, but with a tercile sort on the past PTA and a decile sort on the past return (or 3-by-10). The same qualitative PTA patterns are also evident in this case, similar to results in Table 6. External Appendix A.8 provides details, along with tabular results.

5.3. Main PTA Results for Large-cap, Mid-cap, and Small-cap Firms Separately

Our main results in the prior two subsections evaluate value-weighted portfolio returns, ensuring that our results are not driven solely by small-cap firms. Next, to evaluate whether our main price-anchoring results vary importantly with firm size, we repeat our main PTA analysis (as in Table 6) for large-cap, mid-cap, and small-cap stocks separately.

Table 8 reports the results, classifying firms into large-cap, mid-cap, and small-cap terciles based on market capitalization measured at the end of month $t-2$. Here, we focus on the PTA momentum patterns in monthly intraday returns, given the sizable PTA variation in intraday-return

momentum in our prior results. Panels A, B, and C report on large-cap, mid-cap, and small-cap stocks, respectively.

We find that the intraday-return momentum is decreasing with PTA for all three firm-size groups, with statistically-significant PTA momentum differences for the large-cap and small-cap firms. For large-cap stocks, the high-minus-low intraday-return spread is 2.10% for the low PTA grouping and 0.71% for the high PTA grouping, for a difference of -1.39% (t= -3.40). The comparable PTA-based differences in intraday-return momentum are -0.41% (t= -1.14) for mid-cap stocks and -1.04% (t= -2.62) for small-cap stocks. The risk-adjusted alphas for these momentum differences are also sizable and statistically significant for the large-cap and small-cap stocks. Thus, the role of 52-week price anchoring on intraday-return momentum is pervasive across stocks of different market capitalization.

5.4. Tug-of-War Cross-Reversal Results with PTA

Table 9 reports on the tug-of-war cross-reversal patterns with PTA. Here, we perform the same PTA first-stage sort, but the return evaluation is a cross effect, meaning that: (1) in Panel A, past overnight returns over month $t-1$ are used in the sort, but then we evaluate the subsequent intraday returns over month t; and (2) in Panel B, past intraday returns over month $t-1$ are used in the sort, but then we evaluate the subsequent overnight returns over month t.

Before reporting on the PTA results, we note that cross-reversal behavior is evident in both cases above. When sorting stocks on their past month's overnight returns, then the subsequent average intraday returns are decreasing with the past overnight return (opposite to the momentum in successive monthly overnight returns). And, when sorting stocks on their past month's intraday returns, then the subsequent average overnight returns are decreasing with the past intraday return (opposite to the momentum in successive monthly intraday returns).

For the PTA variation in Panel A, the spread in average intraday returns between high and low past-overnight-return stocks is essentially the same for both high-PTA and low-PTA stocks. For low-PTA stocks (high-PTA stocks), the monthly intraday returns exhibit a cross-reversal, relative to the past month's overnight returns, of -2.25% (-2.26%) for the high-minus-low past-overnight return strategy. Thus, the PTA-based difference in cross-reversal behavior is essentially zero.

However, for the PTA variation in Panel B, the cross-reversal behavior of the monthly overnight returns, relative to the past month's intraday returns, is declining with PTA. There is a monotonic decline in the reversal behavior from the low-PTA stocks (at -2.70%, t=-4.07) to the high-PTA stocks (at -1.37%, t=-3.89). The PTA difference in the cross-reversal behavior is highly reliable at 1.33% (t=2.63) with similar alphas.

Next, shifting to how average returns vary with PTA, we note the following. The average overnight returns are increasing with PTA (Panel B), with statistically significant differences for four of the five past-intraday-return quintiles. And, the average intraday returns are decreasing with PTA (Panel A), with statistically significant differences for two of the five past-intraday-return quintiles. While interesting, this variation is less robust than the comparable INFO-Z variation.

We perform additional analysis to probe robustness and pervasiveness. First, we find similar PTA patterns in a cross-reversal analysis that uses independent sorts on the two conditioning variables, rather than sequential. See External Appendix A.9 for details. Second, we similarly evaluate one-half subperiods for the cross-reversal PTA patterns. For both one-half subperiods, we find that the cross-reversals between this month's overnight returns and last month's intraday returns are weaker (less negative) for high PTA stocks, relative to low-PTA stocks. However, this pattern is marginal and lacks statistical significance over the second-half subperiod. External Appendix A.10 provides detailed results.[11]

5.5. PTA Evidence from Fama-MacBeth Cross-sectional Regressions

To complement the portfolio-sorting evidence, we also estimate monthly Fama–MacBeth regressions in which the dependent variable is either the stock's overnight return or its intraday return in month t. The main explanatory terms are the stock's lagged overnight return and intraday returns over month $t-1$, as well as those lagged returns interacted with a stock's PTA from the end of month $t-2$. Here, our limited goals are to evaluate whether our main PTA findings above are qualitatively consistent in this alternative empirical approach, both without and with additional firm-specific control variables.

Table 10 reports the results. In Panel A, the PTA explanatory terms use continuous measures of PTA. Panel B replaces these variables with high- and low-quintile PTA indicators.

We estimate the same specifications as we did in Section 4.5, except with a stock's PTA replacing its INFO-Z metric: In Panel A, we estimate the following specification:

$$\begin{aligned} R_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(PTA_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\, r^d_{i,t-1} + \beta_4\big(PTA_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_5\, PTA_{i,t-2} + \gamma' CONTROLS_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{3}$$

Panel B replaces the continuous PTA variables with high- and low-quintile PTA dummy variables.

[11] The weaker PTA pattern over the second-half subperiod can largely be attributed to the strong reversal for the past-intraday-losers with a high PTA over this period. The past-intraday-winner stocks exhibit a cross-continuation (rather than cross-reversal), which is a consistent behavior across subperiods. Appendix A.10 provides details.

The corresponding specification is:

$$\begin{aligned} R_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(Hi_PTA_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\big(Lo_PTA_{i,t-2} \times r^n_{i,t-1}\big) \\ + \beta_4\, r^d_{i,t-1} + \beta_5\big(Hi_PTA_{i,t-2} \times r^d_{i,t-1}\big) + \beta_6\big(Lo_PTA_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_7 Hi_PTA_{i,t-2} + \beta_8 Lo_PTA_{i,t-2} + \gamma' CONTROLS + FF48 + \varepsilon_{i,t}. \end{aligned} \tag{4}$$

See the narrative with equations (1) and (2) for more details.

We focus on the PTA results in the table. We begin by discussing the overnight-return results. First, for the monthly overnight stock-return momentum, we find that the relation between a stock's overnight return over month t and its lagged overnight return over month $t-1$ is increasing with PTA. These results are evident both without and with the additional firm-specific control variables; see columns 2 and 3 in both panels. This finding is statistically stronger than the comparable results in the portfolio-sorting analysis. Second, for the cross-reversal results, we find that the reversal in a stock's monthly overnight return, relative to the stock's prior month intraday return, is decreasing with PTA; see columns 2 and 3 in Panel A. Finally, we note that the average overnight returns are higher for the higher PTA stocks for the column-2 results without the additional firm-specific controls, aligning with the results in Table 6. However, this result is not robust to the inclusion of the full set of control variables in Column 3.

We next discuss the intraday-return results. For the monthly intraday stock-return momentum, we find that the relation between a stock's intraday return over month t and its lagged intraday return over month $t-1$ is lower for stocks with a higher PTA (or the momentum in intraday returns is declining with PTA, fitting with our earlier PTA results); see columns 5 and 6 in each panel. For the cross-reversal results, the relation between the monthly intraday returns and lagged overnight returns does not vary with PTA.

In sum, the Fama-MacBeth results are generally consistent with our earlier main PTA results.

5.6. Summary of Main PTA Results

We close this section by summarizing our main 52-week anchoring findings. We focus on the following two PTA results which are reliably evident in both our double-portfolio-sort investigation and our Fama-MacBeth investigation. First, the momentum in a stock's successive monthly intraday returns is decreasing with a stock's PTA. Second, the reversal in a stock's monthly overnight return, relative to the stock's prior month intraday return (the tug-of-war dynamic), is weaker for stocks with a higher PTA. These results are complementary, since both results indicate a declining month-to-month

relation as a stock's PTA increases, referring to both weaker momentum and weaker cross-reversal behavior as PTA increases.

In sum, our collective evidence indicates a material role for 52-week price anchoring in understanding the dynamics of the monthly overnight-versus-intraday returns.

6. Additional Analysis

This section further analyzes and extends our main findings, which are summarized in Sections 4.6 and 5.6. Our goals here are to further characterize and describe the findings, as well as present additional robustness checks. The results from this section will then play a key role in Section 7, where we discuss the implications of our findings for the related literature.

Section 6.1 reports on the correlation between a stock's INFO-Z and PTA metrics, with the low correlation supporting our choice to report on separate analyses in Sections 4 and 5. Section 6.2 evaluates how our key momentum patterns vary across different sentiment states. Section 6.3 evaluates whether our key momentum results are robust to additional control variables that are motivated by the recent overnight-versus-intraday return literature. Finally, to assist in later interpretation, Section 6.5 discusses which categories of stocks are relatively more important in driving the momentum and cross-reversal patterns in our main results.

6.1. Correlation between a Stock's INFO-Z and PTA Metrics

In the prior two sections, we analyzed the role of asymmetric information and 52-week price anchoring in characterizing and understanding the overnight/intraday return anomaly. Our empirical progression in each section was comparable, using our INFO-Z metric in Section 4 and our PTA metric in Section 5. With this approach, the working premise was that the two metrics capture substantially different dimensions of the overnight/intraday return dynamics.

To evaluate this premise, we calculated the cross-sectional correlation of the individual stock's INFO-Z and PTA values for each calendar month in our sample. The correlation between the two metrics is quite modest, with an average (median) correlation for the time-series of monthly correlations equaling 0.199 (0.200). The 25th (75th) percentile of the monthly correlation series was 0.116 (0.276).

Recall that one of the factor inputs into the INFO-Z was a stock's standardized idiosyncratic volatility. If stocks with high idiosyncratic volatility are somewhat more likely to be high-PTA stocks, this could explain the marginal positive correlation. Overall, though, we feel that this marginal

correlation between our INFO-Z and PTA metrics indicates little relation between the two metrics, fitting with our prior empirical approach.

In other work, we also calculate the monthly cross-sectional correlations of our INFO-Z and PTA metrics with a stock's log(size), book-to-market equity ratio, and market beta. As well as their use in the Fama-French 3-factor model, these variables are included as additional level control terms in our previous Fama-MacBeth analysis. Across the months in our sample, we find that: (1) the mean (median) cross-sectional correlations between our INFO-Z metric and a stock's size, book-to-market equity ratio, and beta are -0.508 (-0.514), 0.008 (0.013), and 0.407 (0.419), respectively; and (2) the mean (median) cross-sectional correlations between our PTA metric and these same three terms are 0.043 (0.037), -0.073 (-0.085), and 0.027 (0.024), respectively. These sizable correlations between INFO-Z and size and beta suggest that we evaluate size and beta interactive-return control terms in our Fama-MacBeth work; an analysis which we undertake in Section 6.4.

6.2. Evidence from Investor Sentiment

We next examine how our main momentum results vary across different investor sentiment states. Past literature has shown that sentiment can affect retail investor behavior and that anomalies tend to be stronger in higher sentiment states; see, e.g., Stambaugh, Yu, and Yuan (2012). Further, Akbas et al. (2022) find that their stock return predictability from positive-overnight-return frequency is stronger during high sentiment periods. This investigation also evaluates the pervasiveness of the main momentum patterns across different market states.

Table 11 reports the results for investor sentiment, where we classify months into high-, medium-, and low-sentiment states using Baker–Wurgler sentiment terciles, measured from month $t-2$. Panel A evaluates the asymmetric-information pattern where the momentum in successive monthly overnight returns is stronger for higher INFO-Z stocks. We find that this INFO-Z momentum pattern is evident for all three sentiment states, but it is appreciably weaker for the low-sentiment state. For the high (low) INFO-Z stocks, the momentum spread in average overnight returns between the past-overnight-winner stocks and past-overnight-loser stocks is 3.87% (1.33%) for the high-sentiment state and 4.03% (1.41%) for the mid-sentiment state. Thus, the momentum difference between high and low INFO-Z stocks is over 2.5% greater for both of the two higher sentiment groupings. Conversely, for the low sentiment state, the comparable momentum difference between high and low INFO-Z stocks is only 0.90%. The associated alphas for the INFO-Z momentum differences are also appreciably higher and more statistically reliable for the two higher sentiment states.

Table 11, Panel B, evaluates the price-anchoring pattern where momentum in successive monthly

intraday returns is weaker for higher PTA stocks. We find that this PTA momentum pattern is evident for all three sentiment states, but it is appreciably weaker for the low-sentiment state. For the high (low) PTA stocks, the momentum spread in average intraday returns between the past-intraday-winner stocks and past-intraday-loser stocks is 1.43% (3.12%) for the high-sentiment state and 0.91% (2.80%) for the mid-sentiment state. Thus, the momentum difference between high and low PTA stocks is about -1.7% to -1.9% lower for the two higher sentiment groupings. Conversely, for the low sentiment state, the comparable momentum difference between high and low PTA stocks is only -0.64%. The associated alphas for the PTA momentum differences are also appreciably more negative and more statistically reliable for the two higher sentiment states.

Thus, our findings indicate that: (1) both the INFO-Z-momentum pattern in overnight returns and the PTA-momentum patterns in intraday returns are consistently evident across all three sentiment states (but lack statistical significance for the PTA low-sentiment case), and (2) the momentum patterns tend to be stronger during higher sentiment states, aligning with prior evidence linking anomalies with higher sentiment.

6.3. Controlling for Other Variables suggested by the Economic Channels of Akbas et al. (2022), Lu et al. (2023), and Jones et al. (2024)

We next assess whether our main INFO-Z and PTA momentum results are robust to the inclusion of additional control variables that are suggested by the recent overnight-versus-intraday return literature. We re-evaluate our main INFO-Z and PTA momentum findings with the inclusion of the additional controls, specifically: (1) the pattern where momentum in overnight returns increases with a stock's INFO-Z; and (2) the pattern where momentum in monthly intraday returns decreases with a stock's PTA. Our primary interest is whether the additional control variable suggested by the literature might subsume these main findings. This work extends the Fama-MacBeth specifications in Sections 4.5 and 5.5 with six additional control terms.

6.3.1. Description of Additional Control Variables

We begin by reviewing three related studies and describing the additional control variables that are suggested by these studies. To make the control variables from these earlier studies implementable within a monthly Fama–MacBeth framework in a comparable fashion, each control variable is constructed from stock-level characteristics measured at the end of month $t-1$.

The first control variable is motivated by the opening-liquidity channel in Lu, Malliaris, and Qin (2023), which emphasizes the role of market makers and opening order imbalances under elevated

information asymmetry. For the Lu et al. channel, we construct $Z^{LMQ}_{i,t-1}$ as the monthly average of expected daily opening-half-hour retail market order imbalance for stock i during month $t-1$. This variable is designed to capture the strength of the opening-liquidity-demand channel in a form that is compatible with monthly cross-sectional regressions.

The second control variable is motivated by the return-extrapolation channel in Jones, Pyun, and Wang (2024), which argues that investors extrapolate prior positive daytime returns and thereby generate day–night return patterns. For the Jones et al. channel, we construct $Z^{JPW}_{i,t-1}$ as the monthly average of lagged positive daily intraday returns during month $t-1$, which is intended to proxy for the return-extrapolation mechanism emphasized in that paper.

The LMQ and JPW variables are denoted as Z variables, because in our Fama-MacBeth regression implementation we use standardized versions of the LMQ and JPW variables that have a mean of zero and a standard deviation of one. This treatment makes these variables statistically comparable to our INFO-Z variable.

The third control variable is motivated by the intensity tug-of-war implication in Akbas et al. (2022). Their central idea is that a stock experiences a more intense daily tug of war when it more frequently records a positive overnight return followed by a negative daytime reversal. Akbas et al. show that this pattern predicts higher future returns, primarily through the subsequent overnight component. In our setting, however, the relevant question is different. Our objective is not to ask whether the Akbas et al. variable predicts returns in general, but whether our momentum results remain after controlling for their tug-of-war measure in the same cross-sectional regression. We construct an $ABNR_{i,t-1}$ variable that is high when stock i experiences an unusually large number of days in month $t-1$ with positive overnight returns followed by negative daytime reversals, relative to its own recent history. See External Appendix A.11 for details on the variable construction.

6.3.2. Extended Fama-MacBeth Specification and Results for INFO-Z

We begin by examining the variation in overnight-return momentum with INFO-Z. We estimate an extended version of the Fama-MacBeth equation (1) from Section 4.5, by adding the above LMQ, JPW, and ABNR control variables (the β_9 to β_{11} terms) and these same three variables interacted with the lagged monthly overnight return (the β_3 to β_5 terms):

$$\begin{aligned} r^n_{i,t} = \beta_0 + \beta_1 r^n_{i,t-1} + \beta_2\big(INFO\text{-}Z_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3 \left(Z^{LMQ}_{i,t-1} \times r^n_{i,t-1}\right) + \beta_4 \left(Z^{JPW}_{i,t-1} \times r^n_{i,t-1}\right) \\ \beta_5 \left(ABNR_{i,t-1} \times r^n_{i,t-1}\right) + \beta_6 r^d_{i,t-1} + \beta_7\big(INFO\text{-}Z_{i,t-2} \times r^d_{i,t-1}\big) + \beta_8 INFO\text{-}Z_{i,t-2} \\ + \beta_9 Z^{LMQ}_{i,t-1} + \beta_{10} Z^{JPW}_{i,t-1} + \beta_{11} ABNR_{i,t-1} + \gamma' CONTROLS_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{5}$$

where the three new control variables are defined in Section 6.3.1 and the other terms are defined with equation (1).

Table 12, columns 1 to 3, reports the results. Our primary interest is whether the estimated β_2 on the interactive INFO-Z term remains reliably positive with the additional control variables. In combination with a positive β_1 coefficient, this would indicate a positive partial relation between the successive monthly overnight returns that is increasing with asymmetric information.

Results in Column 2 (without the full set of individual-stock control variables) and Column 3 (with the full set of individual-stock control variables) of Table 12 show that the partial relation between the successive monthly overnight returns is strongly increasing with a stock's INFO-Z. Both of the estimated β_2 coefficients are positive and statistically significant with better than 1% p-values, with their values being marginally greater than the comparable coefficient values in the simpler Table 5 specification. Further, we note that none of the estimated coefficients for the other interactive terms with the lagged overnight return (LMQ, JPW, or ABNR) are statistically significant (the β_3, β_4, and β_5 coefficients).

Further, interestingly, the estimated coefficients on the stand-alone LMQ and JPW terms are positive and highly statistically reliable (the β_9 and β_{10} coefficients). However, the estimated coefficient on the stand-alone INFO-Z term is no longer positive and statistically significant in this more complex specification (β_8). So, the stand-alone LMQ and JPW terms do a better job at predicting the level of the subsequent overnight return than the INFO-Z level variable, but the INFO-Z interactive term ($INFO\text{-}Z_{i,t-2} \times r^n_{i,t-1}$) is the only interactive term that reliably predicts variation in the relation between successive monthly overnight returns.

6.3.3. Extended Fama-MacBeth Specification and Results for PTA

We next examine the variation in momentum for monthly intraday returns with PTA. We estimate an extended version of the Fama-MacBeth equation (3) from Section 5.5, by adding the LMQ, JPW, and ABNR control variables (the β_9 to β_{11} terms) and these same three variables interacted with the lagged monthly intraday return (the β_5 to β_7 terms):

$$\begin{aligned} r^d_{i,t} = \beta_0 &+ \beta_1\, r^n_{i,t-1} + \beta_2\big(PTA_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\, r^d_{i,t-1} + \beta_4\big(PTA_{i,t-2} \times r^d_{i,t-1}\big) \\ &+ \beta_5\left(Z^{LMQ}_{i,t-1} \times r^d_{i,t-1}\right) + \beta_6\left(Z^{JPW}_{i,t-1} \times r^d_{i,t-1}\right) + \beta_7\left(ABNR_{i,t-1} \times r^d_{i,t-1}\right) \\ &+ \beta_8\, PTA_{i,t-2} + \beta_9\, Z^{LMQ}_{i,t-1} + \beta_{10}\, Z^{JPW}_{i,t-1} + \beta_{11}\, ABNR_{i,t-1} \\ &+ \gamma' CONTROLS_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{6}$$

where the three new control variables are defined in Section 6.3.1 and the other terms are defined with equation (3).

Table 12, columns 4 to 6, reports the results. Our primary interest is whether the estimated β_4 coefficient on the interactive PTA term remains negative with the additional control variables. In combination with a positive β_3 coefficient, this would indicate momentum between the successive intraday returns that is declining when stocks have current prices that are high relative to their 52-week price history (a high PTA).

Results in Column 5 (without the full set of individual-stock control variables) and Column 6 (with the full set of individual-stock control variables) of Table 12 show that the partial relation between the successive monthly intraday returns is decreasing with a stock's PTA. The estimated β_4 coefficients are negative and statistically significant with better than 5% p-values, with their values being marginally greater than the comparable coefficient values in the simpler Table 10 specification. Further, we again note that none of the estimated coefficients for the other interactive terms with the lagged intraday return (LMQ, JPW, or ABNR) are statistically significant (β_5, β_6, or β_7).

However, the estimated coefficients on the stand-alone JPW and ABNR terms are highly statistically significant (the β_{10} and β_{11} coefficients). For the JPW term, the negative coefficient for the intraday returns in columns 4 to 6 fits with the positive coefficient on the overnight returns in columns 1 to 3, in a cross tug-of-war sense.

In sum, the additional stand-alone control variables (the JPW and ABNR level terms here) do a better job at predicting the level of the subsequent intraday returns than PTA. However, the PTA interactive term ($PTA_{i,t-2} \times r^d_{i,t-1}$) is the only interactive term that reliably predicts variation in the relation between successive monthly intraday returns.

6.4. Additional Interactive-Return Control Variables

Finally, it is a natural question whether the conditional roles of our INFO-Z and PTA metrics in our setting would remain reliably evident when including other well-motivated interactive-return terms. We briefly evaluate this issue in Appendix A.12. There, we include three additional interactive-return control terms; interacting with a stock's log(market capitalization), book-to-market equity ratio, and market beta. As well as the role of these three terms in the ubiquitous Fama-French 3-factor model, this choice is suggested by the cross-sectional correlations reported at the end of Section 6.1. There we reported a stock's INFO-Z metric is substantially negatively (positively) correlated with its size (beta), implying that smaller-cap and higher-beta stocks tend to reflect greater asymmetric information, relative to larger-cap and lower-beta stocks. With the inclusion of these additional three

interactive-return control terms, we find that our principal conditional results with the INFO-Z and PTA interactive-return terms remain reliably evident and tend to be even stronger. For brevity, detailed results and a further justification of our empirical choices are provided in Appendix A.12.

6.5. Key Corner Portfolios in our Double-Portfolio-Sort Results

Our main analysis features a 5 x 5 matrix of average excess value-weighted stock portfolio returns over month t (either monthly overnight returns or monthly intraday returns), formed from sorts on either the INFO-Z or PTA value from month $t-2$ and the lagged return-of-interest from month $t-1$. Thus, in our featured results on how momentum and cross-reversals vary with either INFO-Z or PTA, the four corner portfolios in this 5 x 5 matrix are especially important. Accordingly, in this subsection, we discuss which of the four corner portfolios are relatively more important in establishing the key momentum and cross-reversal variations in our main findings. Then, we factor these observations into our next section, in terms of relating our findings to the existing literature.

6.5.1. Variation in Overnight Momentum and Cross-Reversals with INFO-Z

We begin by discussing our featured dynamic results with asymmetric information; where higher INFO-Z stocks exhibit both stronger momentum in successive monthly overnight returns and stronger cross-reversals in the negative relation between the current month's intraday return and the prior month's overnight return. For both these patterns, we note that the bottom right-hand corner portfolio, composed of high INFO-Z and past-overnight-winner stocks, is particularly important. For the key momentum results in Table 1, Panel A, the average excess overnight return for this corner portfolio is relatively very high at 3.47% (versus 0.97% for the corner portfolio with low INFO-Z and past overnight-return winners), which is substantially behind the result of stronger momentum for high INFO-Z stocks. And, for the key cross-reversal result in Table 4, Panel A, the average excess intraday return for this corner portfolio of high-INFO-Z, past-overnight-winners is relatively very low at -2.13% (versus -0.24% for the corner portfolio with low INFO-Z and past-overnight-return winners), which is substantially behind the result of stronger cross-reversals for the high INFO-Z stocks.[12] These corner-portfolio patterns are qualitatively consistent in our robustness analysis.

[12]By comparison, the average excess overnight return across the 25 portfolios in Table 1, Panel A, is +0.86% per month, and the average excess intraday return across the 25 portfolios in Table 4, Panel A, is -0.09% per month.

6.5.2. Variation in Intraday Momentum and Cross-Reversals with PTA

We next discuss our featured dynamic results with 52-week price anchoring; where high-PTA stocks exhibit both weaker momentum in successive monthly intraday returns and weaker cross-reversals in the negative relation between the current monthly overnight return and the prior month's intraday return. For both these patterns, we note that the bottom right-hand corner portfolio, composed of the high PTA and past-intraday-winner stocks, is particularly important. For the key momentum results in Table 6, Panel B, the average excess intraday return for this corner portfolio is relatively low at -0.50% (versus 0.49% for the corner portfolio with low PTA and past-intraday-return winners), which is substantially behind the result of weaker momentum for the high PTA stocks. All the other past-intraday-winner portfolios in Table 6, Panel B, column 5, have sizably positive average excess returns (which is consistent with momentum behavior). And, for the key cross-reversal result in Table 9, Panel B, the average excess overnight return for this corner portfolio of high-PTA, past-intraday-winner stocks is relatively quite high at 1.25% (versus -0.33% for the corner portfolio with low PTA and past intraday-return-winner stocks), which is substantially behind the result of weaker cross-reversals for the high PTA stocks. All the other past-intraday-return winner portfolios in Table 9, Panel B, column 5, have negative average returns which is consistent with cross-reversal behavior).[13] These corner-portfolio comparative results are qualitatively consistent in our related robustness analysis.

6.5.3. Summary Remarks on the Key Corner Portfolios

Thus, past relative-winner stocks are more influential in establishing both our main INFO-Z and PTA conditional patterns. Specifically, we refer to: (a) for our INFO-Z results, stocks with a relatively high overnight return over month $t-1$ and a high INFO-Z as of the end of month $t-2$; and (b) for our PTA results, stocks with a relatively high intraday return over month $t-1$ and a high PTA as of the end of month $t-2$. We consider these stylized facts when discussing implications of our findings in Section 7.

Further, we argue that past-winner stocks, especially those with a relatively high PTA, seem likely to have greater liquidity, on average, relative to past-loser, low-PTA stocks (which are likely to have a lower market capitalization and/or be in a more financially-distressed state).[14] If so, then

[13] By comparison, the average excess intraday return across the 25 portfolios in Table 6, Panel B, is -0.037% per month, and the average excess overnight return across the 25 portfolios in Table 9, Panel B, is +0.73% per month.

[14] Date from Chen et al. (2026) indicate that portfolios of high-PTA stocks tend to be much larger (in market capitalization) than comparable portfolios of low-PTA stocks (over three times larger, on average, for quintile-PTA sorts in their sample).

our corner-portfolio results also suggest that highly illiquidity stocks are not overly influential in establishing our primary conditional findings. This liquidity premise is also supported by our size-segmented analysis, which indicated that our main conditional findings remain strongly evident for stocks in the top-third of market capitalization, and by our focus on value-weighted portfolio returns in our empirical analyses.

7. Implications of our Main Findings for the Related Literature

We next turn to a discussion of what our collective evidence suggests for the literatures on overnight-versus-intraday stocks returns, the cross-sectional pricing of a stock's asymmetric information environment, and stock pricing influences from 52-week price anchoring. Our primary contribution is to the overnight-versus-intraday return literature. Our results show that the familiar momentum and cross-reversal patterns vary systematically across stocks, and that two economically-motivated conditioning variables are associated with different sides of the intraday/overnight return cycle. INFO-Z is especially informative about how prior overnight returns are propagated, whereas PTA is especially informative about how prior intraday returns are propagated.

Our limited empirical work does not provide a complete model of the intraday-overnight tug of war behavior, nor does it attempt to identify all firm characteristics that might condition the return dynamics. Rather, we view our evidence as documenting robust empirical heterogeneity in this setting, which bears on existing explanations and provides guidance for future work in this area.

7.1. Implications for the Variation in Overnight Momentum and Cross-Reversals with a Stock's Asymmetric-Information Environment

Our strongest empirical results involve INFO-Z. Stocks with a higher INFO-Z exhibit both stronger momentum in successive monthly overnight returns and a stronger cross-reversal between last month's overnight return and this month's intraday return.

The broad direction of this evidence is economically intuitive because asymmetric information is already fundamental to several explanations in the overnight–intraday literature. For example, Lou et al. (2019) emphasize heterogeneity between overnight and intraday investor clienteles, while Lu et al. (2023) explicitly model information asymmetry around the open. Our contribution is therefore not to argue that information asymmetry should matter in this setting, but to quantify a strong and pervasive cross-sectional pattern that had not been directly documented: a stock's information environment is especially informative about the *propagation of prior overnight returns* into both

return components next month.

This distinction is particularly relevant to Lu et al. (2023). Their heterogeneous-liquidity-provider model emphasizes elevated information asymmetry and a high price of liquidity near the open, followed by greater competition in liquidity provision during the day. Our results are consistent with that intuition, but the empirical object is different. INFO-Z predicts the strength of month-to-month overnight momentum and the accompanying intraday cross-reversal. Moreover, the INFO-Z interaction remains strongly significant after controlling for a variable motivated by the Lu et al. opening-liquidity channel (Table 12). We interpret this as evidence that INFO-Z contains incremental information about overnight-return propagation, rather than as a direct test or rejection of the Lu et al. mechanism.

Our results also help distinguish INFO-Z from the tug-of-war intensity measure of Akbas et al. (2022). Their measure predicts future returns, especially through the overnight component, whereas our focus is on cross-sectional variation in the relation between prior and subsequent return components. In Table 12, the INFO-Z interaction with lagged overnight returns remains strongly significant after controlling for the Akbas et al. measure and its interaction. Thus, the two variables appear to capture different aspects of the overnight–intraday return behavior.

Bogousslavsky (2021) emphasizes institutional constraints and the greater cost of carrying positions overnight. It seems plausible that such influences would be greater for stocks that face a greater asymmetric-information environment, suggesting that our INFO-Z results provide some support for the Bogousslavsky perspective. However, from Section 6.5.1, recall that past-winner stocks are more influential in establishing the conditional INFO-Z patterns, but such constraints would seem to be more impactful on past loser stocks. Thus, our corner-portfolio observations seem to be odds with this constraint perspective.

Finally, the oversized role of past-winner stocks in establishing the conditional INFO-Z patterns seems consistent with the asymmetric extrapolation perspective from Jones et al. (2024). This is because extrapolative buying pressure should tend to be stronger following positive returns than extrapolative selling pressure following negative returns, due to short selling limitations. It seems likely that such a dynamic would be stronger for stocks that face a higher asymmetric-information environment. We view this as a useful interpretation rather than a separately identified mechanism.

In sum, our INFO-Z findings confirm an important role for asymmetric information in understanding the overnight-versus-intraday return behavior. We believe that the strong and pervasive INFO-Z heterogeneity in the return behavior, along with the specific association with prior overnight-return propagation, are important new empirical facts. Further, the larger role of past-winner stocks

in establishing the heterogeneity provides an additional empirical restriction on explanations of why asymmetric-information differences are closely tied to prior overnight-return propagation.

7.2. Implications for the Variation in Intraday Momentum and Cross-Reversals with a Stock's 52-week Price Anchoring State

The PTA dynamic patterns tend to be somewhat weaker than the INFO-Z patterns, but they are arguably more surprising in the context of the existing overnight–intraday literature. Higher-PTA stocks exhibit weaker momentum in successive monthly intraday returns and a weaker overnight cross-reversal following the prior month's intraday return. Thus, PTA is especially informative about how prior *intraday* returns are propagated into the next month's return components.

52-week price anchoring is not emphasized in the key overnight–intraday studies that motivate our analysis. The PTA evidence therefore introduces a new conditioning dimension into this literature. It also provides a natural link to the return-extrapolation perspective of Jones et al. (2024). Investors may evaluate recent price changes relative to salient reference values such as the 52-week price range. For a stock whose price already appears expensive relative to that range, extrapolative demand following strong prior intraday performance may be tempered because the stock is closer to a salient upper reference point. If extrapolative demand contributes to intraday momentum, this logic predicts weaker subsequent intraday momentum for high-PTA stocks, as we find.

This interpretation is also consistent with the asymmetry in our corner portfolios. High PTA especially weakens the continuation associated with past intraday winners, and high-PTA, past-intraday-winner stocks are particularly important for both the intraday-momentum and overnight cross-reversal results. This outsized winner influence fits naturally with an extrapolative-beliefs interpretation, while again falling short of a direct test of that mechanism. Recall that the Jones et al. (2024) paper argues that extrapolative demand should be relatively stronger for past winners (rather than past-losers), due to short selling constraints.[15]

Other PTA channels might also contribute. Low-PTA stocks seem likely to reflect higher distress, informational risk, or liquidity-provision costs (on average), which could interact with the institutional-constraint channel of Bogousslavsky (2021) or the heterogeneous-liquidity-provider channel of Lu et al. (2023). We do not attempt to distinguish fully among these possibilities. Importantly, however, the key PTA interaction remains significant after controlling for variables motivated by Lu et al. (2023), Jones et al. (2024), and Akbas et al. (2022). We therefore view PTA as containing

[15]Relatedly, when decomposing ETF returns, Liu et al. (2026) find that average overnight returns are significantly positive but average intraday returns are negative. Their analysis concludes that the pattern is primarily driven by demand shocks from retail investors and limited supply from arbitrageurs.

incremental information about intraday-return propagation, rather than as a proxy that is readily subsumed by these existing variables.

The broader implication for the 52-week anchoring literature is that salient price history appears to matter not only for conventional close-to-close return predictability, but also for how prior returns carry forward across the day/night trading cycle. This extends the empirical reach of price anchoring into a setting where its relevance had not previously been emphasized.

7.3. Implications for the Variation in Average Stock Returns with INFO-Z

The market microstructure literature has debated whether a stock's asymmetric-information environment is priced in the cross section, often using the PIN proposed in Easley et al. (1996) and Easley et al. (2002). Easley et al. (2002), Easley et al. (2010), and Kelly and Ljungqvist (2012) provide evidence consistent with information risk being priced, whereas Duarte and Young (2009) and Lai et al. (2014) cast doubt on the pricing of PIN risk. More recent studies find that particular components of informed trading can be priced (Brennan et al. (2016); Chen et al. (2024)).

Our evidence adds a time-of-day dimension to this debate. We find a striking time-of-day pattern in average returns: Stocks with a higher INFO-Z earn higher average excess overnight returns but lower average excess intraday returns. Recall that INFO-Z includes PIN as an important input, and the main patterns remain similar when PIN is examined separately (in place of our INFO-Z composite). The higher overnight returns of high-INFO-Z stocks are consistent with the overnight period carrying greater illiquidity and informational risk. At the same time, lower intraday returns partly offset this differential during the trading day. Conventional close-to-close returns can therefore mask economically meaningful and opposing cross-sectional return differences across the two portions of the trading cycle.

We view this pricing implication as secondary to the paper's main contribution to the overnight–intraday literature, but it broadens the relevance of the INFO-Z evidence. Cross-sectional variation in asymmetric information is associated not only with the dynamics of return propagation; it also predicts a striking contrasting variation in average overnight returns (positively) and average intraday returns (negatively).

8. Conclusions

We document substantial cross-sectional heterogeneity in overnight–intraday return behavior along two distinct dimensions. First, a stock's asymmetric-information environment, measured by INFO-Z,

is especially informative about how the prior month's *overnight* returns propagate into the subsequent month's overnight and intraday returns. Higher INFO-Z stocks exhibit stronger overnight momentum and a stronger intraday cross-reversal following prior overnight returns. Second, a stock's 52-week price-anchoring expensiveness state, measured by PTA, is especially informative about how the prior month's *intraday* returns propagate into the subsequent month's intraday and overnight returns. Higher PTA stocks exhibit weaker intraday momentum and a weaker overnight cross-reversal following prior intraday returns. These patterns remain evident across alternative portfolio sorts and Fama–MacBeth regressions and survive controls motivated by recent explanations.

The strong INFO-Z dynamic patterns provide support and granularity to proposed economic channels for the intraday-versus-overnight tug of war that rely heavily on asymmetric information. The PTA dynamic patterns introduce a new conditioning dimension to the literature that further assists in interpreting the phenomena. The PTA conditioning especially suggests empirical support to explanations that rely on extrapolative beliefs.

For both our main INFO-Z and PTA dynamic findings, stocks that were past-winners with high INFO-Z, or high PTA, are relatively more important in establishing our featured conditional patterns. This finding further characterizes the phenomena and also suggests that highly illiquidity stocks are not overly influential in our findings.

In a tertile evaluation of market sentiment states, we find that both the INFO-Z-momentum pattern in overnight returns and the PTA-momentum patterns in intraday returns are consistently evident across all three sentiment states. However, the momentum patterns tend to be stronger during higher sentiment states, aligning with prior evidence linking anomalies with higher sentiment.

Finally, we also find that higher INFO-Z stocks earn higher average excess overnight returns but lower average excess intraday returns, adding a time-of-day perspective to the literature on asymmetric-information risk. The PTA findings, in turn, extend the 52-week price-anchor literature by linking salient price history to the propagation of returns across the intraday/overnight cycle.

Taken together, our results support a focused interpretation of our paper. We do not claim that INFO-Z and PTA exhaust the possible sources of conditional variation in overnight–intraday returns, and our empirical design does not uniquely identify the mechanisms behind the patterns. Instead, we contribute by documenting two robust, economically-motivated dimensions of heterogeneity with a striking directional contrast: asymmetric information is most closely associated with the propagation of prior overnight returns, while 52-week price anchoring is most closely associated with the propagation of prior intraday returns. Thus, our collective evidence provides useful empirical restrictions for future work in this area.

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Table 1: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on INFO-Z

This table analyzes momentum in both monthly overnight stock returns and monthly intraday stock returns, conditional on a stock's INFO-Z. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on lagged INFO-Z and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their 3-factor INFO-Z ranking as of the close of month $t-2$ using NYSE breakpoints. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.11	-0.04	0.21	0.49	0.97	1.08	1.11	1.09
	(-0.52)	(-0.20)	(1.36)	(2.86)	(3.40)	(3.99)	(4.11)	(3.99)
INFO-Z 2	-0.02	0.06	0.23	0.64	1.34	1.36	1.43	1.35
	(-0.07)	(0.24)	(0.86)	(2.66)	(4.38)	(4.29)	(4.60)	(4.46)
INFO-Z 3	0.06	0.46	0.77	1.17	1.92	1.86	1.60	1.51
	(0.19)	(1.81)	(3.65)	(4.60)	(5.08)	(4.13)	(4.31)	(4.34)
INFO-Z 4	0.21	0.75	0.87	1.32	2.22	2.01	1.91	1.86
	(0.71)	(3.23)	(3.37)	(4.32)	(5.28)	(5.20)	(5.03)	(5.24)
INFO-Z High	0.38	0.84	1.34	1.95	3.47	3.09	2.75	2.74
	(1.19)	(2.71)	(4.91)	(6.38)	(5.84)	(5.06)	(5.04)	(5.10)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	0.49	0.88	1.13	1.46	2.50	2.01	1.64	1.65
	(2.46)	(5.07)	(4.94)	(5.86)	(6.26)	(5.05)	(4.76)	(4.87)
α_{FF6+2}	0.55	0.81	1.00	1.31	2.20			
	(2.88)	(4.57)	(4.22)	(5.45)	(6.24)			
α_{Q+2}	0.56	0.80	1.03	1.29	2.21			
	(2.95)	(4.03)	(4.38)	(5.62)	(6.37)			

Table 1: (continued)

Panel B: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.46	0.39	0.18	0.48	0.82	1.28	1.09	1.25
	(-1.74)	(2.91)	(0.99)	(2.45)	(4.28)	(4.73)	(4.76)	(4.98)
INFO-Z 2	-0.71	0.11	0.16	0.64	0.66	1.37	1.15	1.24
	(-2.77)	(0.47)	(0.74)	(2.91)	(3.20)	(5.30)	(4.91)	(5.03)
INFO-Z 3	-1.05	-0.19	0.23	0.16	0.57	1.62	1.23	1.21
	(-2.26)	(-0.67)	(0.95)	(0.74)	(1.91)	(3.62)	(3.16)	(3.28)
INFO-Z 4	-1.64	-0.29	0.11	-0.41	0.10	1.75	1.61	1.77
	(-3.65)	(-0.94)	(0.39)	(-1.40)	(0.40)	(4.28)	(4.54)	(4.57)
INFO-Z High	-2.15	-0.62	-0.59	-0.53	-0.36	1.79	1.45	1.57
	(-3.67)	(-1.70)	(-1.99)	(-1.51)	(-1.30)	(3.53)	(3.35)	(3.49)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-1.68	-1.01	-0.77	-1.01	-1.18	0.51	0.36	0.32
	(-3.56)	(-3.00)	(-3.01)	(-3.13)	(-3.83)	(1.27)	(0.91)	(0.83)
α_{FF6+2}	-1.45	-0.94	-0.57	-0.74	-1.09			
	(-3.79)	(-4.40)	(-2.96)	(-3.25)	(-4.42)			
α_{Q+2}	-1.39	-0.95	-0.60	-0.62	-1.07			
	(-3.80)	(-4.31)	(-3.10)	(-2.57)	(-4.22)			

Table 2: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on INFO-Z: One-half Subperiods

This table repeats the momentum analysis in Table 1, but with separate estimations for one-half subperiods. Panel A reports on overnight returns and Panel B on intraday returns. Panels A.1 and B.1 report on the 1997:01-2009:06 subperiod and Panels A.2 and B.2 on the 2009:07-2021:12 subperiod.

Panel A: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

Panel A.1: 1997:01-2009:06	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.82	-0.72	-0.17	0.22	0.88	1.71	1.87	1.79
	(-3.57)	(-2.96)	(-1.13)	(0.88)	(1.86)	(4.00)	(4.72)	(4.27)
INFO-Z 2	-0.82	-0.60	-0.31	0.35	1.53	2.35	2.47	2.35
	(-2.39)	(-1.85)	(-0.82)	(0.93)	(2.72)	(5.18)	(6.09)	(5.78)
INFO-Z 3	-0.83	0.05	0.32	1.06	2.47	3.30	2.81	2.67
	(-2.44)	(0.14)	(1.15)	(2.73)	(3.79)	(5.05)	(5.78)	(5.61)
INFO-Z 4	-0.54	0.29	0.60	1.10	2.68	3.22	3.21	3.05
	(-1.44)	(1.23)	(1.61)	(2.24)	(3.75)	(5.72)	(6.56)	(6.14)
INFO-Z High	-0.45	0.19	1.19	2.01	4.25	4.70	4.38	4.25
	(-1.36)	(0.48)	(2.93)	(4.37)	(4.36)	(4.95)	(5.50)	(5.21)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	0.38	0.91	1.37	1.79	3.37	2.99	2.51	2.46
	(1.89)	(3.67)	(3.77)	(4.61)	(5.60)	(5.03)	(5.07)	(5.11)
α_{FF6+2}	0.60	0.95	1.29	1.73	3.11			
	(2.11)	(4.57)	(3.35)	(4.79)	(6.42)			
α_{Q+2}	0.54	0.91	1.29	1.63	3.00			
	(2.18)	(3.58)	(3.42)	(4.70)	(6.13)			

Panel A.2: 2009:07-2021:12	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.60	0.63	0.59	0.76	1.05	0.45	0.39	0.39
	(2.88)	(2.65)	(2.77)	(3.71)	(3.35)	(2.23)	(2.01)	(2.02)
INFO-Z 2	0.78	0.72	0.76	0.94	1.15	0.37	0.32	0.30
	(2.78)	(3.24)	(2.69)	(3.21)	(4.92)	(2.11)	(1.82)	(1.73)
INFO-Z 3	0.94	0.86	1.22	1.27	1.37	0.43	0.33	0.34
	(3.07)	(2.82)	(4.60)	(3.91)	(4.36)	(1.94)	(1.51)	(1.48)
INFO-Z 4	0.96	1.21	1.14	1.53	1.77	0.81	0.63	0.68
	(2.79)	(3.44)	(3.25)	(4.27)	(4.41)	(4.23)	(2.99)	(3.64)
INFO-Z High	1.21	1.48	1.48	1.89	2.69	1.49	1.17	1.18
	(2.95)	(3.87)	(4.13)	(4.73)	(4.48)	(4.25)	(4.22)	(4.33)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	0.61	0.85	0.89	1.13	1.64	1.04	0.78	0.78
	(1.82)	(3.61)	(3.47)	(4.09)	(4.25)	(3.29)	(2.92)	(3.08)
α_{FF6+2}	0.50	0.64	0.75	0.87	1.28			
	(1.97)	(2.70)	(3.57)	(3.79)	(5.20)			
α_{Q+2}	0.57	0.67	0.82	0.93	1.35			
	(2.24)	(2.57)	(3.61)	(3.86)	(5.20)			

Table 2: (continued)

Panel B: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Intdy Return

Panel B.1: 1997:01-2009:06	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-1.05	0.27	-0.10	0.38	0.92	1.97	1.65	1.87
	(-2.69)	(1.64)	(-0.36)	(1.18)	(3.14)	(4.95)	(5.01)	(5.33)
INFO-Z 2	-1.17	-0.38	0.14	0.75	0.65	1.82	1.57	1.65
	(-3.22)	(-1.01)	(0.36)	(2.03)	(2.06)	(5.05)	(5.14)	(4.94)
INFO-Z 3	-1.89	-0.43	0.24	0.28	0.72	2.61	1.98	1.80
	(-2.43)	(-0.94)	(0.61)	(0.86)	(1.49)	(3.91)	(3.03)	(3.12)
INFO-Z 4	-2.78	-0.75	-0.07	-0.67	0.27	3.04	2.75	2.97
	(-4.29)	(-1.57)	(-0.16)	(-1.30)	(0.64)	(5.99)	(6.08)	(5.98)
INFO-Z High	-3.56	-1.17	-0.76	-0.73	-0.23	3.34	2.63	2.69
	(-3.93)	(-1.97)	(-1.55)	(-1.24)	(-0.48)	(5.38)	(5.38)	(5.17)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-2.51	-1.44	-0.65	-1.11	-1.15	1.37	0.99	0.82
	(-3.41)	(-2.66)	(-1.54)	(-2.16)	(-2.09)	(2.75)	(1.96)	(1.66)
α_{FF6+2}	-2.15	-1.19	-0.41	-0.62	-1.16			
	(-4.11)	(-4.25)	(-1.22)	(-1.86)	(-3.14)			
α_{Q+2}	-1.97	-1.16	-0.35	-0.50	-1.15			
	(-4.08)	(-4.16)	(-1.05)	(-1.51)	(-2.95)			

Panel B.2: 2009:07-2021:12	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.13	0.51	0.47	0.59	0.72	0.60	0.44	0.51
	(0.51)	(2.53)	(2.50)	(2.69)	(3.07)	(2.81)	(2.29)	(2.71)
INFO-Z 2	-0.25	0.61	0.18	0.53	0.66	0.91	0.61	0.72
	(-0.94)	(3.14)	(0.86)	(2.19)	(2.60)	(3.02)	(2.25)	(2.41)
INFO-Z 3	-0.21	0.05	0.23	0.04	0.42	0.64	0.38	0.49
	(-0.63)	(0.21)	(0.78)	(0.16)	(1.20)	(1.49)	(1.21)	(1.32)
INFO-Z 4	-0.51	0.18	0.29	-0.16	-0.06	0.45	0.43	0.51
	(-1.41)	(0.65)	(1.10)	(-0.69)	(-0.21)	(1.38)	(1.69)	(1.77)
INFO-Z High	-0.73	-0.07	-0.42	-0.32	-0.48	0.25	0.05	0.04
	(-1.87)	(-0.22)	(-1.36)	(-0.95)	(-1.81)	(0.56)	(0.12)	(0.11)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-0.85	-0.58	-0.89	-0.91	-1.21	-0.35	-0.39	-0.47
	(-1.84)	(-1.68)	(-3.08)	(-2.40)	(-4.35)	(-0.70)	(-0.80)	(-0.97)
α_{FF6+2}	-0.60	-0.56	-0.62	-0.86	-0.99			
	(-1.53)	(-1.84)	(-2.68)	(-2.69)	(-3.47)			
α_{Q+2}	-0.56	-0.50	-0.62	-0.76	-1.02			
	(-1.40)	(-1.53)	(-2.58)	(-2.24)	(-3.72)			

Table 3: Momentum in Overnight Stock Returns: Partitioned by Firm Size and INFO-Z

This table reports on a similar momentum analysis for monthly overnight stock returns, as in Table 1, Panel A, but separately for stocks partitioned into large-cap, mid-cap, and small-cap terciles, based on a stock's market capitalization (size) measured at month $t-2$. Following from the strong overnight-return results in Table 1, here we report on monthly overnight-returns only. For each size tercile, stocks are first sorted into quintiles on their INFO-Z measured at the close of month $t-2$, and then within each quintile on past one-month overnight returns in month $t-1$, using NYSE breakpoints. For brevity, we report only the low and high past-return quintiles and their high-minus-low spread portfolios, along with excess returns and alphas. See Table 1 for other tabular and estimation details. The sample period is 1997–2021.

Panel A: Large-cap stocks	Past Ovnt Returns		High - Low		
	Low	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.22	0.88	1.10	1.21	1.18
	(-0.99)	(3.40)	(3.74)	(4.00)	(3.97)
INFO-Z High	0.57	3.14	2.57	2.28	2.24
	(2.06)	(5.55)	(4.69)	(4.62)	(4.82)
High-Low INFO-Z Strategies					
Excess	0.80	2.26	1.47	1.07	1.06
	(3.83)	(6.04)	(4.26)	(3.40)	(3.53)
α_{FF6+2}	0.84	1.90			
	(3.45)	(5.38)			
α_{Q+2}	0.85	1.91			
	(3.45)	(5.65)			
Panel B: Mid-cap stocks	Past Ovnt Returns		High - Low		
INFO-Z Low	-0.23	0.96	1.19	1.07	1.09
	(-0.89)	(3.48)	(5.10)	(4.83)	(5.02)
INFO-Z High	0.31	2.96	2.66	2.56	2.52
	(0.93)	(6.16)	(4.99)	(5.28)	(5.11)
High-Low INFO-Z Strategies					
Excess	0.53	2.00	1.47	1.49	1.43
	(3.17)	(5.11)	(3.55)	(3.96)	(3.68)
α_{FF6+2}	0.47	1.96			
	(2.75)	(5.03)			
α_{Q+2}	0.49	1.92			
	(2.80)	(4.76)			
Panel C: Small-cap stocks	Past Ovnt Returns		High - Low		
INFO-Z Low	-0.21	1.12	1.33	1.25	1.23
	(-0.65)	(3.98)	(5.16)	(5.39)	(5.52)
INFO-Z High	-0.00	3.02	3.02	2.83	2.80
	(-0.01)	(7.14)	(5.05)	(5.39)	(5.51)
High-Low INFO-Z Strategies					
Excess	0.20	1.89	1.69	1.58	1.57
	(0.80)	(6.24)	(4.02)	(4.19)	(4.21)
α_{FF6+2}	0.24	1.82			
	(0.88)	(6.10)			
α_{Q+2}	0.22	1.79			
	(0.79)	(6.04)			

Table 4: Cross-reversals in Monthly Overnight & Intraday Stock Returns, Conditional on INFO-Z

This table reports the cross-reversal tug-of-war INFO-Z results, the counterpart results to Table 1. Panel A (B) reports the average value-weighted excess intraday(overnight) returns in month t for 25 (5 by 5) portfolios sequentially sorted on a stock's lagged INFO-Z and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their 3-factor INFO-Z ranking as of the close of month $t-2$ using NYSE breakpoints. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.82	0.53	0.44	0.08	-0.24	-1.06	-1.12	-1.17
	(4.56)	(2.99)	(2.55)	(0.42)	(-1.02)	(-4.93)	(-4.70)	(-4.61)
INFO-Z 2	0.89	0.54	0.46	-0.00	-0.57	-1.46	-1.60	-1.53
	(3.99)	(2.62)	(2.03)	(-0.02)	(-2.20)	(-4.55)	(-4.81)	(-4.71)
INFO-Z 3	1.00	0.37	-0.05	-0.11	-1.09	-2.09	-2.06	-2.00
	(4.22)	(1.49)	(-0.16)	(-0.44)	(-3.53)	(-7.62)	(-6.82)	(-7.43)
INFO-Z 4	0.53	0.15	-0.02	-0.68	-1.76	-2.28	-2.57	-2.60
	(1.83)	(0.52)	(-0.08)	(-2.09)	(-4.35)	(-6.32)	(-5.87)	(-5.74)
INFO-Z High	0.65	-0.15	-0.64	-1.24	-2.13	-2.79	-3.03	-2.91
	(1.85)	(-0.48)	(-1.74)	(-3.29)	(-4.66)	(-7.27)	(-8.42)	(-7.93)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-0.17	-0.68	-1.08	-1.33	-1.89	-1.73	-1.91	-1.74
	(-0.51)	(-2.10)	(-3.19)	(-3.73)	(-4.91)	(-6.15)	(-7.25)	(-6.44)
α_{FF6+2}	0.01	-0.53	-0.94	-1.06	-1.90			
	(0.05)	(-2.53)	(-4.12)	(-4.60)	(-7.83)			
α_{Q+2}	-0.01	-0.47	-0.91	-1.04	-1.75			
	(-0.05)	(-2.23)	(-3.80)	(-4.85)	(-7.19)			

Table 4: (continued)

Panel B: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	1.38	0.54	0.31	-0.01	-0.39	-1.77	-1.70	-1.78
	(5.23)	(2.75)	(1.84)	(-0.04)	(-1.38)	(-5.46)	(-5.51)	(-5.58)
INFO-Z 2	1.57	0.83	0.45	0.10	-0.13	-1.69	-1.55	-1.62
	(4.74)	(3.20)	(1.94)	(0.38)	(-0.42)	(-5.06)	(-4.62)	(-5.02)
INFO-Z 3	1.99	1.18	0.91	0.55	0.07	-1.91	-1.73	-1.69
	(5.32)	(4.47)	(2.91)	(1.89)	(0.24)	(-4.27)	(-4.56)	(-4.69)
INFO-Z 4	2.19	1.29	0.89	0.77	0.51	-1.68	-1.65	-1.69
	(5.79)	(4.50)	(2.81)	(2.47)	(1.49)	(-3.94)	(-4.08)	(-4.17)
INFO-Z High	3.34	1.87	1.47	1.25	1.03	-2.31	-2.22	-2.31
	(5.80)	(4.65)	(3.49)	(3.67)	(2.46)	(-4.31)	(-4.44)	(-4.52)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	1.96	1.33	1.17	1.26	1.42	-0.54	-0.52	-0.52
	(5.23)	(4.49)	(3.24)	(5.58)	(4.77)	(-1.74)	(-1.66)	(-1.64)
α_{FF6+2}	1.83	1.05	1.14	1.17	1.31			
	(5.04)	(4.11)	(3.36)	(6.22)	(4.70)			
α_{Q+2}	1.82	1.13	1.06	1.10	1.30			
	(5.05)	(4.36)	(3.37)	(6.34)	(4.70)			

Table 5: INFO-Z Fama-MacBeth Regressions

This table reports Fama-MacBeth regressions of monthly stock overnight or intraday returns on a stock's lagged INFO-Z as well as the interactions with lagged overnight (ret_night) and lagged intraday returns (ret_day). See equations (1) and (2). In Panel A, INFO-Z explanatory terms use the continuous INFO-Z variable. In Panel B, the INFO-Z explanatory terms use INFO-Z dummy variables with high- and low-quintile dummy variables. Columns (1) to (3) report results where the dependent variable is the monthly overnight return from month t. Columns (4) to (6) report results where the dependent variable is monthly intraday return from month t. We control for the following lagged firm characteristic variables: the exponentially weighted moving average overnight/intraday returns with a half-life of 60 months, the lagged 12-month cumulative stock return, market capitalization, book-to-market ratio, 12-month market beta computed using daily returns with three lags, 12-month stock turnover, return-on-equity, asset growth, equity issuance, and discretionary accruals; all as of month $t-2$. We also include the industry fixed effects, using the Fama-French 48-industry definitions. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period spans from 1997 to 2021.

Panel A: Fama-MacBeth regressions with continuous INFO-Z variable

	[1]	[2]	[3]	[4]	[5]	[6]
Dependent Variable:	Overnight Returns	Overnight Returns	Overnight Returns	Intraday Returns	Intraday Returns	Intraday Returns
Intercept	0.39	1.02	-0.07	0.50	-0.16	0.94
	(2.25)	(4.69)	(0.30)	(1.73)	(0.48)	(2.50)
ret night	8.03	8.44	7.05	-9.42	-9.30	-7.74
	(7.70)	(7.23)	(8.37)	(-9.64)	(-7.65)	(-7.31)
INFO-Z × ret night		3.35	2.63		-1.70	-0.83
		(4.39)	(4.14)		(-1.72)	(-0.89)
ret day	-5.42	-5.77	-5.60	3.35	3.70	3.02
	(-5.48)	(-5.68)	(-5.71)	(3.87)	(4.14)	(3.78)
INFO-Z × ret day		-1.07	-1.20		1.59	1.62
		(-2.41)	(-2.81)		(2.28)	(2.42)
INFO-Z		0.91	0.27		-0.93	-0.22
		(6.77)	(2.28)		(-4.04)	(-1.01)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.07	0.08	0.11	0.11	0.13	0.10

Table 5: (continued)

Panel B: Fama-MacBeth regressions with INFO-Z-quintile dummy variables

	[1]	[2]	[3]	[4]	[5]	[6]
Dependent	Overnight	Overnight	Overnight	Intraday	Intraday	Intraday
Variable:	Returns	Returns	Returns	Returns	Returns	Returns
Intercept	0.39	0.36	-0.29	0.50	0.55	1.15
	(2.25)	(1.94)	(-1.48)	(1.73)	(1.75)	(3.60)
ret night	8.03	6.16	5.20	-9.42	-9.08	-7.56
	(7.70)	(7.07)	(7.97)	(-9.64)	(-9.19)	(-9.04)
Hi INFO-Z × ret night		2.90	2.23			
		(4.28)	(3.66)			
Lo INFO-Z × ret night		-0.02	-0.45			
		(-0.04)	(-0.89)			
ret day	-5.42	-5.44	-5.23	3.35	2.24	1.71
	(-5.48)	(-5.55)	(-5.60)	(3.87)	(2.72)	(2.23)
Hi INFO-Z × ret day					1.79	1.60
					(2.94)	(2.83)
Lo INFO-Z × ret day					-0.12	0.04
					(-0.22)	(0.08)
Hi INFO-Z		0.54	0.13		-0.47	-0.01
		(5.50)	(1.65)		(-3.17)	(-0.09)
Lo INFO-Z		-0.19	-0.04		0.09	-0.03
		(-3.60)	(-0.76)		(0.94)	(-0.43)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.07	0.08	0.10	0.11	0.12	0.16

Table 6: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on PTA

This table analyzes momentum in both monthly overnight stock returns and monthly intraday stock returns, conditional on a stock's 52-week PTA metric. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on a stock's lagged 52-week PTA and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their PTA ranking as of the close of month $t-2$ using NYSE breakpoints. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight (intraday) returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.01	0.02	0.42	0.74	1.82	1.81	1.71	1.64
	(0.03)	(0.08)	(1.88)	(2.61)	(4.24)	(4.49)	(4.21)	(4.27)
PTA 2	-0.04	-0.15	0.15	0.53	0.98	1.02	0.97	0.91
	(-0.15)	(-0.64)	(0.78)	(2.37)	(3.75)	(3.58)	(3.69)	(3.63)
PTA 3	-0.03	-0.01	0.27	0.35	0.94	0.97	0.96	0.95
	(-0.10)	(-0.03)	(1.28)	(1.69)	(4.00)	(3.06)	(3.12)	(3.15)
PTA 4	-0.04	0.18	0.45	0.86	1.19	1.23	1.16	1.14
	(-0.19)	(0.69)	(2.31)	(4.85)	(5.03)	(5.00)	(5.43)	(5.31)
PTA High	0.59	0.88	1.16	1.79	3.08	2.49	2.33	2.25
	(2.12)	(3.60)	(5.32)	(6.88)	(6.74)	(5.52)	(5.54)	(5.59)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.58	0.86	0.74	1.05	1.26	0.68	0.63	0.60
	(2.73)	(4.89)	(4.16)	(5.25)	(3.46)	(1.98)	(1.93)	(1.81)
α_{FF6+2}	0.62	0.80	0.65	0.83	1.25			
	(2.58)	(4.70)	(3.81)	(4.27)	(3.78)			
α_{Q+2}	0.57	0.78	0.64	0.82	1.17			
	(2.40)	(4.41)	(4.27)	(3.85)	(3.15)			

Table 6: (continued)

Panel B: Monthly Intdy Returns: Conditional on PTA & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-1.94	-0.25	0.36	0.49	0.49	2.42	2.31	2.46
	(-3.10)	(-0.69)	(1.06)	(1.76)	(1.56)	(4.30)	(4.63)	(4.42)
PTA 2	-0.52	0.39	0.60	0.68	0.94	1.45	1.32	1.42
	(-1.61)	(1.66)	(2.66)	(3.07)	(4.39)	(4.42)	(4.20)	(4.47)
PTA 3	-0.78	0.18	0.50	0.59	1.03	1.81	1.50	1.64
	(-2.08)	(0.83)	(3.01)	(2.67)	(5.64)	(4.51)	(4.45)	(4.54)
PTA 4	-0.81	0.05	0.08	0.40	0.71	1.52	1.25	1.37
	(-2.89)	(0.29)	(0.35)	(2.06)	(3.04)	(4.99)	(4.37)	(4.87)
PTA High	-1.54	-0.54	-0.83	-0.70	-0.50	1.04	0.86	0.94
	(-3.82)	(-2.27)	(-2.71)	(-2.83)	(-2.26)	(3.03)	(3.11)	(3.48)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.40	-0.28	-1.19	-1.19	-0.99	-1.38	-1.46	-1.52
	(1.08)	(-0.77)	(-3.03)	(-3.83)	(-3.32)	(-3.72)	(-3.68)	(-3.34)
α_{FF6+2}	0.08	-0.71	-1.61	-1.62	-1.38			
	(0.20)	(-2.24)	(-3.95)	(-4.96)	(-4.73)			
α_{Q+2}	0.01	-0.93	-1.71	-1.65	-1.51			
	(0.04)	(-2.49)	(-3.78)	(-4.76)	(-5.07)			

Table 7: Momentum in Monthly Overnight & Intraday Stock Returns, conditional on PTA: One-half Subperiods

This table repeats the momentum analysis in Table 6, but with separate estimations for one-half subperiods. Panel A reports on overnight returns and Panel B on intraday returns. Panels A.1 and B.1 report on the 1997:01-2009:06 subperiod and Panels A.2 and B.2 on the 2009:07-2021:12 subperiod.

Panel A: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Ovnt Return

Panel A.1: 1997:01-2009:06	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-0.71	-0.52	0.16	0.47	2.53	3.24	3.12	2.99
	(-1.79)	(-1.97)	(0.58)	(1.10)	(3.69)	(6.20)	(5.75)	(5.51)
PTA 2	-0.80	-0.85	-0.28	0.26	1.02	1.82	1.78	1.62
	(-2.71)	(-2.67)	(-1.40)	(0.79)	(2.44)	(4.57)	(5.18)	(4.73)
PTA 3	-0.85	-0.66	-0.23	0.02	1.05	1.89	1.86	1.85
	(-2.83)	(-2.22)	(-0.77)	(0.05)	(2.52)	(4.09)	(4.17)	(4.29)
PTA 4	-0.68	-0.60	-0.01	0.75	1.24	1.92	1.76	1.77
	(-2.56)	(-2.08)	(-0.03)	(3.17)	(3.24)	(5.50)	(6.07)	(6.34)
PTA High	-0.17	0.28	0.82	1.71	3.57	3.75	3.61	3.43
	(-0.57)	(0.89)	(2.71)	(4.66)	(4.59)	(5.62)	(6.48)	(6.35)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.54	0.80	0.66	1.24	1.04	0.50	0.49	0.45
	(1.54)	(3.01)	(2.14)	(4.93)	(1.53)	(0.81)	(0.97)	(0.87)
α_{FF6+2}	0.37	0.61	0.58	1.00	0.86			
	(1.11)	(2.37)	(1.78)	(4.83)	(1.46)			
α_{Q+2}	0.35	0.53	0.63	0.96	0.80			
	(1.12)	(1.88)	(2.20)	(4.34)	(1.27)			

Panel A.2: 2009:07-2021:12	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.73	0.56	0.67	1.01	1.10	0.38	0.28	0.30
	(2.10)	(1.61)	(2.06)	(2.98)	(2.61)	(1.94)	(1.17)	(1.20)
PTA 2	0.73	0.55	0.58	0.80	0.94	0.21	0.14	0.18
	(3.09)	(2.30)	(2.04)	(2.84)	(2.98)	(0.92)	(0.63)	(0.81)
PTA 3	0.80	0.65	0.76	0.69	0.84	0.04	0.08	0.09
	(2.65)	(2.56)	(3.45)	(3.03)	(3.67)	(0.22)	(0.41)	(0.45)
PTA 4	0.59	0.95	0.90	0.97	1.14	0.55	0.52	0.49
	(2.25)	(3.57)	(3.65)	(3.69)	(4.04)	(2.86)	(3.06)	(3.10)
PTA High	1.35	1.48	1.50	1.87	2.58	1.23	1.06	1.05
	(4.00)	(5.28)	(5.22)	(4.96)	(5.94)	(4.19)	(3.20)	(2.94)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.63	0.92	0.82	0.86	1.48	0.85	0.78	0.75
	(2.66)	(4.30)	(4.80)	(2.88)	(4.90)	(2.53)	(1.81)	(1.60)
α_{FF6+2}	0.88	0.98	0.71	0.62	1.65			
	(2.87)	(3.73)	(4.72)	(2.15)	(5.38)			
α_{Q+2}	0.82	0.98	0.68	0.59	1.57			
	(2.55)	(3.77)	(4.82)	(1.90)	(4.42)			

Table 7: (continued)

Panel B: Monthly Intdy Returns: Conditional on PTA & Prior Month's Intdy Return

Panel B.1: 1997:01-2009:06	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-3.38	-0.81	0.54	0.81	0.68	4.06	3.74	3.82
	(-3.41)	(-1.41)	(0.91)	(1.72)	(1.27)	(4.65)	(5.04)	(4.45)
PTA 2	-1.17	0.15	0.23	0.70	0.99	2.16	1.94	2.03
	(-2.31)	(0.38)	(0.64)	(1.83)	(2.93)	(4.32)	(4.10)	(4.08)
PTA 3	-1.60	0.08	0.62	0.39	1.13	2.74	2.38	2.48
	(-2.82)	(0.21)	(2.42)	(1.06)	(4.61)	(4.56)	(4.80)	(4.62)
PTA 4	-1.49	-0.11	-0.13	0.38	0.87	2.36	1.99	2.08
	(-3.66)	(-0.40)	(-0.35)	(1.34)	(2.11)	(5.55)	(4.72)	(5.08)
PTA High	-2.47	-1.09	-1.36	-0.99	-0.49	1.98	1.59	1.57
	(-3.86)	(-2.94)	(-3.09)	(-3.10)	(-1.46)	(4.28)	(5.03)	(5.28)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.90	-0.28	-1.90	-1.79	-1.17	-2.08	-2.16	-2.25
	(1.59)	(-0.46)	(-3.48)	(-4.22)	(-2.42)	(-3.20)	(-3.41)	(-2.77)
α_{FF6+2}	0.50	-0.79	-2.52	-2.34	-1.66			
	(0.81)	(-1.64)	(-4.42)	(-5.16)	(-3.44)			
α_{Q+2}	0.28	-1.28	-2.84	-2.57	-1.97			
	(0.42)	(-2.38)	(-4.68)	(-5.41)	(-4.38)			

Panel B.2: 2009:07-2021:12	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-0.50	0.30	0.19	0.17	0.29	0.79	0.66	0.79
	(-1.04)	(0.83)	(0.58)	(0.65)	(1.06)	(2.30)	(2.37)	(2.65)
PTA 2	0.13	0.64	0.97	0.65	0.88	0.75	0.65	0.74
	(0.53)	(2.67)	(4.94)	(3.09)	(3.45)	(2.46)	(2.54)	(2.76)
PTA 3	0.05	0.29	0.38	0.79	0.93	0.88	0.57	0.66
	(0.17)	(1.46)	(1.86)	(3.47)	(3.53)	(2.76)	(1.99)	(2.06)
PTA 4	-0.13	0.22	0.28	0.43	0.56	0.69	0.44	0.57
	(-0.55)	(0.97)	(1.56)	(1.69)	(2.79)	(2.75)	(2.19)	(2.69)
PTA High	-0.62	0.02	-0.30	-0.41	-0.51	0.10	-0.01	0.09
	(-2.16)	(0.12)	(-0.91)	(-1.20)	(-1.79)	(0.32)	(-0.04)	(0.33)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.11	-0.28	-0.48	-0.58	-0.81	-0.69	-0.67	-0.70
	(-0.25)	(-0.69)	(-0.98)	(-1.54)	(-2.44)	(-2.12)	(-1.96)	(-2.13)
α_{FF6+2}	-0.46	-0.65	-0.68	-0.90	-1.12			
	(-1.64)	(-1.82)	(-1.98)	(-2.80)	(-4.12)			
α_{Q+2}	-0.44	-0.63	-0.58	-0.81	-1.14			
	(-1.44)	(-1.63)	(-1.53)	(-2.52)	(-3.98)			

Table 8: Momentum in Intraday Stock Returns: Partitioned by Firm Size and PTA

This table reports on a similar momentum analysis for monthly intraday stock returns, as in Table 6, Panel B, but separately for stocks partitioned into large-cap, mid-cap, and small-cap terciles, based on a stock's market capitalization (size) measured at month $t-2$. Following from the strong intraday-return results in Table 6, here we report on monthly intraday-returns only. For each size tercile, stocks are first sorted into quintiles on their 52-week PTA measured at the close of month $t-2$, and then within each quintile on past one-month intraday returns in month $t-1$, using NYSE breakpoints. For brevity, we report only the low and high past-return quintiles and their high-minus-low spread portfolios, along with excess returns and alphas. See Table 6 for other tabular and estimation details. The sample period is 1997–2021.

Panel A: Large-cap stocks	Past Intrdy Returns		High - Low		
	Low	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-1.32	0.77	2.10	1.99	2.06
	(-2.49)	(2.98)	(4.25)	(4.77)	(4.56)
PTA High	-1.49	-0.78	0.71	0.61	0.65
	(-3.79)	(-2.83)	(2.23)	(2.41)	(2.48)
High-Low PTA Strategies					
Excess	-0.18	-1.55	-1.39	-1.38	-1.41
	(-0.54)	(-5.13)	(-3.40)	(-3.34)	(-3.26)
α_{FF6+2}	-0.50	-1.87			
	(-1.32)	(-6.26)			
α_{Q+2}	-0.62	-2.01			
	(-1.68)	(-6.25)			
Panel B: Mid-cap stocks	Past Intrdy Returns		High - Low		
PTA Low	-1.19	0.17	1.36	1.09	1.16
	(-2.00)	(0.52)	(2.85)	(3.05)	(3.03)
PTA High	-1.27	-0.32	0.95	0.78	0.84
	(-3.61)	(-1.22)	(2.79)	(2.33)	(2.52)
High-Low PTA Strategies					
Excess	-0.08	-0.49	-0.41	-0.31	-0.32
	(-0.21)	(-1.61)	(-1.14)	(-0.92)	(-0.92)
α_{FF6+2}	-0.64	-0.95			
	(-2.07)	(-3.84)			
α_{Q+2}	-0.74	-1.06			
	(-1.95)	(-3.30)			
Panel C: Small-cap stocks	Past Intrdy Returns		High - Low		
PTA Low	-1.77	0.21	1.98	1.69	1.75
	(-3.07)	(0.53)	(4.32)	(4.11)	(4.38)
PTA High	-0.86	0.08	0.94	0.86	0.98
	(-2.18)	(0.21)	(2.40)	(2.47)	(2.77)
High-Low PTA Strategies					
Excess	0.91	-0.13	-1.04	-0.82	-0.77
	(1.94)	(-0.35)	(-2.62)	(-2.11)	(-2.15)
α_{FF6+2}	0.43	-0.40			
	(1.09)	(-1.46)			
α_{Q+2} 0.26	-0.51				
	(0.72)	(-1.81)			

Table 9: Cross-reversals in Monthly Overnight & Intraday Stock Returns, Conditional on PTA

This table reports the cross-reversal tug-of-war results for the 52-week PTA, the counterpart results to Table 6. Panel A (B) reports the average value-weighted excess intraday(overnight) returns in month t for 25 (5 by 5) portfolios sequentially sorted on a stock's lagged PTA and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their PTA ranking as of the close of month $t-2$ using NYSE breakpoints. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Intdy Returns: Conditional on PTA & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.81	0.61	0.17	-0.39	-1.44	-2.25	-2.18	-2.21
	(1.98)	(1.87)	(0.49)	(-1.31)	(-3.52)	(-6.00)	(-5.56)	(-5.65)
PTA 2	0.86	0.72	0.76	0.52	-0.24	-1.10	-1.16	-1.15
	(3.64)	(2.98)	(3.58)	(2.06)	(-0.89)	(-4.31)	(-4.04)	(-4.03)
PTA 3	0.91	0.60	0.45	0.22	-0.25	-1.16	-1.26	-1.21
	(3.95)	(3.26)	(2.35)	(0.93)	(-0.98)	(-3.68)	(-4.22)	(-3.97)
PTA 4	0.67	0.48	0.26	-0.25	-0.63	-1.31	-1.36	-1.34
	(3.54)	(2.31)	(1.25)	(-1.24)	(-3.03)	(-6.88)	(-6.85)	(-6.30)
PTA High	0.40	-0.38	-0.68	-0.92	-1.86	-2.26	-2.40	-2.32
	(1.57)	(-1.58)	(-3.01)	(-3.09)	(-4.85)	(-6.72)	(-6.93)	(-6.62)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.40	-0.99	-0.85	-0.53	-0.42	-0.02	-0.22	-0.11
	(-1.16)	(-2.81)	(-2.52)	(-1.63)	(-1.57)	(-0.06)	(-0.58)	(-0.29)
α_{FF6+2}	-0.86	-1.12	-1.25	-0.69	-1.08			
	(-2.58)	(-3.52)	(-4.38)	(-2.05)	(-3.29)			
α_{Q+2}	-1.00	-1.35	-1.39	-0.81	-1.11			
	(-2.66)	(-3.22)	(-4.70)	(-2.20)	(-3.61)			

Table 9: (continued)

Panel B: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	2.37	1.14	0.46	-0.06	-0.33	-2.70	-2.67	-2.72
	(4.52)	(3.78)	(1.43)	(-0.22)	(-0.97)	(-4.07)	(-3.91)	(-3.84)
PTA 2	1.49	0.60	0.26	-0.01	-0.54	-2.03	-1.98	-2.03
	(5.06)	(2.99)	(1.15)	(-0.04)	(-1.64)	(-5.12)	(-5.55)	(-5.69)
PTA 3	1.55	0.61	0.06	-0.06	-0.41	-1.96	-1.83	-1.88
	(5.21)	(2.80)	(0.24)	(-0.24)	(-1.39)	(-5.31)	(-5.35)	(-5.47)
PTA 4	1.45	0.69	0.44	0.31	-0.07	-1.52	-1.36	-1.42
	(4.68)	(3.97)	(2.20)	(1.52)	(-0.22)	(-3.83)	(-3.69)	(-3.70)
PTA High	2.62	1.62	1.58	1.23	1.25	-1.37	-1.18	-1.22
	(6.72)	(5.80)	(5.53)	(4.67)	(3.74)	(-3.89)	(-3.59)	(-3.82)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.25	0.48	1.13	1.30	1.58	1.33	1.49	1.49
	(0.55)	(2.49)	(5.06)	(7.55)	(6.36)	(2.63)	(2.67)	(2.56)
α_{FF6+2}	0.13	0.41	1.12	1.21	1.62			
	(0.29)	(2.30)	(6.21)	(8.07)	(7.53)			
α_{Q+2}	0.05	0.37	1.02	1.19	1.55			
	(0.10)	(1.91)	(5.10)	(8.23)	(7.61)			

Table 10: 52-week PTA Fama-MacBeth Regressions

This table reports Fama-MacBeth regressions of monthly stock overnight or intraday returns on a stock's lagged PTA as well as the interactions with lagged overnight (ret_night) and lagged intraday returns (ret_day). See equations (3) and (4). In Panel A, the PTA explanatory terms use the continuous PTA variable. In Panel B, the PTA explanatory terms use PTA dummy variables with high- and low-quintile dummy variables. Columns (1) to (3) report results where the dependent variable is the monthly overnight return from month t. Columns (4) to (6) report results where the dependent variable is monthly intraday return from month t. We control for the following lagged firm characteristic variables: the exponentially weighted moving average overnight/intraday returns with a half-life of 60 months, the lagged 12-month cumulative stock return, market capitalization, book-to-market ratio, 12-month market beta computed using daily returns with three lags, 12-month stock turnover, return-on-equity, asset growth, equity issuance, and discretionary accruals; all as of month $t-2$. We also include the industry fixed effects, using the Fama-French 48-industry definitions. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period spans from 1997 to 2021.

Panel A: Fama-MacBeth regressions with continuous PTA variable

	[1]	[2]	[3]	[4]	[5]	[6]
Dependent Variable:	Overnight Returns	Overnight Returns	Overnight Returns	Intraday Returns	Intraday Returns	Intraday Returns
Intercept	0.39	0.40	-0.32	0.50	0.55	1.26
	(2.25)	(2.26)	(1.63)	(1.73)	(1.90)	(3.96)
ret night	8.03	7.36	5.85	-9.42	-9.23	-7.60
	(7.70)	(7.59)	(8.53)	(-9.64)	(-9.96)	(-9.84)
PTA × ret night		1.60	1.35		-0.23	0.16
		(3.27)	(3.01)		(-0.50)	(0.32)
ret day	-5.42	-5.74	-5.55	3.35	3.48	2.71
	(-5.48)	(-5.52)	(-5.62)	(3.87)	(3.96)	(3.50)
PTA × ret day		1.80	1.76		-1.02	-0.85
		(5.89)	(6.47)		(-2.41)	(-2.07)
PTA		0.22	-0.31		-0.17	0.45
		(3.11)	(-3.14)		(-1.31)	(3.05)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.07	0.08	0.10	0.11	0.13	0.16

Table 10: (continued)

Panel B: Fama-MacBeth regressions with PTA-quintile dummy variables

	[1]	[2]	[3]	[4]	[5]	[6]
Dependent	Overnight	Overnight	Overnight	Intraday	Intraday	Intraday
Variable:	Returns	Returns	Returns	Returns	Returns	Returns
Intercept	0.39	0.21	-0.40	0.50	0.72	1.29
	(2.25)	(1.15)	(-1.97)	(1.73)	(2.38)	(3.89)
ret night	8.03	6.93	5.86	-9.42	-9.11	-7.52
	(7.70)	(7.39)	(8.37)	(-9.64)	(-9.49)	(-9.10)
Hi PTA × ret night		2.50	1.68			
		(4.24)	(3.38)			
Lo PTA × ret night		-0.23	-0.35			
		(-0.30)	(-0.47)			
ret day	-5.42	-5.44	-5.27	3.35	2.82	2.39
	(-5.48)	(-5.46)	(-5.55)	(3.87)	(3.20)	(2.90)
Hi PTA × ret day					-0.28	-0.49
					(-0.44)	(-0.82)
Lo PTA × ret day					1.80	1.37
					(2.27)	(1.84)
Hi PTA		0.65	0.16		-0.62	-0.07
		(9.45)	(3.72)		(-5.14)	(-0.86)
Lo PTA		0.49	0.37		-0.38	-0.26
		(3.88)	(3.68)		(-2.37)	(-2.73)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.07	0.08	0.10	0.11	0.13	0.16

Table 11: Investor sentiment and Momentum with INFO-Z and 52-week PTA

This table reports conditional overnight-momentum results with INFO-Z and conditional intraday-momentum results with PTA for different investor sentiment states. We classify each month as a high, medium, or low Baker-Wurgler sentiment month, as of month $t-2$ using sample terciles. We then report the results for each sentiment state using our previous double-sorting method; Panel A for INFO-Z with the Table 1 method, and Panel B for PTA with the Table 6 method. To conserve space, we report only the low and high past-return quintiles and the corresponding high-minus-low spread portfolios, along with excess returns and alphas. The sample period is 1997–2021. Other tabular and estimation details are as in Tables 1 and 6.

Panel A: Investor sentiment, INFO-Z, and overnight stock returns

Panel A.1: High Sentiment	Past Ovnt Returns		High - Low		
	Low	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.85	0.48	1.33	1.68	1.51
	(-2.59)	(1.37)	(4.02)	(4.17)	(4.43)
INFO-Z High	-0.53	3.35	3.87	3.21	3.27
	(-1.20)	(4.06)	(4.81)	(5.18)	(5.74)
High-Low INFO-Z Strategies					
Excess	0.32	2.87	2.54	1.54	1.76
	(0.89)	(4.16)	(3.98)	(3.63)	(4.06)
α_{FF6+2}	1.05	2.59			
	(3.42)	(6.14)			
α_{Q+2}	0.76	2.52			
	(2.53)	(5.93)			
Panel A.2: Medium Sentiment	Past Ovnt Returns		High - Low		
INFO-Z Low	0.25	1.65	1.41	1.08	1.28
	(0.63)	(3.14)	(2.47)	(2.92)	(2.60)
INFO-Z High	0.49	4.52	4.03	3.29	3.80
	(0.92)	(3.88)	(3.30)	(3.97)	(3.70)
High-Low INFO-Z Strategies					
Excess	0.24	2.86	2.62	2.22	2.52
	(0.87)	(3.83)	(3.66)	(4.07)	(4.06)
α_{FF6+2}	0.23	2.45			
	(0.89)	(4.36)			
α_{Q+2}	0.23	2.75			
	(0.73)	(4.18)			
Panel A.3: Low Sentiment	Past Ovnt Returns		High - Low		
INFO-Z Low	0.27	0.76	0.49	0.53	0.58
	(1.67)	(3.71)	(2.60)	(2.49)	(2.68)
INFO-Z High	1.17	2.55	1.39	1.09	1.07
	(4.14)	(5.93)	(4.41)	(3.63)	(3.72)
High-Low INFO-Z Strategies					
Excess	0.90	1.80	0.90	0.55	0.50
	(2.93)	(5.83)	(3.95)	(2.32)	(2.45)
α_{FF6+2}	0.90	1.46			
	(2.32)	(4.58)			
α_{Q+2}	0.83	1.33			
	(2.40)	(4.58)			

Table 11: (continued)

Panel B: Investor sentiment, 52-week PTA, and intraday stock returns

Panel B.1: High Sentiment	Past Intrdy Returns		High - Low		
	Low	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-2.78	0.34	3.12	2.39	2.58
	(-2.64)	(0.78)	(3.35)	(4.63)	(4.95)
PTA High	-1.93	-0.51	1.43	0.92	1.26
	(-2.49)	(-1.28)	(2.13)	(2.30)	(2.82)
High-Low PTA Strategies					
Excess	0.85	-0.84	-1.69	-1.47	-1.33
	(1.47)	(-1.95)	(-4.09)	(-5.50)	(-5.39)
α_{FF6+2}	-0.20	-1.68			
	(-0.56)	(-4.30)			
α_{Q+2}	-0.33	-1.66			
	(-0.81)	(-3.63)			

Panel B.2: Medium Sentiment	Past Intrdy Returns		High - Low		
PTA Low	-3.41	-0.61	2.80	2.24	2.78
	(-3.45)	(-1.17)	(3.27)	(2.96)	(3.17)
PTA High	-2.15	-1.24	0.91	0.06	0.33
	(-2.43)	(-2.39)	(1.61)	(0.12)	(0.76)
High-Low PTA Strategies					
Excess	1.27	-0.62	-1.89	-2.18	-2.44
	(2.46)	(-1.16)	(-3.32)	(-2.98)	(-3.09)
α_{FF6+2}	0.64	-1.55			
	(1.32)	(-2.68)			
α_{Q+2}	1.16	-1.28			
	(1.89)	(-2.23)			

Panel B.3: Low Sentiment	Past Intrdy Returns		High - Low		
PTA Low	0.39	1.80	1.41	1.07	1.58
	(0.57)	(3.15)	(1.82)	(2.32)	(2.19)
PTA High	-0.45	0.32	0.77	0.54	0.87
	(-1.01)	(1.02)	(1.61)	(1.27)	(2.36)
High-Low PTA Strategies					
Excess	-0.84	-1.47	-0.64	-0.53	-0.71
	(-1.89)	(-3.14)	(-0.99)	(-1.22)	(-1.12)
α_{FF6+2}	-0.23	-0.76			
	(-0.53)	(-1.74)			
α_{Q+2}	-0.75	-1.46			
	(-1.43)	(-4.20)			

Table 12: Fama-MacBeth Regressions with Additional Control Variables

This table reports on the extended Fama-MacBeth regressions as given by equation (5) for the INFO-Z specification and by equation (6) for the PTA specification. These specifications extend the earlier Fama-MacBeth results in Table 5 and Table 10 by adding six additional control variables, which are motivated by Lu et al. (2023), Jones et al. (2024), and Akbas et al. (2022). Section 6.3.1 explains these additional explanatory terms. The LMQ, JPW, and ABNR notation for these additional terms refer to the Lu et al. (2023), Jones et al. (2024), and Akbas et al. (2022) papers, respectively. The specifications include both the stand-alone LMQ, JPW, and ABNR variables and the variables interacted with the lagged return of interest. Columns (1) to (3) report the estimation results for three variations of the INFO-Z equation (5), with the monthly overnight returns from month t as the dependent variable. Columns (4) to (6) report the estimation results for three variations of the PTA equation (6), with the monthly intraday returns from month t as the dependent variable. The control variables indicated by the 'Control row' are the following lagged firm characteristic variables: the exponentially weighted moving average overnight/intraday returns with a half-life of 60 months, the lagged 12-month cumulative stock return, market capitalization, book-to-market ratio, 12-month market beta computed using daily returns with three lags, 12-month stock turnover, return-on-equity, asset growth, equity issuance, and discretionary accruals; all as of month $t-2$. We also include the industry fixed effects, using the Fama-French 48-industry definitions. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period spans from 1997 to 2021.

[Tabular Results on the next page]

Table 12: (continued)

Dependent Variable:	[1] Overnight Returns	[2] Overnight Returns	[3] Overnight Returns	[4] Intraday Returns	[5] Intraday Returns	[6] Intraday Returns
Intercept	0.68	0.73	-0.07	0.20	0.22	0.89
	(3.39)	(3.31)	(-0.31)	(0.65)	(0.72)	(2.98)
ret night	6.76	8.00	7.17	-9.10	-8.94	-7.87
	(7.37)	(6.81)	(7.25)	(-9.35)	(-9.80)	(-9.75)
INFO-Z × ret night		3.41	3.00			
		(3.84)	(3.63)			
PTA × ret night					-0.36	-0.17
					(-0.76)	(-0.38)
Z_LMQ × ret night	0.36	0.30	0.38			
	(1.41)	(1.28)	(1.59)			
Z_JPW × ret night	0.17	-0.42	-0.51			
	(0.77)	(-1.44)	(-1.74)			
ABNR × ret night	-0.20	-0.42	-0.40			
	(-0.64)	(-1.34)	(-1.38)			
ret day	-9.21	-8.96	-7.50	7.25	7.46	4.84
	(-6.95)	(-6.48)	(-6.41)	(5.95)	(5.96)	(5.31)
INFO-Z × ret day		-1.01	-1.07			
		(-2.34)	(-2.47)			
PTA × ret day					-1.04	-0.96
					(-2.37)	(-2.25)
Z_LMQ × ret day				0.22	0.22	0.22
				(1.30)	(1.30)	(1.34)
Z_JPW × ret day				-0.09	0.05	0.24
				(-0.42)	(0.24)	(1.19)
ABNR × ret day				-0.21	-0.13	-0.14
				(-0.94)	(-0.56)	(-0.60)
INFO-Z		0.10	-0.13			
		(1.03)	(-1.19)			
PTA					-0.17	0.32
					(-1.36)	(2.20)
Z_LMQ	0.06	0.06	0.05	0.00	0.00	0.00
	(3.26)	(3.16)	(3.02)	(0.00)	(0.00)	(0.13)
Z_JPW	0.71	0.65	0.39	-0.70	-0.73	-0.39
	(6.06)	(5.31)	(5.88)	(-5.31)	(-5.44)	(-5.55)
ABNR	-0.03	-0.01	0.01	0.16	0.16	0.12
	(-1.10)	(-0.38)	(0.65)	(3.61)	(3.67)	(3.25)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.09	0.09	0.11	0.14	0.15	0.17

A. External Appendices: Additional Analysis and Robustness

The external appendices report details on additional analysis and robustness results that are only summarized in the main text of the paper, *Heterogeneity in Overnight-versus-Intraday Stock-Return Behavior with Asymmetric Information and 52-week Price Anchoring*. Each appendix below begins by referring back to the section in the main paper that it supports.

Appendices A.1 to A.5 report additional results related to asymmetric information. First, Appendix A.1 reports on a momentum analysis comparable to the main Table 1, but with independent sorts on the two conditioning variables (rather than sequential). Second, Appendix A.2 shows that our main findings remain robust to an alternative sorting procedure based on a 3×10 double-sort portfolio design. Third, Appendix A.3 shows that our main INFO-Z results are similarly evident when using a simpler INFO-Z variable, which does not control for the liquidity variables. Fourth, Appendix A.4 shows that our main momentum results with asymmetric information are similar but modestly weaker, when using any of the three inputs to our composite INFO-Z variable, in place of our INFO-Z composite variable. Finally, Appendix A.5 shows that the cross tug-of-war results, as in main Table 4, are similar when using independent sorts on the two conditioning variables (rather than sequential).

Appendices A.7 to A.9 report additional results related to our 52-week price anchoring metric, PTA. First, Appendix A.7 reports on a momentum analysis comparable to the main Table 6, but with independent sorts on the two conditioning variables (rather than sequential). Second, Appendix A.8 shows that our main findings remain robust to an alternative sorting procedure based on a 3×10 double-sort portfolio design. Finally, Appendix A.9 shows that the cross tug-of-war results, as in main Table 9, are similar when using independent sorts on the two conditioning variables (rather than sequential).

A.1. Momentum INFO-Z Results: Alternative Independent Double-Sorting

This appendix supports Section 4.2.1 by providing additional details on the alternative independent double-sorting momentum evaluation with INFO-Z. Our intent is to see whether the same qualitative conclusions also emerge under an alternative independent sorting design for the two conditioning variables, INFO-Z and last month's monthly overnight returns or intraday returns.

We perform this independent-sorting analysis and find qualitatively consistent results. Appendix Table A1 reports the tabular details on this robustness exercise. In Panel A (B), we form 25 value-weighted portfolios from independent 5×5 sorts on a stock's lagged INFO-Z and lagged overnight

(intraday) returns. At the end of month $t-1$, stocks are sorted into quintiles based on INFO-Z measured at the close of month $t-2$ using NYSE breakpoints. Independently, they are also sorted into quintiles based on their past one-month overnight or intraday returns in month $t-1$, again using NYSE breakpoints. The reported portfolios are formed from the intersections of these two independent sorts.

Panel A of Table A1 shows that the positive relation between INFO-Z and overnight momentum remains clearly visible under independent sorting. When INFO-Z is in the highest quintile, the stocks in the highest past-overnight-return quintile have an average monthly overnight excess return of 3.14%, compared with 0.42% for the corresponding lowest-past-overnight-return quintile. The resulting high-minus-low spread is therefore 2.72% per month. At the low end of the INFO-Z distribution, the same spread is 1.18%. Thus, the high-minus-low overnight spread increases from 1.18% in the lowest INFO-Z quintile to 2.72% in the highest INFO-Z quintile, a difference of 1.54% ($t = 4.28$) with alphas for this difference of 1.15% ($t = 3.86$) and 1.19% ($t = 4.06$).

Next, Panel B of Table A1 reports on the comparable momentum analysis of the monthly intraday returns with the independent sorting procedure. Consistent with our main Table 1, we again find that there is no reliable difference in the month-to-month intraday-return momentum with INFO-Z.

Finally, we note that the average overnight returns are increasing with INFO-Z (Panel A), with statistically significant differences for all five past-overnight-return quintiles. And, the average intraday returns are decreasing in INFO-Z (Panel B), with statistically significant differences for four of the five past-intraday-return quintiles.

We conclude that our main asymmetric-information results are robust with the alternative independent double-sorting procedure.

A.2. Momentum INFO-Z Results: Alternative 3-by-10 Double-Sorted Portfolio

This appendix supports Section 4.2.2, by providing details on the INFO-Z results with the alternative 3-by-10 double sorting procedure. Appendix Table A2 reports on this robustness check using a 3×10 double-sorting design. Specifically, at the end of month $t-1$, we first sort stocks into terciles based on their INFO-Z ranking measured at the close of month $t-2$ using NYSE breakpoints. Within each tercile, we then sort stocks into deciles based on their past one-month overnight or intraday returns in month $t-1$. This procedure generates 30 value-weighted portfolios and allows us to examine whether our main findings remain visible under a finer partition of the return dimension.

Panel A of Table A2 shows that the positive relation between INFO-Z and overnight momentum remains strong under this alternative sorting scheme. The high-minus-low overnight-return spread is

1.36% for low-INFO-Z stocks, 1.98% for medium-INFO-Z stocks, and 3.43% for high-INFO-Z stocks. Thus, the strength of the overnight momentum rises monotonically with information asymmetry. The same pattern is visible in the factor-adjusted alphas.

Panel B of Table A2 similarly evaluates the monthly intraday momentum with the 3-by-10 double sort. Consistent with our earlier results with the double-quintile sorts, we find that there is not a reliable difference in the intraday momentum with variation in INFO-Z.

Finally, we note that the average overnight returns are increasing with INFO-Z (Panel A), with statistically significant differences for four of the five past-overnight-return quintiles. And, the average intraday returns are decreasing in INFO-Z (Panel B), with statistically significant differences for all five past-intraday-return quintiles.

We conclude that our main asymmetric-information results are robust with this alternative double-sort granularity.

A.3. Momentum INFO-Z Results: A Simpler INFO-Z Variable

This appendix reports details for results only summarized in Section 4.2.3. Our featured INFO-Z variable controls for two common illiquidity variables; a stock's Amihud's (2002) PIM and turnover. Here, we evaluate a simpler INFO-Z that is defined simply as the average Z-score of the three asymmetric information variables (PIN, spread, and idiosyncratic volatility) without the illiquidity controls.

Appendix Table A3 repeats the empirical exercise of Table 1, but with this alternative simpler INFO-Z construct. Using our double-quintile sorting analysis, we find similar qualitative results with this simpler INFO-Z variable. Panel A shows that the monthly overnight momentum is increasing with the simpler INFO-Z (difference in momentum of 1.93% ($t = 4.77$) between the high and low INFO-Z quintiles), with sizable alphas for this difference too. Conversely, Panel B shows relatively little variation in the intraday momentum with INFO-Z.

Finally, we note that the average overnight returns are increasing with INFO-Z (Panel A), with statistically significant differences for all five past-overnight-return quintiles. And, the average intraday returns are decreasing in INFO-Z (Panel B), with statistically significant differences for all five past-intraday-return quintiles.

We conclude that our main asymmetric-information results are robust with the simpler INFO-Z.

A.4. Momentum INFO-Z Results: PIN, Spread, and Idiosyncratic Volatility

This appendix reports details for results only summarized in Section 4.2.4. Here, rather than our composite 3-factor INFO-Z variable, we similarly evaluate single-factor INFO-Z variables for each of these three inputs separately. We re-estimate different versions of Table 1 for each of the single-factor INFO-Z variables; a stock's PIN, spread, and idiosyncratic volatility.

Appendix Tables A4, A5, and A6 report the results, for the single-factor PIN, spread, and idiosyncratic volatility versions of the INFO-Z variable. For each of the three single-factor INFO-Z variables, we find results quite similar to our composite 3-factor INFO-Z. Unsurprisingly, the variation in return patterns with the single-factor INFO-Z variables are generally somewhat more modest than that for our 3-factor composite INFO-Z variable.

For each single-factor INFO-Z, we find that: (1) the monthly overnight momentum is statistically-reliably increasing with the single INFO-Z factor, and (2) for each past-return quintile, the average excess overnight (intraday) returns is increasing (decreasing) with the single INFO-Z factor, generally with statistically-significant differences between the high- and low-INFO-Z-factor. These patterns are broadly similar for each single-factor INFO-Z variable.

The idiosyncratic-volatility INFO-Z metric has the largest INFO-Z difference in overnight momentum at 1.54% ($t = 4.71$) versus 1.47% ($t = 4.62$) for the PIN version and 1.36% ($t = 5.54$) for the spread version. Alpha differences are also sizable and highly statistically reliable.

Finally, in terms of the higher average returns for each past-return quintile for higher INFO-Z stocks relative to lower INFO-Z stocks, the PIN has the largest average difference with high statistical significance for this PIN-based difference in average excess returns for each past-return quintile.

A.5. Cross-Reversal INFO-Z Results: Alternative Independent Sorts

This appendix reports details on results only summarized near the end of Section 4.4. Appendix Table A7 reports on a similar cross-reversal INFO-Z analysis as in Table 4, but with independent double-quintile sorts on both the conditioning variables.

We highlight the following two patterns that are quite similar for both our earlier sequential-sorting method and the independent sorting method here. First, in Panel A, we note the cross-reversal behavior between this month's intraday returns and last month's overnight returns is increasing (become more negative) as INFO-Z increases. For the high INFO-Z stocks, the difference in average intraday returns between past overnight-return winners and overnight-return losers is -2.43% (t=-7.39). Conversely, for the low INFO-Z stocks, the difference in average intraday returns between past

overnight-return winners and overnight-return losers is -1.11% (t=-4.26). The difference is -1.32% (t=-5.34). Thus, the monthly intraday returns exhibit an appreciable cross-reversal, relative to the prior month's overnight returns, with this cross-reversal behavior increasing with INFO-Z.

Next, we note that the average overnight returns are increasing with INFO-Z (Panel B), with statistically significant differences between the high and low INFO-Z groupings for all five past-intraday-return quintiles. And, the average intraday returns are decreasing in INFO-Z (Panel A), with the differences between the high and low INFO-Z groupings being statistically significant with 1% p-value for three of the five past-intraday-return quintiles.

A.6. Cross-Reversal INFO-Z Results: One-Half Subperiod Results

This appendix reports details on results only summarized near the end of Section 4.4. Appendix Table A8 reports on a similar cross-reversal INFO-Z analysis as in Table 4, but with our double-quintile-sorting analyses conducted over each one-half subperiod.

Both subperiods exhibit the same featured INFO-Z result. The cross-reversal between this month's intraday return and last month's overnight return becomes stronger as INFO-Z increases. We find that: (a) for the first one-half subperiod, the difference in average monthly intraday returns between past-overnight-winner stocks and past-overnight-loser stocks is -1.67% for the low INFO-Z stocks and -4.26% for the high INFO-Z stocks; and (b) for the second one-half subperiod, this same difference in average monthly intraday returns is -0.45% for the low INFO-Z stocks and -1.32% for the high INFO-Z stocks. While the results are stronger in the first-half subperiod, these differences in cross-reversals and their related alphas have p-values better than 0.1% for both one-half subperiods.

A.7. PTA Results: With Alternative Independent Double-Sorting

This appendix reports details for results only summarized in Section 5.2.1. We repeat the analysis as in Table 6, but with independent sorting on the two conditioning variables (PTA and the past-return quintile).

Appendix Table A9 reports the results. First, we find that the negative relation between PTA and the intraday momentum remains strongly evident with the independent sorting. With the independent sorting, the difference in intraday momentum between the high PTA stocks and low PTA stocks is -1.15% (t=-3.10). The alphas for this PTA difference are quite similar. Interestingly, for the independent sorting, the overnight momentum is increasing with PTA, to a larger and more statistically-reliable degree than seen in Table 6 with the sequential sorts.

Next, across the past-return quintiles, we again find that the average overnight returns are

increasing with PTA, while the average intraday returns tend to decline with PTA. For the overnight returns across the five past-return quintiles, the average returns for the high-PTA stocks is always statistically-reliably greater than the average returns for the low-PTA stocks with an average difference of 0.89%. Conversely, for the intraday returns across the five past-return quintiles, the average returns for the high-PTA stocks is less than the average returns for the low-PTA stocks for 4 of the 5 cases with an average difference of -0.67%, with this difference being statistically significant for 3 of the 5 cases.

In sum, we conclude that our main PTA results are robust with the alternative independent double-sorting procedure.

A.8. PTA Results: With Alternative 3 by 10 Double-Portfolio Sorts

This appendix reports details for results only summarized in Section 5.2.2. Our main results are based on double-quintile sorting on a stock's lagged PTA and the prior month's overnight or intraday return. To evaluate robustness further, here we perform a similar empirical exercise, but with a tercile sort on the past PTA and a decile sort on the past return (or 3-by-10).

Appendix Table A10 reports the results. We find results that are qualitatively comparable to our earlier PTA results. First, the intraday momentum is declining with a stock's PTA. Second, however, the overnight momentum is increasing with a stock's PTA, but to a lesser extent than the momentum decline with PTA. Third, the average of the monthly overnight returns are increasing with a stock's PTA. Finally, the average monthly intraday returns tend to be lower for higher-PTA stocks.

We conclude that our primary PTA results are robust to this different double-sort granularity.

A.9. PTA Cross Tug-of-War Results: With Independent Sorts

This appendix reports details on results only summarized near the end of Section 5.4. Appendix Table A11 reports on the same cross-reversal PTA analysis, as in Table 9, but with independent sorts on both the conditioning variables. We again find that the cross-reversals between the monthly overnight return and last month's intraday return is decreasing (becoming less negative) with a stock's PTA.

Specifically, for the PTA variation in Panel B of Table A11 using our double-quintile sort analysis, the cross-reversal behavior of the monthly overnight returns, relative to the past month's intraday returns, is declining with a stock's PTA. There is a nearly monotonic decline in the cross-reversal behavior from the low-PTA stocks (at -2.50%, t=-4.22) to the high-PTA stocks (at -1.30%, t = -3.79).

The PTA difference in the cross-reversal behavior is highly reliable at 1.20% (t=3.05) with similar alphas.

A.10. Cross-Reversal PTA Results: One-Half Subperiod Results

This appendix reports details on results only summarized in Section 5.4. Appendix Table A12 reports on a similar cross-reversal PTA analysis as in Table 9, but with analyses conducted over each one-half subperiod.

For the first-half subperiod over 1997:01 to 2009:06, the cross-reversal between last month's intraday return and this month's overnight return becomes appreciably weaker (less negative) as PTA increases; which aligns with our full-sample results in Table 9. Using our double-quintile sort analysis, the difference in average monthly overnight returns between past-intraday-winner stocks and past-intraday-loser stocks for this month's intraday returns is -4.91% for the low PTA stocks and -2.38% for the high PTA stocks for the first-half subperiod. This difference in cross-reversal, and the related alphas, have p-values better than 1%.

For the second-half subperiod over 2009:07 to 2021:12, the cross-reversal between last month's intraday returns and this month's overnight returns is also smaller for the high-PTA stocks, relative to the low-PTA stocks. However, the difference is marginal and lacks statistical significance. Consistent with our full-sample results and first-half subperiod results, we do note that the overnight returns for the past-intraday-winner stocks with a high PTA do not exhibit a cross-reversal, but rather exhibit a sizable cross-continuation with an average excess overnight return of 1.82% for that corner portfolio. However, for the second-half subperiod, the average overnight returns for the past-intraday-loser stocks with a high PTA are especially high at 2.18% (relative to the other past-intraday-loser stocks over this subperiod), indicating a relatively strong cross-reversal. Thus, for this second-half subperiod, the strong cross-reversal for the past-intraday-loser, high-PTA corner portfolio of stocks acts to mitigate the usual decline in overall cross-reversal behavior as PTA increases.

A.11. Construction of the $ABNR_{i,t-1}$ Variable

This appendix supports Section 6.3.1 by providing details on the construction of the $ABNR_{i,t-1}$ variable.

To construct the Akbas et al. proxy in a form that is compatible with our monthly Fama–MacBeth design, let

$$NR_{i,d} = \mathbf{1}\Big(r^n_{i,d} > 0 \text{ and } r^d_{i,d} < 0\Big),$$

where $r^n_{i,d}$ is stock i's overnight return on day d and $r^d_{i,d}$ is its intraday return on day d. We then compute the monthly frequency of such negative daytime reversals in month $t-1$ as

$$NR_FREQ_{i,t-1} = \frac{1}{N_{i,t-1}} \sum_{d \in t-1} NR_{i,d},$$

where $N_{i,t-1}$ denotes the number of trading days for stock i in month $t-1$. Following Akbas et al., we define the abnormal reversal-frequency measure as

$$ABNR_{i,t-1} = \frac{NR_FREQ_{i,t-1}}{\frac{1}{12}\sum_{s=1}^{12} NR_FREQ_{i,t-1-s}}.$$

Thus, $ABNR_{i,t-1}$ is high when stock i experiences an unusually large number of days in month $t-1$ with positive overnight returns followed by negative daytime reversals, relative to its own recent history.

A.12. Fama-MacBeth Model with Other Interactive-Return Control Variables

This appendix supports Section 6.4 by providing details on additional Fama-MacBeth analysis that evaluates extra interactive-return control terms. Here, we evaluate three additional interactive-return control terms; interacting with a stock's log of market capitalization (size), book-to-market equity ratio (BM), and market beta (β). In addition to their role in the well-known Fama-French 3-factor model, these terms are suggested by the cross-sectional correlations reported at the end of Section 6.1. Recall that we found that a stock's INFO-Z metric is substantially negatively correlated with its size and positively correlated with its market beta, implying that smaller-cap and higher-beta stocks tend to reflect greater asymmetric information, relative to larger-cap and lower-beta stocks.[16]

Specifically, we estimate a specification that is structured as in Table 12 but where the LMQ, JPQ, and ABNR interactive-return terms are replaced by size, BM, and beta interactive-return terms. For ease of coefficient interpretation in this setting, we use the Z-standardized transformation of the size, BM, and β variables, what have a mean of zero and standard deviation of one. These three control variables were included previously in the stand-alone level control terms only. For this investigation, we report results that exclude stocks with a negative book-to-market equity ratio, but our empirical results are similar without this exclusion.

[16]This evaluation of additional interactive-return control terms is representative, rather than comprehensive. From the list of extra control variables at the end of Section 3, other candidate control variables for the additional interactive-return evaluation are either controls on longer-horizon past returns, longer-horizon turnover, or lower-frequency accounting-related or firm-event related variables. Our view is that these other terms would likely be less useful in our conditional dynamic framework.

We begin by examining the variation in overnight-return momentum with INFO-Z by estimating the following specification:

$$\begin{aligned} r^n_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(\textit{INFO-Z}_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\, \big(Z^{Size}_{i,t-2} \times r^n_{i,t-1}\big) + \beta_4\, \big(Z^{BM}_{i,t-2} \times r^n_{i,t-1}\big) \\ \beta_5\, \big(Z^{\beta}_{i,t-2} \times r^n_{i,t-1}\big) + \beta_6\, r^d_{i,t-1} + \beta_7\big(\textit{INFO-Z}_{i,t-2} \times r^d_{i,t-1}\big) + \beta_8\, \textit{INFO-Z}_{i,t-2} \\ + \beta_9\, Z^{Size}_{i,t-2} + \beta_{10}\, Z^{BM}_{i,t-2} + \beta_{11}\, Z^{\beta}_{i,t-2} + \gamma' \textit{CONTROLS}_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{A.1}$$

where the three new interactive-return control terms (the β_3 to β_5 terms) are defined above and the other terms are defined with equation (1).

Appendix Table A13, columns 1 to 3, reports the results. Our primary interest is whether the estimated β_2 on the interactive INFO-Z term remains reliably positive with the additional control interactive-return terms. In combination with a positive β_1 coefficient, this would indicate a positive partial relation between the successive monthly overnight returns that is increasing with asymmetric information. Secondarily, we are interested in whether any of the new interactive-return terms are statistically significant.

Results in Column 2 (without the full set of individual-stock control variables) and Column 3 (with the full set of individual-stock control variables) of Table A13 show that the partial relation between the successive monthly overnight returns is still reliably increasing with a stock's INFO-Z. Both of the estimated β_2 coefficients are positive and statistically significant, with their values being marginally greater than the comparable coefficient values in the simpler Table 5 specification. Further, the new additional interactive-return terms add little (the β_3, β_4, and β_5 terms), with statistically-insignificant estimated coefficients for all the cases except for the size interactive-return term in column 2. We conclude that our principal INFO-Z conditional patterns remain with the inclusion of these additional interactive-return terms.

We next re-examine the variation in momentum for monthly intraday returns with PTA by estimating the following specification with the three new interactive-return terms:

$$\begin{aligned} r^d_{i,t} = \beta_0 + \beta_1\, r^n_{i,t-1} + \beta_2\big(PTA_{i,t-2} \times r^n_{i,t-1}\big) + \beta_3\, r^d_{i,t-1} + \beta_4\big(PTA_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_5\, \big(Z^{Size}_{i,t-2} \times r^d_{i,t-1}\big) + \beta_6\, \big(Z^{BM}_{i,t-2} \times r^d_{i,t-1}\big) + \beta_7\, \big(Z^{\beta}_{i,t-2} \times r^d_{i,t-1}\big) \\ + \beta_8\, PTA_{i,t-2} + \beta_9\, Z^{Size}_{i,t-2} + \beta_{10}\, Z^{BM}_{i,t-2} + \beta_{11}\, Z^{\beta}_{i,t-2} \\ + \gamma' \textit{CONTROLS}_{i,t-2} + FF48 + \varepsilon_{i,t}, \end{aligned} \tag{A.2}$$

where the three new interactive-return control terms (the β_5 to β_7 terms) are defined above and the other terms are defined with equation (3).

Table A13, columns 4 to 6, reports the results. Our primary interest is whether the estimated β_4 coefficient on the interactive PTA term remains negative with the additional control interactive-return terms. In combination with a positive β_3 coefficient, this would indicate momentum between the successive intraday returns that is declining when stocks have current prices that are high relative to their 52-week price history (a high PTA). Secondarily, we are interested in whether any of the new interactive-return terms are statistically significant.

Results in Column 5 (without the full set of individual-stock control variables) and Column 6 (with the full set of individual-stock control variables) of Appendix Table A13 show that the partial relation between the successive monthly intraday returns is decreasing with a stock's PTA. The estimated β_4 coefficients are negative and statistically significant, with their values being appreciably greater (more negative) than the comparable coefficient values in the simpler Table 10 specification.

For the PTA case, the new additional interactive-return terms with size and BM add little (the β_5 and β_6 terms), with estimated coefficients that lack statistical significance in columns 5 and 6 at better than a 5% p-value. However, interestingly, the estimated coefficient on the beta interactive-return term (the β_7 term) is sizably positive and statistically significant.

Overall, we conclude that our principal PTA conditional patterns remain with the inclusions of these three additional interactive-return control terms.

Table A1: INFO-Z Momentum Results with Independently Double-sorted Portfolios

This appendix table analyzes momentum in both monthly overnight stock returns and monthly intraday stock returns conditional on a stock's INFO-Z, similar to Table 1 but with independent sorts on INFO-Z and the lagged return. Panel A (B) of this appendix table reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios independently sorted on the lagged INFO-Z and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are sorted into quintiles based on their INFO-Z ranking as of the close of month $t-2$ using NYSE breakpoints. In addition, stocks are independently sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. Portfolios are formed using the intersection of the two independent sorts. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.10	-0.09	0.36	0.55	1.08	1.18	1.25	1.18
	(-0.46)	(-0.37)	(2.19)	(2.96)	(3.53)	(3.96)	(4.37)	(4.07)
INFO-Z 2	-0.03	0.07	0.30	0.71	1.37	1.39	1.43	1.37
	(-0.09)	(0.30)	(1.16)	(2.85)	(4.37)	(4.20)	(4.31)	(4.27)
INFO-Z 3	0.04	0.52	0.82	1.14	1.94	1.90	1.66	1.59
	(0.12)	(2.28)	(4.07)	(4.50)	(5.19)	(4.16)	(4.39)	(4.51)
INFO-Z 4	0.26	0.69	0.86	1.27	2.17	1.91	1.80	1.73
	(0.91)	(2.83)	(3.31)	(4.28)	(5.26)	(5.21)	(5.06)	(5.17)
INFO-Z High	0.42	0.83	1.20	1.84	3.14	2.72	2.40	2.38
	(1.36)	(2.57)	(4.25)	(6.33)	(5.64)	(4.93)	(4.90)	(4.94)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	0.52	0.92	0.84	1.29	2.06	1.54	1.15	1.19
	(2.83)	(5.22)	(4.05)	(5.20)	(5.96)	(4.28)	(3.86)	(4.06)
α_{FF6+2}	0.60	0.85	0.67	1.18	1.76			
	(3.44)	(4.48)	(3.27)	(4.69)	(6.18)			
α_{Q+2}	0.59	0.85	0.70	1.15	1.78			
	(3.37)	(4.21)	(3.51)	(4.81)	(6.44)			

Table A1: (continued)

Panel B: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.69	0.08	0.22	0.44	1.03	1.72	1.50	1.73
	(-2.17)	(0.41)	(1.21)	(2.09)	(4.82)	(4.87)	(5.26)	(5.26)
INFO-Z 2	-0.79	0.04	0.18	0.55	0.84	1.62	1.36	1.45
	(-2.83)	(0.17)	(0.92)	(2.62)	(4.18)	(5.27)	(4.88)	(5.24)
INFO-Z 3	-1.01	-0.19	0.12	0.19	0.59	1.60	1.19	1.19
	(-2.26)	(-0.66)	(0.52)	(0.91)	(1.93)	(3.49)	(2.94)	(3.10)
INFO-Z 4	-1.52	-0.33	0.04	-0.38	0.05	1.57	1.45	1.61
	(-3.38)	(-1.12)	(0.14)	(-1.28)	(0.21)	(4.01)	(4.07)	(4.18)
INFO-Z High	-1.83	-0.32	-0.60	-0.46	-0.32	1.52	1.18	1.28
	(-3.39)	(-0.98)	(-1.82)	(-1.36)	(-1.19)	(3.42)	(3.15)	(3.40)
High-Low PTH Strategies	Low	2	3	4	High			
Excess	-1.14	-0.40	-0.83	-0.90	-1.34	-0.20	-0.32	-0.45
	(-2.73)	(-1.27)	(-3.07)	(-2.83)	(-4.27)	(-0.60)	(-0.86)	(-1.21)
α_{FF6+2}	-0.90	-0.37	-0.70	-0.69	-1.22			
	(-2.66)	(-1.56)	(-2.60)	(-3.14)	(-5.00)			
α_{Q+2}	-0.78	-0.33	-0.76	-0.60	-1.22			
	(-2.32)	(-1.42)	(-3.00)	(-2.60)	(-4.81)			

Table A2: INFO-Z Momentum Results using 3-by-10 Double-Sorted Portfolios

This appendix table reports on robustness for the INFO-Z momentum results by evaluating different granularity for the double-sorted portfolios. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 30 (3 by 10) portfolios sequentially sorted on a stock's lagged INFO-Z and lagged overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into terciles based on their INFO-Z ranking as of the close of month $t-2$ using NYSE breakpoints. Within each tercile, stocks are further sorted into deciles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Decile					High - Low		
	Low	2	5	9	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.13	-0.09	0.10	0.66	1.23	1.36	1.38	1.33
	(-0.55)	(-0.38)	(0.53)	(3.21)	(3.86)	(4.09)	(4.33)	(4.05)
INFO-Z Medium	-0.02	0.19	0.56	1.44	1.97	1.98	1.95	1.86
	(-0.06)	(0.69)	(2.28)	(4.65)	(4.76)	(4.16)	(4.52)	(4.63)
INFO-Z High	0.20	0.43	1.07	2.30	3.62	3.43	3.06	3.07
	(0.59)	(1.50)	(3.59)	(6.49)	(5.81)	(5.05)	(4.90)	(4.82)
High-Low INFO-Z Strategies	Low	2	5	9	High			
Excess	0.33	0.52	0.98	1.64	2.40	2.07	1.68	1.74
	(1.54)	(3.11)	(4.43)	(7.01)	(5.90)	(4.79)	(4.32)	(4.33)
α_{FF6+2}	0.42	0.50	0.84	1.51	2.11			
	(1.96)	(2.78)	(3.75)	(6.97)	(5.37)			
α_{Q+2}	0.42	0.50	0.90	1.47	2.16			
	(2.08)	(2.81)	(4.01)	(7.17)	(5.39)			

Table A2: (continued)

Panel B: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Decile					High - Low		
	Low	2	5	9	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.93	-0.11	0.20	0.55	1.17	2.10	1.83	2.00
	(-2.68)	(-0.49)	(1.10)	(3.07)	(5.71)	(5.15)	(5.90)	(5.77)
INFO-Z Medium	-1.34	-0.47	0.11	0.61	0.50	1.84	1.55	1.63
	(-3.07)	(-1.52)	(0.52)	(2.32)	(1.73)	(4.54)	(4.58)	(4.50)
INFO-Z High	-2.70	-1.29	-0.34	-0.46	-0.07	2.63	2.30	2.48
	(-4.21)	(-2.82)	(-1.23)	(-1.71)	(-0.26)	(4.06)	(4.23)	(4.40)
High-Low INFO-Z Strategies	Low	2	5	9	High			
Excess	-1.77	-1.18	-0.54	-1.02	-1.24	0.53	0.48	0.48
	(-3.47)	(-3.47)	(-1.87)	(-3.59)	(-4.61)	(1.04)	(0.93)	(0.93)
α_{FF6+2}	-1.60	-0.84	-0.37	-0.93	-1.12			
	(-3.63)	(-2.80)	(-1.31)	(-3.28)	(-4.34)			
α_{Q+2}	-1.58	-0.80	-0.38	-0.93	-1.10			
	(-3.48)	(-2.80)	(-1.33)	(-3.31)	(-4.25)			

Table A3: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on a simpler INFO-Z without Illiquidity Controls

As a robustness check on the INFO-Z momentum analysis, this appendix table repeats the analysis in our main Table 1 but with a simple (or raw) INFO-Z variable, defined simply as the average of the Z-scores of the three inputs: a stocks' PIN, spread, and idiosyncratic volatility. So, here, we do not regress out the INFO-Z's relation to the two stock liquidity variables (referring to Amihud's (2002) Price-Impact-Measure (PIM) and turnover), as we did for our main results. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on the lagged simple INFO-Z and lagged monthly overnight (intraday) returns. All other estimation and tabular details are as for Table 1.

Panel A: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.21	-0.09	0.19	0.48	0.91	1.12	1.14	1.13
	(-0.88)	(-0.44)	(1.16)	(2.81)	(3.97)	(4.84)	(4.93)	(4.81)
INFO-Z 2	0.03	0.24	0.25	0.76	1.12	1.08	1.07	0.99
	(0.13)	(1.11)	(0.98)	(2.89)	(3.48)	(3.27)	(3.41)	(3.17)
INFO-Z 3	0.09	0.29	0.74	1.02	1.77	1.67	1.55	1.57
	(0.34)	(1.13)	(3.43)	(4.93)	(5.57)	(4.64)	(4.50)	(4.48)
INFO-Z 4	0.27	0.59	0.99	1.32	2.32	2.05	1.94	1.81
	(0.90)	(2.38)	(3.82)	(4.33)	(6.29)	(5.83)	(5.59)	(5.64)
INFO-Z High	0.49	0.88	1.56	2.07	3.54	3.05	2.76	2.78
	(1.45)	(2.72)	(5.50)	(5.77)	(6.03)	(5.21)	(5.05)	(5.07)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	0.70	0.97	1.37	1.59	2.63	1.93	1.61	1.65
	(3.19)	(5.16)	(5.96)	(5.20)	(5.89)	(4.77)	(4.21)	(4.22)
α_{FF6+2}	0.76	0.88	1.31	1.49	2.38			
	(3.57)	(5.04)	(5.70)	(5.03)	(5.65)			
α_{Q+2}	0.76	0.87	1.32	1.44	2.41			
	(3.61)	(4.46)	(5.92)	(4.90)	(5.67)			

Table A3: (continued)

Panel B: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	-0.28	0.42	0.27	0.42	0.84	1.12	1.06	1.19
	(-1.27)	(2.96)	(1.88)	(2.05)	(4.35)	(5.59)	(4.59)	(4.94)
INFO-Z 2	-0.77	0.08	0.16	0.59	0.71	1.48	1.30	1.43
	(-2.76)	(0.43)	(0.66)	(3.07)	(3.30)	(4.50)	(4.80)	(4.88)
INFO-Z 3	-1.01	-0.00	0.11	0.44	0.79	1.80	1.24	1.30
	(-2.37)	(-0.01)	(0.44)	(1.72)	(2.49)	(3.55)	(3.47)	(3.53)
INFO-Z 4	-1.47	-0.65	0.08	-0.12	0.13	1.59	1.39	1.52
	(-3.47)	(-1.73)	(0.32)	(-0.49)	(0.53)	(4.16)	(4.32)	(4.51)
INFO-Z High	-2.32	-0.83	-0.52	-0.51	-0.50	1.82	1.58	1.70
	(-3.82)	(-2.16)	(-1.60)	(-1.62)	(-1.73)	(3.66)	(3.56)	(3.72)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-2.03	-1.25	-0.79	-0.92	-1.34	0.70	0.52	0.51
	(-3.90)	(-3.70)	(-2.67)	(-3.43)	(-4.24)	(1.68)	(1.25)	(1.27)
α_{FF6+2}	-1.70	-1.12	-0.58	-0.78	-1.19			
	(-4.19)	(-5.04)	(-2.82)	(-3.43)	(-4.74)			
α_{Q+2}	-1.69	-1.13	-0.56	-0.68	-1.19			
	(-4.07)	(-4.67)	(-2.79)	(-2.78)	(-4.60)			

Table A4: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on PIN

In this appendix table, we repeat a similar analysis as in Table 1, but conditional on a stock's standardized PIN only (rather than the composite INFO-Z). Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on lagged PIN and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their PIN ranking as of the close of month $t-2$ using NYSE breakpoints. As we did for the composite INFO-Z, we use the residual PIN that controls for liquidity effects. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on PIN & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PIN Low	-0.11	-0.08	0.31	0.50	1.06	1.17	1.24	1.20
	(-0.49)	(-0.36)	(1.66)	(2.41)	(3.59)	(4.29)	(4.63)	(4.65)
PIN 2	0.04	0.26	0.46	0.91	1.53	1.48	1.47	1.39
	(0.17)	(0.96)	(2.37)	(4.31)	(4.17)	(3.75)	(3.59)	(3.52)
PIN 3	0.01	0.29	0.66	1.01	1.78	1.77	1.59	1.56
	(0.02)	(1.25)	(3.24)	(4.05)	(4.12)	(3.73)	(3.91)	(3.91)
PIN 4	0.02	0.47	0.76	1.08	2.26	2.24	2.19	2.16
	(0.08)	(2.21)	(3.11)	(4.19)	(5.89)	(5.57)	(5.91)	(5.81)
PIN High	0.45	0.64	1.00	1.84	3.10	2.64	2.33	2.30
	(1.91)	(2.62)	(4.96)	(7.22)	(5.98)	(5.01)	(4.81)	(5.03)
High-Low PIN Strategies	Low	2	3	4	High			
Excess	0.56	0.72	0.69	1.33	2.03	1.47	1.10	1.10
	(4.18)	(5.38)	(3.65)	(6.00)	(6.04)	(4.62)	(3.43)	(3.81)
α_{FF6+2}	0.66	0.73	0.65	1.44	1.76			
	(3.70)	(4.58)	(2.86)	(6.19)	(4.93)			
α_{Q+2}	0.64	0.73	0.66	1.34	1.75			
	(3.61)	(4.66)	(2.90)	(5.71)	(5.25)			

Table A4: (continued)

Panel B: Monthly Intdy Returns: Conditional on PIN & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PIN low	-0.44	0.39	0.15	0.37	0.89	1.34	1.22	1.30
	(-1.55)	(2.35)	(0.82)	(1.86)	(4.94)	(4.21)	(4.23)	(4.45)
PIN 2	-0.86	-0.25	0.12	0.43	0.64	1.50	1.10	1.28
	(-2.48)	(-0.86)	(0.54)	(1.99)	(2.76)	(3.76)	(3.45)	(3.60)
PIN 3	-1.19	0.01	-0.09	0.30	0.49	1.69	1.35	1.46
	(-2.93)	(0.03)	(-0.41)	(1.38)	(1.96)	(4.12)	(4.47)	(5.07)
PIN 4	-1.38	-0.17	0.10	0.14	0.22	1.61	1.29	1.48
	(-2.85)	(-0.61)	(0.37)	(0.64)	(0.89)	(3.55)	(3.36)	(3.46)
PIN 5	-2.22	-0.45	-0.19	-0.56	-0.52	1.69	1.49	1.54
	(-4.07)	(-1.44)	(-0.63)	(-1.68)	(-1.52)	(4.05)	(3.56)	(3.76)
High-Low PIN Strategies	Low	2	3	4	High			
Excess	-1.77	-0.84	-0.34	-0.93	-1.42	0.36	0.27	0.25
	(-4.15)	(-2.91)	(-1.27)	(-3.19)	(-3.88)	(0.85)	(0.56)	(0.52)
α_{FF6+2}	-1.65	-0.83	-0.51	-0.83	-1.39			
	(-4.41)	(-3.07)	(-2.21)	(-4.03)	(-3.97)			
α_{Q+2}	-1.67	-0.89	-0.47	-0.75	-1.42			
	(-4.45)	(-3.11)	(-1.72)	(-3.27)	(-4.03)			

Table A5: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on Spread

In this appendix table, we repeat a similar analysis as in Table 1, but conditional on a stock's standardized spread only (rather than the composite INFO-Z). Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on lagged Spread and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their Spread ranking as of the close of month $t-2$ using NYSE breakpoints. As we did for the composite INFO-Z, we use the residual Spread that controls for liquidity effects. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on Spread & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
Sprd Low	0.03	0.02	0.27	0.52	1.31	1.29	1.24	1.22
	(0.13)	(0.13)	(1.91)	(3.79)	(3.93)	(3.21)	(3.22)	(3.29)
Sprd 2	-0.14	0.15	0.38	0.69	1.25	1.39	1.31	1.24
	(-0.53)	(0.72)	(1.78)	(3.54)	(4.88)	(4.83)	(5.02)	(5.24)
Sprd 3	0.13	0.26	0.47	0.89	1.30	1.17	1.22	1.17
	(0.41)	(0.88)	(1.85)	(3.80)	(4.00)	(3.77)	(4.14)	(3.94)
Sprd 4	0.15	0.44	0.59	1.22	1.81	1.67	1.51	1.48
	(0.51)	(1.49)	(2.49)	(4.64)	(4.91)	(4.22)	(4.15)	(4.15)
Sprd High	0.41	0.74	1.06	1.69	3.06	2.65	2.51	2.45
	(1.24)	(2.36)	(3.89)	(4.89)	(6.24)	(5.60)	(5.86)	(5.68)
High-Low Spread Strategies	Low	2	3	4	High			
Excess	0.38	0.72	0.79	1.17	1.75	1.36	1.27	1.23
	(1.42)	(3.08)	(3.96)	(3.75)	(6.61)	(5.54)	(5.22)	(4.76)
α_{FF6+2}	0.35	0.63	0.68	1.06	1.61			
	(1.44)	(3.08)	(3.50)	(3.72)	(5.87)			
α_{Q+2}	0.35	0.58	0.65	0.96	1.59			
	(1.43)	(2.67)	(3.36)	(3.35)	(5.75)			

Table A5: (continued)

Panel B: Monthly Intdy Returns: Conditional on Spread & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
Sprd Low	-0.29	0.14	0.26	0.27	0.42	0.71	0.62	0.69
	(-0.91)	(0.60)	(1.38)	(1.33)	(1.63)	(3.36)	(2.83)	(2.57)
Sprd 2	-0.66	0.43	0.16	0.43	0.67	1.33	1.17	1.24
	(-2.36)	(2.49)	(0.78)	(2.15)	(3.29)	(4.73)	(4.35)	(4.51)
Sprd 3	-0.79	0.11	0.19	0.27	0.37	1.16	1.01	1.13
	(-2.29)	(0.51)	(0.99)	(1.19)	(1.32)	(3.39)	(3.37)	(3.45)
Sprd 4	-1.13	-0.31	-0.28	0.05	0.35	1.48	1.13	1.21
	(-2.88)	(-1.09)	(-1.13)	(0.22)	(1.46)	(3.60)	(3.33)	(3.62)
Sprd High	-2.10	-0.79	-0.62	-0.10	0.06	2.16	1.86	2.00
	(-3.69)	(-2.19)	(-2.02)	(-0.33)	(0.20)	(4.09)	(4.35)	(4.55)
High-Low Spread Strategies	Low	2	3	4	High			
Excess	-1.81	-0.93	-0.88	-0.37	-0.37	1.45	1.24	1.30
	(-4.05)	(-3.15)	(-2.91)	(-1.18)	(-1.03)	(2.78)	(2.81)	(2.92)
α_{FF6+2}	-1.48	-0.82	-0.65	-0.22	-0.23			
	(-3.53)	(-3.43)	(-2.08)	(-0.63)	(-0.69)			
α_{Q+2}	-1.47	-0.78	-0.62	-0.19	-0.17			
	(-3.82)	(-3.01)	(-2.06)	(-0.59)	(-0.50)			

Table A6: Momentum in Monthly Overnight & Intraday Stock Returns, Conditional on Idiosyncratic Volatility

In this appendix table, we repeat a similar analysis as in Table 1, but conditional on a stock's standardized idiosyncratic volatility only (rather than the composite INFO-Z). Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios sequentially sorted on the lagged Idiosyncratic Volatility (IVol) and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into quintiles based on their IVol ranking as of the close of month $t-2$ using NYSE breakpoints. As we did for the composite INFO-Z, we use the residual IVol that controls for liquidity effects. Within each quintile, stocks are further sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on IVol & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
IVol Low	-0.12	0.13	0.31	0.54	1.21	1.34	1.27	1.23
	(-0.51)	(0.71)	(1.64)	(3.49)	(4.09)	(4.48)	(4.60)	(4.53)
IVol 2	-0.10	-0.03	0.34	0.77	1.46	1.56	1.36	1.34
	(-0.37)	(-0.11)	(1.65)	(3.01)	(4.63)	(4.96)	(4.27)	(4.25)
IVol 3	0.00	0.25	0.55	1.01	1.70	1.70	1.58	1.55
	(0.02)	(0.96)	(2.43)	(4.71)	(5.20)	(4.65)	(4.60)	(4.95)
IVol 4	0.25	0.52	0.72	1.07	1.76	1.50	1.42	1.36
	(0.90)	(2.20)	(2.90)	(3.76)	(4.01)	(3.71)	(4.04)	(3.94)
IVol High	0.25	0.78	1.23	1.64	3.13	2.88	2.69	2.64
	(0.71)	(2.56)	(5.06)	(5.59)	(6.14)	(5.05)	(5.29)	(5.12)
High-Low IVol Strategies	Low	2	3	4	High			
Excess	0.37	0.65	0.91	1.10	1.92	1.54	1.42	1.40
	(1.57)	(3.12)	(4.77)	(4.93)	(6.15)	(4.71)	(4.60)	(4.43)
α_{FF6+2}	0.42	0.65	0.74	1.16	1.84			
	(1.96)	(3.48)	(3.63)	(5.52)	(6.09)			
α_{Q+2}	0.46	0.69	0.71	1.12	1.86			
	(2.16)	(3.64)	(3.64)	(5.48)	(6.09)			

Table A6: (continued)

Panel B: Monthly Intdy Returns: Conditional on IVol & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
IVol Low	-0.53	0.36	0.07	0.39	0.72	1.25	1.06	1.23
	(-1.75)	(2.26)	(0.34)	(1.77)	(3.66)	(3.80)	(3.42)	(3.64)
IVol 2	-0.81	0.25	0.36	0.37	0.88	1.68	1.52	1.56
	(-2.90)	(1.08)	(1.61)	(1.55)	(3.80)	(4.56)	(5.50)	(5.59)
IVol 3	-1.19	-0.01	-0.11	0.39	0.40	1.58	1.32	1.42
	(-3.11)	(-0.03)	(-0.41)	(1.69)	(1.52)	(4.41)	(4.24)	(4.46)
IVol 4	-1.14	-0.11	0.06	0.07	0.55	1.69	1.39	1.54
	(-2.64)	(-0.43)	(0.21)	(0.31)	(2.21)	(3.93)	(3.85)	(4.04)
IVol 5	-2.06	-0.84	-0.42	-0.30	-0.30	1.76	1.50	1.60
	(-3.60)	(-2.12)	(-1.43)	(-1.04)	(-1.03)	(3.68)	(3.79)	(4.12)
High-Low IVol Strategies	Low	2	3	4	High			
Excess	-1.53	-1.20	-0.49	-0.69	-1.02	0.52	0.43	0.37
	(-3.40)	(-3.51)	(-1.87)	(-2.94)	(-3.27)	(1.24)	(1.07)	(0.94)
α_{FF6+2}	-1.28	-1.04	-0.17	-0.51	-0.84			
	(-3.52)	(-4.07)	(-0.77)	(-2.30)	(-2.53)			
α_{Q+2}	-1.23	-1.02	-0.17	-0.48	-0.86			
	(-3.63)	(-3.73)	(-0.76)	(-2.23)	(-2.62)			

Table A7: Cross-reversals in Monthly Overnight & Overnight Stock Returns, Conditional on INFO-Z: Independent Sorts

This appendix table reports the cross-tug-of-war INFO-Z results, similar to Table 4 but with independent sorts on the two conditioning variables. Panel A (B) reports the average value-weighted excess intraday(overnight) returns in month t for 25 (5 by 5) portfolios independently sorted on a stock's lagged INFO-Z and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are sorted into quintiles based on their 3-factor INFO-Z ranking as of the close of month $t-2$ using NYSE breakpoints. In addition, stocks are independently sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. Portfolios are formed using the intersection of the two independent sorts. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021.

Panel A: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.83	0.45	0.43	0.01	-0.28	-1.11	-1.27	-1.23
	(4.00)	(2.54)	(2.76)	(0.06)	(-1.05)	(-4.26)	(-4.55)	(-4.09)
INFO-Z 2	0.95	0.50	0.29	-0.13	-0.69	-1.64	-1.79	-1.72
	(4.30)	(2.37)	(1.42)	(-0.66)	(-2.50)	(-5.00)	(-5.14)	(-5.23)
INFO-Z 3	1.05	0.51	-0.06	-0.16	-1.12	-2.17	-2.10	-2.04
	(4.34)	(1.95)	(-0.25)	(-0.58)	(-3.50)	(-7.14)	(-6.43)	(-7.28)
INFO-Z 4	0.52	0.13	-0.06	-0.71	-1.61	-2.14	-2.42	-2.45
	(1.88)	(0.42)	(-0.23)	(-2.26)	(-4.14)	(-6.00)	(-5.78)	(-5.68)
INFO-Z High	0.57	-0.06	-0.64	-1.12	-1.86	-2.43	-2.65	-2.59
	(1.60)	(-0.20)	(-1.72)	(-2.89)	(-4.38)	(-7.39)	(-8.34)	(-8.12)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-0.26	-0.51	-1.07	-1.13	-1.58	-1.32	-1.38	-1.35
	(-0.82)	(-1.65)	(-3.08)	(-2.86)	(-4.29)	(-5.34)	(-5.48)	(-5.21)
α_{FF6+2}	-0.13	-0.41	-0.78	-0.82	-1.51			
	(-0.66)	(-2.11)	(-4.34)	(-2.98)	(-6.75)			
α_{Q+2}	-0.11	-0.40	-0.77	-0.78	-1.46			
	(-0.52)	(-1.99)	(-3.83)	(-3.24)	(-6.52)			

Table A7: (continued)

Panel B: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	1.55	0.78	0.25	-0.01	-0.44	-1.99	-1.84	-1.95
	(5.22)	(4.19)	(1.26)	(-0.04)	(-1.47)	(-5.11)	(-5.11)	(-5.19)
INFO-Z 2	1.66	0.84	0.43	0.08	-0.06	-1.72	-1.53	-1.59
	(4.61)	(3.19)	(1.95)	(0.28)	(-0.20)	(-4.67)	(-4.16)	(-4.58)
INFO-Z 3	1.97	1.20	0.94	0.48	0.08	-1.89	-1.70	-1.68
	(5.27)	(4.74)	(3.09)	(1.63)	(0.25)	(-4.24)	(-4.48)	(-4.60)
INFO-Z 4	2.12	1.27	0.89	0.78	0.58	-1.54	-1.51	-1.58
	(5.95)	(4.35)	(2.96)	(2.66)	(1.61)	(-3.91)	(-3.87)	(-3.98)
INFO-Z High	3.04	1.74	1.23	1.36	1.05	-1.99	-1.86	-1.93
	(5.60)	(4.16)	(3.02)	(3.97)	(2.62)	(-4.16)	(-4.21)	(-4.43)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	1.49	0.97	0.98	1.36	1.49	0.00	-0.01	0.02
	(4.84)	(3.34)	(3.17)	(6.67)	(5.11)	(0.00)	(-0.05)	(0.10)
α_{FF6+2}	1.34	0.67	1.01	1.32	1.33			
	(4.46)	(2.97)	(3.49)	(7.19)	(4.94)			
α_{Q+2}	1.31	0.72	0.97	1.25	1.33			
	(4.61)	(3.38)	(3.44)	(7.21)	(5.12)			

Table A8: Cross-reversals in Monthly Overnight & Intraday Stock Returns, Conditional on INFO-Z: One-half Subperiods

This appendix table repeats the cross-reversal analysis in Table 4, but with separate estimations for one-half subperiods. Panel A reports on monthly intraday returns, conditional on last month's overnight returns and INFO-Z; and Panel B reports on monthly overnight returns, conditional on last month's intraday returns and INFO-Z. Panels A.1 and B.1 report on the 1997:01-2009:06 subperiod and Panels A.2 and B.2 on the 2009:07-2021:12 subperiod.

Panel A: Monthly Intdy Returns: Conditional on INFO-Z & Prior Month's Ovnt Return

Panel A.1: 1997:01-2009:06	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.93	0.59	0.11	-0.20	-0.74	-1.67	-1.80	-1.92
	(3.45)	(1.94)	(0.47)	(-0.64)	(-2.17)	(-6.54)	(-5.32)	(-5.87)
INFO-Z 2	1.09	0.40	0.37	-0.03	-1.28	-2.37	-2.58	-2.54
	(2.92)	(1.13)	(0.98)	(-0.08)	(-3.41)	(-4.97)	(-5.47)	(-5.63)
INFO-Z 3	1.36	0.34	-0.33	-0.20	-1.59	-2.96	-2.86	-2.76
	(3.59)	(0.93)	(-0.75)	(-0.50)	(-3.08)	(-8.19)	(-6.53)	(-7.41)
INFO-Z 4	0.57	0.11	-0.07	-1.26	-2.83	-3.40	-3.86	-3.95
	(1.14)	(0.22)	(-0.16)	(-2.49)	(-4.82)	(-6.53)	(-6.02)	(-5.64)
INFO-Z High	0.89	-0.23	-0.89	-1.86	-3.36	-4.26	-4.35	-4.30
	(1.42)	(-0.45)	(-1.35)	(-2.92)	(-5.01)	(-10.86)	(-10.61)	(-10.09)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-0.03	-0.82	-0.99	-1.66	-2.62	-2.59	-2.55	-2.39
	(-0.06)	(-1.50)	(-1.65)	(-2.58)	(-4.31)	(-6.77)	(-6.18)	(-5.83)
α_{FF6+2}	0.16	-0.39	-0.85	-1.32	-2.39			
	(0.54)	(-1.02)	(-2.45)	(-4.07)	(-6.86)			
α_{Q+2}	0.19	-0.24	-0.75	-1.16	-2.20			
	(0.60)	(-0.65)	(-2.09)	(-3.74)	(-6.37)			

Panel A.2: 2009:07-2021:12	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	0.71	0.47	0.77	0.37	0.26	-0.45	-0.43	-0.41
	(3.27)	(2.61)	(3.78)	(1.62)	(1.14)	(-2.02)	(-2.18)	(-1.97)
INFO-Z 2	0.69	0.68	0.56	0.02	0.13	-0.56	-0.56	-0.52
	(3.03)	(3.37)	(2.79)	(0.09)	(0.67)	(-2.59)	(-2.53)	(-2.36)
INFO-Z 3	0.63	0.40	0.24	-0.03	-0.58	-1.21	-1.20	-1.22
	(2.59)	(1.25)	(0.76)	(-0.08)	(-2.41)	(-5.54)	(-4.00)	(-4.22)
INFO-Z 4	0.48	0.19	0.02	-0.09	-0.69	-1.17	-1.22	-1.18
	(1.83)	(0.73)	(0.08)	(-0.34)	(-2.36)	(-5.41)	(-4.04)	(-3.66)
INFO-Z High	0.41	-0.07	-0.40	-0.63	-0.91	-1.32	-1.73	-1.64
	(1.35)	(-0.20)	(-1.34)	(-2.16)	(-3.01)	(-4.92)	(-5.74)	(-5.85)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	-0.30	-0.53	-1.16	-1.00	-1.17	-0.87	-1.30	-1.23
	(-0.94)	(-1.60)	(-3.78)	(-3.96)	(-3.35)	(-3.73)	(-5.19)	(-5.22)
α_{FF6+2}	-0.00	-0.59	-0.97	-0.73	-1.30			
	(-0.00)	(-3.05)	(-3.62)	(-2.44)	(-5.22)			
α_{Q+2}	0.02	-0.54	-0.94	-0.77	-1.22			
	(0.06)	(-2.62)	(-3.23)	(-2.77)	(-4.69)			

Table A8: (continued)

Panel B: Monthly Ovnt Returns: Conditional on INFO-Z & Prior Month's Intdy Return

Panel B.1: 1997:01-2009:06	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	1.48	0.26	-0.10	-0.50	-1.28	-2.76	-2.72	-2.79
	(3.31)	(0.83)	(-0.49)	(-2.04)	(-3.68)	(-5.90)	(-6.48)	(-6.29)
INFO-Z 2	1.76	0.60	0.17	-0.60	-0.88	-2.64	-2.55	-2.55
	(3.05)	(1.30)	(0.46)	(-1.56)	(-2.05)	(-5.64)	(-5.41)	(-5.37)
INFO-Z 3	2.69	1.23	0.60	-0.06	-0.64	-3.32	-2.99	-2.90
	(4.47)	(2.61)	(1.11)	(-0.15)	(-1.38)	(-5.08)	(-5.50)	(-5.64)
INFO-Z 4	2.95	1.42	0.52	0.39	-0.28	-3.22	-3.15	-3.19
	(5.38)	(2.86)	(1.06)	(0.76)	(-0.56)	(-5.81)	(-6.08)	(-5.82)
INFO-Z High	4.32	1.85	1.37	0.90	0.30	-4.02	-3.77	-3.82
	(4.35)	(3.16)	(1.86)	(1.56)	(0.47)	(-5.24)	(-4.97)	(-4.69)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	2.83	1.59	1.47	1.40	1.58	-1.26	-1.05	-1.03
	(4.60)	(3.86)	(2.41)	(3.64)	(3.12)	(-2.50)	(-1.87)	(-1.77)
α_{FF6+2}	2.69	1.59	1.46	1.31	1.64			
	(4.56)	(4.46)	(2.70)	(4.28)	(3.62)			
α_{Q+2}	2.59	1.54	1.27	1.22	1.56			
	(4.33)	(4.43)	(2.47)	(4.39)	(3.48)			

Panel B.2: 2009:07-2021:12	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
INFO-Z Low	1.27	0.82	0.72	0.48	0.50	-0.78	-0.64	-0.67
	(4.63)	(3.86)	(3.60)	(2.04)	(1.97)	(-4.72)	(-4.32)	(-3.91)
INFO-Z 2	1.38	1.06	0.72	0.81	0.63	-0.75	-0.48	-0.55
	(4.25)	(5.01)	(2.82)	(3.05)	(2.03)	(-2.71)	(-1.97)	(-1.95)
INFO-Z 3	1.29	1.12	1.21	1.16	0.79	-0.50	-0.40	-0.41
	(3.80)	(4.71)	(4.18)	(3.43)	(2.41)	(-2.13)	(-2.11)	(-1.81)
INFO-Z 4	1.43	1.16	1.27	1.16	1.29	-0.14	-0.11	-0.12
	(3.42)	(4.09)	(3.37)	(3.39)	(3.64)	(-0.73)	(-0.64)	(-0.72)
INFO-Z High	2.36	1.88	1.58	1.60	1.75	-0.61	-0.69	-0.81
	(5.62)	(3.43)	(3.76)	(4.94)	(4.02)	(-2.12)	(-2.53)	(-2.54)
High-Low INFO-Z Strategies	Low	2	3	4	High			
Excess	1.09	1.06	0.86	1.11	1.26	0.17	-0.05	-0.14
	(4.61)	(2.61)	(2.37)	(5.05)	(4.19)	(0.65)	(-0.19)	(-0.46)
α_{FF6+2}	0.99	0.54	0.80	1.03	0.94			
	(4.46)	(2.61)	(2.73)	(5.66)	(3.65)			
α_{Q+2}	1.08	0.67	0.83	1.05	0.94			
	(3.95)	(2.60)	(2.80)	(5.88)	(3.56)			

Table A9: 52-week PTA Momentum Results with Independently Double-sorted Portfolios

This appendix table analyzes momentum in both monthly overnight stock returns and monthly intraday stock returns conditional on a stock's PTA, similar to Table 6 but with independent sorts on a stock's PTA and lagged return. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 25 (5 by 5) portfolios independently sorted on a stock's lagged PTA and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are sorted into quintiles based on their PTA ranking as of the close of month $t-2$ using NYSE breakpoints. In addition, stocks are independently sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. Portfolios are formed using the intersection of the two independent sorts. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.05	-0.07	0.41	0.53	1.64	1.59	1.51	1.45
	(0.19)	(-0.30)	(1.77)	(2.07)	(4.47)	(4.60)	(4.25)	(4.41)
PTA 2	-0.04	-0.10	0.19	0.54	0.98	1.01	0.95	0.90
	(-0.15)	(-0.42)	(1.03)	(2.44)	(3.75)	(3.60)	(3.69)	(3.60)
PTA 3	-0.07	0.07	0.23	0.41	1.04	1.10	1.10	1.08
	(-0.23)	(0.28)	(1.11)	(1.96)	(3.75)	(3.19)	(3.24)	(3.23)
PTA 4	-0.08	0.17	0.53	0.89	1.29	1.36	1.25	1.23
	(-0.31)	(0.67)	(2.90)	(4.78)	(4.56)	(4.33)	(4.59)	(4.48)
PTA High	0.58	0.91	1.01	1.51	3.00	2.42	2.25	2.17
	(2.07)	(3.47)	(4.58)	(6.53)	(6.79)	(5.57)	(5.52)	(5.65)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.52	0.98	0.60	0.98	1.36	0.84	0.74	0.72
	(2.80)	(5.68)	(3.09)	(5.43)	(4.37)	(2.83)	(2.73)	(2.65)
α_{FF6+2}	0.57	0.96	0.47	0.80	1.31			
	(2.54)	(5.44)	(2.70)	(4.37)	(4.95)			
α_{Q+2}	0.52	0.92	0.52	0.79	1.24			
	(2.46)	(4.95)	(3.14)	(4.17)	(4.19)			

Table A9: (continued)

Panel B: Monthly Intdy Returns: Conditional on PTA & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-1.52	0.04	0.29	0.25	0.56	2.08	2.00	2.05
	(-2.58)	(0.10)	(0.85)	(0.90)	(2.01)	(3.96)	(4.48)	(4.34)
PTA 2	-0.70	0.33	0.64	0.58	1.01	1.71	1.58	1.72
	(-2.14)	(1.27)	(2.91)	(2.56)	(4.53)	(5.46)	(5.03)	(5.44)
PTA 3	-0.84	0.11	0.39	0.65	1.02	1.85	1.51	1.65
	(-2.25)	(0.44)	(2.30)	(3.08)	(5.70)	(4.62)	(4.33)	(4.40)
PTA 4	-0.90	-0.13	0.13	0.41	0.76	1.66	1.43	1.54
	(-2.91)	(-0.72)	(0.63)	(1.99)	(3.95)	(5.30)	(4.67)	(5.03)
PTA High	-1.52	-0.50	-0.62	-0.54	-0.58	0.93	0.77	0.88
	(-4.02)	(-2.09)	(-2.65)	(-2.22)	(-2.40)	(2.97)	(3.00)	(3.41)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.01	-0.54	-0.91	-0.78	-1.14	-1.15	-1.23	-1.17
	(0.02)	(-1.44)	(-2.46)	(-2.52)	(-4.02)	(-3.10)	(-3.23)	(-3.01)
α_{FF6+2}	-0.31	-0.88	-1.55	-1.23	-1.53			
	(-0.82)	(-2.35)	(-4.48)	(-3.80)	(-6.10)			
α_{Q+2}	-0.46	-1.03	-1.71	-1.22	-1.64			
	(-1.31)	(-2.50)	(-4.38)	(-3.47)	(-6.05)			

Table A10: PTA Momentum Results using 3-by-10 Double-Sorted Portfolios

This appendix table reports on robustness for the PTA momentum results by evaluating different granularity for the double-sorted portfolios. Panel A (B) reports the average value-weighted excess overnight (intraday) returns in month t for 30 (3 by 10) portfolios sequentially sorted on a stock's lagged 52-week PTA and lagged overnight (intraday) returns. At the end of month $t-1$, stocks are first sorted into terciles based on their PTA ranking as of the close of month $t-2$ using NYSE breakpoints. Within each tercile, stocks are further sorted into deciles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021. Please see Section 3 for details of the sample selection criteria and variable description.

Panel A: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Ovnt Return

	Past Overnight Return Decile					High - Low		
	Low	2	5	9	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-0.06	-0.22	0.35	1.09	1.83	1.88	1.83	1.85
	(-0.20)	(-0.74)	(1.74)	(3.59)	(4.19)	(3.89)	(3.91)	(3.87)
PTA Medium	-0.06	-0.01	0.20	0.66	1.15	1.21	1.19	1.15
	(-0.20)	(-0.05)	(0.87)	(3.40)	(4.16)	(3.11)	(3.23)	(3.16)
PTA High	0.14	0.51	0.86	2.22	3.03	2.89	2.62	2.58
	(0.48)	(1.90)	(3.98)	(6.79)	(6.51)	(6.00)	(6.08)	(6.32)
High-Low PTA Strategies	Low	2	5	9	High			
Excess	0.19	0.72	0.51	1.13	1.20	1.01	0.80	0.73
	(1.10)	(3.66)	(2.74)	(4.66)	(3.95)	(3.40)	(2.77)	(2.17)
α_{FF6+2}	0.35	0.79	0.51	1.08	1.15			
	(1.78)	(4.18)	(2.71)	(3.99)	(4.50)			
α_{Q+2}	0.29	0.74	0.47	1.15	1.01			
	(1.40)	(3.80)	(2.59)	(4.56)	(3.02)			

Table A10: (continued)

Panel B: Monthly Intdy Returns: Conditional on PTA & Prior Month's Intdy Return

	Past Intraday Return Decile					High - Low		
	Low	2	5	9	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	-2.28	-0.72	0.77	0.69	0.82	3.10	2.92	3.12
	(-3.29)	(-1.69)	(2.27)	(2.53)	(2.73)	(4.75)	(4.75)	(4.58)
PTA Medium	-1.08	-0.32	0.40	0.77	1.13	2.21	1.80	1.91
	(-2.69)	(-1.13)	(2.38)	(3.41)	(5.80)	(5.30)	(5.23)	(5.11)
PTA High	-1.83	-0.95	-0.18	-0.25	-0.19	1.63	1.41	1.57
	(-4.34)	(-3.17)	(-0.83)	(-1.08)	(-0.75)	(3.98)	(4.12)	(4.65)
High-Low PTA Strategies	Low	2	5	9	High			
Excess	0.45	-0.23	-0.96	-0.94	-1.02	-1.47	-1.51	-1.55
	(1.08)	(-0.60)	(-2.63)	(-3.45)	(-3.53)	(-2.92)	(-2.80)	(-2.49)
α_{FF6+2}	0.18	-0.64	-1.28	-1.30	-1.33			
	(0.35)	(-1.92)	(-3.33)	(-4.48)	(-4.65)			
α_{Q+2}	0.13	-0.71	-1.36	-1.48	-1.42			
	(0.24)	(-1.93)	(-3.04)	(-5.14)	(-4.94)			

Table A11: Cross-reversals in Monthly Overnight & Intraday Stock Returns, Conditional on PTA: Independent Sorts

This appendix table reports the cross-reversal PTA results, similar to Table 9 but with independent sorts on the two conditioning variables. Panel A (B) reports the average value-weighted excess intraday(overnight) returns in month t for 25 (5 by 5) portfolios independently sorted on a stock's lagged PTA and lagged monthly overnight (intraday) returns. At the end of month $t-1$, stocks are sorted into quintiles based on their PTA ranking as of the close of month $t-2$ using NYSE breakpoints. In addition, stocks are independently sorted into quintiles based on their past one-month overnight or intraday returns using NYSE breakpoints in month $t-1$. Portfolios are formed using the intersection of the two independent sorts. We also report the high-minus-low portfolio excess returns and alphas based on an extended Fama-French five-factor model (2015) and an extended version of Hou, Xue, and Zhang's (2015) Q-factor. The Fama-French five-factor model is extended by adding the momentum factor from Carhart (1997) and the two behavioral factors from Daniel et al. (2020). The Q-factor model is also extended by adding the same two behavioral factors. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period is from 1997 to 2021.

Panel A: Monthly Intdy Returns: Conditional on PTA & Prior Month's Ovnt Return

	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.82	0.54	0.13	-0.14	-1.20	-2.03	-1.93	-1.97
	(2.12)	(1.63)	(0.38)	(-0.46)	(-3.34)	(-5.92)	(-5.29)	(-5.27)
PTA 2	0.81	0.75	0.73	0.49	-0.29	-1.10	-1.17	-1.12
	(3.26)	(3.07)	(3.33)	(2.06)	(-1.04)	(-4.12)	(-3.76)	(-3.57)
PTA 3	1.04	0.56	0.44	0.12	-0.39	-1.43	-1.49	-1.45
	(4.44)	(3.12)	(1.95)	(0.52)	(-1.32)	(-3.78)	(-4.18)	(-4.15)
PTA 4	0.73	0.45	0.17	-0.29	-0.65	-1.37	-1.41	-1.39
	(3.62)	(2.26)	(0.97)	(-1.66)	(-2.58)	(-6.11)	(-6.13)	(-5.65)
PTA High	0.48	-0.28	-0.61	-0.82	-1.74	-2.21	-2.33	-2.28
	(1.77)	(-1.15)	(-2.72)	(-2.95)	(-4.86)	(-6.89)	(-7.02)	(-6.83)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.35	-0.82	-0.74	-0.68	-0.53	-0.19	-0.40	-0.31
	(-0.99)	(-2.55)	(-2.18)	(-2.12)	(-2.03)	(-0.63)	(-1.12)	(-0.86)
α_{FF6+2}	-0.72	-1.01	-0.96	-0.84	-1.12			
	(-2.27)	(-3.69)	(-3.03)	(-2.41)	(-3.50)			
α_{Q+2}	-0.88	-1.15	-1.17	-0.96	-1.19			
	(-2.32)	(-3.25)	(-4.07)	(-2.50)	(-3.94)			

Table A11: (continued)

Panel B: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Intdy Return

	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	2.21	0.78	0.20	0.06	-0.29	-2.50	-2.42	-2.49
	(4.83)	(2.90)	(0.73)	(0.18)	(-0.88)	(-4.22)	(-4.21)	(-4.27)
PTA 2	1.47	0.62	0.28	-0.12	-0.50	-1.97	-1.93	-1.96
	(5.19)	(2.72)	(1.41)	(-0.47)	(-1.57)	(-5.24)	(-5.69)	(-5.84)
PTA 3	1.59	0.82	0.09	-0.07	-0.39	-1.98	-1.85	-1.92
	(5.14)	(3.92)	(0.42)	(-0.30)	(-1.25)	(-5.23)	(-5.18)	(-5.38)
PTA 4	1.59	0.77	0.41	0.23	0.02	-1.58	-1.38	-1.45
	(4.25)	(4.70)	(2.15)	(0.93)	(0.06)	(-3.46)	(-3.31)	(-3.34)
PTA High	2.48	1.64	1.48	1.17	1.18	-1.30	-1.13	-1.18
	(6.51)	(5.40)	(5.30)	(4.05)	(3.38)	(-3.79)	(-3.44)	(-3.78)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.26	0.86	1.27	1.11	1.47	1.20	1.29	1.31
	(0.84)	(4.60)	(8.28)	(5.06)	(5.70)	(3.05)	(3.01)	(3.04)
α_{FF6+2}	0.19	0.79	1.16	1.26	1.48			
	(0.62)	(3.99)	(7.75)	(6.84)	(6.51)			
α_{Q+2}	0.12	0.72	1.16	1.16	1.43			
	(0.35)	(3.32)	(8.13)	(5.73)	(6.62)			

Table A12: Cross-reversals in Monthly Overnight & Intraday Stock Returns, Conditional on PTA: One-half Subperiods

This appendix table repeats the cross-reversal analysis in Table 9, but with separate estimations for one-half subperiods. Panel A reports on intraday returns, conditional on last month's overnight returns and PTA; and Panel B on overnight returns, conditional on last month's intraday returns and PTA. Panels A.1 and B.1 report on the 1997:01-2009:06 subperiod and Panels A.2 and B.2 on the 2009:07-2021:12 subperiod.

Panel A: Monthly Intdy Returns: Conditional on PTA & Prior Month's Ovnt Return

Panel A.1: 1997:01-2009:06	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	1.01	0.76	-0.03	-0.61	-2.50	-3.50	-3.43	-3.51
	(1.37)	(1.45)	(-0.05)	(-1.27)	(-3.93)	(-6.97)	(-6.47)	(-6.61)
PTA 2	0.68	0.89	0.87	0.36	-0.83	-1.51	-1.70	-1.69
	(1.70)	(2.35)	(2.30)	(0.88)	(-1.89)	(-3.58)	(-3.89)	(-4.06)
PTA 3	1.06	0.75	0.15	0.09	-0.95	-2.01	-2.04	-2.01
	(3.31)	(2.49)	(0.54)	(0.22)	(-2.69)	(-4.97)	(-5.00)	(-4.73)
PTA 4	0.79	0.54	0.11	-0.59	-1.14	-1.93	-1.97	-2.05
	(2.67)	(1.51)	(0.36)	(-1.85)	(-4.39)	(-9.42)	(-7.61)	(-7.92)
PTA High	0.42	-0.60	-0.83	-1.45	-2.92	-3.34	-3.45	-3.34
	(0.91)	(-1.70)	(-2.71)	(-3.17)	(-5.35)	(-7.71)	(-6.72)	(-6.31)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.59	-1.37	-0.80	-0.84	-0.43	0.16	-0.01	0.17
	(-0.93)	(-2.41)	(-1.60)	(-1.95)	(-0.95)	(0.28)	(-0.02)	(0.25)
α_{FF6+2}	-1.13	-1.60	-1.30	-1.02	-1.15			
	(-2.01)	(-3.54)	(-3.39)	(-2.15)	(-2.09)			
α_{Q+2}	-1.54	-2.08	-1.76	-1.33	-1.37			
	(-2.37)	(-3.28)	(-5.39)	(-2.83)	(-2.50)			

Panel A.2: 2009:07-2021:12	Past Overnight Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	0.61	0.46	0.37	-0.17	-0.38	-0.99	-0.84	-0.89
	(2.07)	(1.26)	(0.96)	(-0.54)	(-1.54)	(-4.92)	(-3.06)	(-2.82)
PTA 2	1.04	0.55	0.66	0.68	0.34	-0.69	-0.52	-0.49
	(4.58)	(1.91)	(3.79)	(2.43)	(1.83)	(-2.98)	(-1.69)	(-1.59)
PTA 3	0.76	0.46	0.74	0.35	0.44	-0.32	-0.45	-0.39
	(2.48)	(2.32)	(3.34)	(1.37)	(2.02)	(-0.99)	(-1.76)	(-1.49)
PTA 4	0.56	0.42	0.41	0.09	-0.12	-0.68	-0.73	-0.69
	(2.83)	(2.01)	(1.69)	(0.49)	(-0.56)	(-3.81)	(-4.20)	(-3.97)
PTA High	0.39	-0.16	-0.52	-0.40	-0.80	-1.19	-1.35	-1.27
	(1.75)	(-0.52)	(-1.65)	(-1.33)	(-2.85)	(-4.74)	(-5.30)	(-4.88)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.22	-0.62	-0.89	-0.22	-0.42	-0.20	-0.51	-0.38
	(-0.81)	(-1.49)	(-1.94)	(-0.47)	(-1.40)	(-0.66)	(-1.27)	(-0.93)
α_{FF6+2}	-0.56	-0.66	-1.14	-0.35	-1.07			
	(-2.07)	(-1.92)	(-3.19)	(-0.94)	(-3.48)			
α_{Q+2}	-0.56	-0.70	-1.06	-0.29	-0.94			
	(-1.94)	(-2.03)	(-2.64)	(-0.68)	(-3.28)			

Table A12: (continued)

Panel B: Monthly Ovnt Returns: Conditional on PTA & Prior Month's Intdy Return

Panel B.1: 1997:01-2009:06	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	3.54	1.23	0.03	-0.61	-1.38	-4.91	-4.92	-4.97
	(4.28)	(2.77)	(0.06)	(-1.73)	(-3.08)	(-5.26)	(-5.07)	(-4.82)
PTA 2	1.67	0.36	-0.11	-0.60	-1.59	-3.26	-3.19	-3.20
	(3.63)	(1.14)	(-0.36)	(-1.81)	(-4.47)	(-5.96)	(-6.82)	(-6.71)
PTA 3	1.62	0.46	-0.58	-0.72	-1.30	-2.92	-2.77	-2.76
	(3.07)	(1.27)	(-1.67)	(-2.30)	(-3.55)	(-5.09)	(-5.32)	(-5.26)
PTA 4	1.81	0.43	-0.02	-0.24	-0.90	-2.71	-2.50	-2.56
	(3.37)	(1.66)	(-0.07)	(-1.03)	(-2.38)	(-4.77)	(-4.99)	(-4.72)
PTA High	3.06	1.59	1.41	0.80	0.68	-2.38	-2.20	-2.15
	(4.51)	(3.71)	(2.95)	(1.91)	(1.26)	(-4.44)	(-4.89)	(-4.52)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	-0.48	0.36	1.38	1.40	2.05	2.53	2.72	2.81
	(-0.59)	(1.31)	(5.10)	(6.29)	(5.12)	(2.94)	(3.00)	(2.96)
α_{FF6+2}	-0.69	0.11	1.30	1.33	2.03			
	(-0.86)	(0.47)	(4.75)	(6.90)	(6.11)			
α_{Q+2}	-0.85	0.10	1.21	1.24	1.96			
	(-0.98)	(0.41)	(4.54)	(6.18)	(6.17)			

Panel B.2: 2009:07-2021:12	Past Intraday Return Quintile					High - Low		
	Low	2	3	4	High	Excess	α_{FF6+2}	α_{Q+2}
PTA Low	1.20	1.05	0.89	0.48	0.71	-0.49	-0.36	-0.39
	(2.74)	(2.60)	(2.37)	(1.17)	(2.45)	(-1.40)	(-1.37)	(-1.30)
PTA 2	1.32	0.83	0.62	0.58	0.51	-0.81	-0.77	-0.81
	(3.58)	(3.59)	(2.12)	(1.99)	(1.50)	(-2.95)	(-4.20)	(-3.57)
PTA 3	1.49	0.76	0.70	0.60	0.49	-1.00	-0.87	-0.89
	(5.30)	(3.21)	(3.26)	(2.28)	(1.77)	(-4.31)	(-4.45)	(-3.75)
PTA 4	1.10	0.95	0.90	0.86	0.77	-0.33	-0.17	-0.21
	(3.96)	(4.57)	(3.59)	(3.56)	(2.29)	(-1.30)	(-0.70)	(-0.76)
PTA High	2.18	1.64	1.75	1.67	1.82	-0.35	-0.10	-0.15
	(6.41)	(4.57)	(5.54)	(6.05)	(5.51)	(-2.11)	(-0.41)	(-0.58)
High-Low PTA Strategies	Low	2	3	4	High			
Excess	0.98	0.60	0.87	1.19	1.12	0.14	0.26	0.25
	(3.11)	(2.31)	(2.55)	(4.70)	(5.27)	(0.43)	(0.79)	(0.75)
α_{FF6+2}	0.99	0.74	0.92	1.06	1.25			
	(3.35)	(2.92)	(3.99)	(4.22)	(5.19)			
α_{Q+2}	0.96	0.72	0.81	1.03	1.21			
	(2.92)	(2.78)	(2.94)	(3.94)	(5.30)			

Table A13: Fama-MacBeth Regressions with Additional Interactive-Return Control Variables

This appendix table reports on a second set of extended Fama-MacBeth regressions with additional interactive-return control terms. Here, our primary interest is whether the INFO-Z and PTA interactive-return terms retain their empirical importance when adding interactive-return terms with additional key control variables. Specifically, we estimate a specification that is structured like that in Table 12 but where the LMQ, JPQ, and ABNR interactive-return terms are replaced by other interactive-return terms where the stock-specific interactive variables are now the natural log of a stock's market capitalization (size), the book-to-market equity ratio (BM), and the market beta (β). For ease of coefficient interpretation on the new interactive-return terms, we use the Z-standardized transformation of the size, BM, and beta variables (that have a mean of zero and standard deviation of one) in the interactive terms. These three control variables were included previously in the level control terms only. See equation (A.1) for the INFO-Z specification and equation (A.2) for the PTA specification. Columns (1) to (3) report the estimation results for three variations of the INFO-Z equation (A.1), with the monthly overnight returns from month t as the dependent variable. Columns (4) to (6) report the estimation results for three variations of the PTA equation (A.2), with the monthly intraday returns from month t as the dependent variable. In the table, the additional control variables in the 'Control' row are as listed for Table 12. We also include the industry fixed effects, using the Fama-French 48-industry definitions. In parentheses, we report t-statistics computed using Newey–West (1987) standard errors, corrected for serial dependence with 12 lags. The sample period spans from 1997 to 2021.

[Tabular Results on the next page]

Table A13: (continued)

Dependent Variable:	[1] Overnight Returns	[2] Overnight Returns	[3] Overnight Returns	[4] Intraday Returns	[5] Intraday Returns	[6] Intraday Returns
Intercept	0.34	0.53	0.06	0.33	0.34	0.57
	(1.59)	(2.17)	(0.21)	(0.78)	(0.77)	(1.28)
ret night	7.57	9.55	8.41	-9.55	-9.16	-8.03
	(8.79)	(8.03)	(7.78)	(-10.20)	(-9.29)	(-8.69)
INFO-Z × ret night		3.89	2.75			
		(2.71)	(2.01)			
PTA × ret night					-0.57	-0.13
					(-0.55)	(-0.13)
LnSize × ret night	0.37	1.09	0.61			
	(0.80)	(2.17)	(1.31)			
BM × ret night	0.47	0.28	0.11			
	(1.33)	(0.81)	(0.31)			
Beta × ret night	0.62	0.08	-0.20			
	(1.44)	(0.18)	(-0.43)			
ret day	-5.25	-5.87	-5.66	1.75	1.95	1.92
	(-5.80)	(-6.11)	(-5.80)	(1.91)	(2.12)	(2.06)
INFO-Z × ret day		-1.97	-1.58			
		(-2.40)	(-2.02)			
PTA × ret day					-2.13	-1.88
					(-3.60)	(-2.97)
LnSize × ret day				-0.09	-0.12	-0.16
				(-0.19)	(-0.24)	(-0.33)
BM × ret day				0.66	0.49	0.37
				(2.30)	(1.67)	(1.18)
Beta × ret day				1.19	1.10	1.13
				(2.55)	(2.37)	(2.50)
INFO-Z		0.30	-0.02			
		(2.11)	(-0.13)			
PTA					0.07	0.54
					(0.38)	(2.27)
LnSize	0.04	0.08	0.03	-0.19	-0.18	-0.20
	(0.86)	(1.61)	(0.49)	(-1.95)	(-2.00)	(-2.20)
BM	0.05	0.03	0.01	-0.05	-0.02	-0.02
	(1.88)	(1.51)	(0.23)	(-0.97)	(-0.31)	(-0.48)
Beta	0.45	0.40	0.24	-0.41	-0.40	-0.22
	(5.80)	(4.99)	(4.61)	(-3.47)	(-3.37)	(-2.54)
Controls	No	No	Yes	No	No	Yes
FF48	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R2	0.10	0.11	0.12	0.15	0.16	0.18